Deye RWA Platform Product and Operating Manual

Internal test edition, 12 November 2026

This manual explains how the four participant roles use wallets to complete applications, valuation, fundraising, settlement, monthly distributions and exits. Equipment remains the owner's property. Investors acquire shares in finite revenue rights, with no fixed coupon or guaranteed repayment of their original investment. All payments use demonstration funds.

The instructions cover dynamic guidance, household wallet payments, monthly distributions and the separate self-hosted Oracle test environment. The unified site has three environments, two on Base Sepolia with Chain ID 84532 and one private demo on Chain ID 20261008. Screenshots come from isolated tests and may show different contracts, balances or dates. Verify the environment, product and wallet before acting.

Every illustrated workflow step uses the English interface. The monthly payment and reconciliation images were captured for this release; unchanged issuance, exit and closure images retain the verified English captures. Station names use English display translations without changing energy observations. These examples do not imply that the shared product has reached the illustrated stage.

The payment release is tested through two consecutive months of household payments, receipt confirmation and distributions on an isolated chain. The browser test uses dedicated test keys to sign actual transactions outside the browser and does not claim renewed Rabby extension certification. Advanced transfer, redemption, servicing and expiry instructions retain their validated business rules and must follow the current window conditions. This remains demonstration software. Real tariff settlement, bank clearing, production identity compliance and a full security audit are not complete.

Contents

1 Testing approaches and their purposes **7**

1.1 Selecting a test path 9

2 Product states and role handoffs **10**

2.1 Application, admission and issuance sequence 11

2.2 Monthly bills and household payment 12

2.3 Monthly valuation and investor handoff 13

3 System composition and architecture **14**

3.1 System composition 14

3.2 Business architecture 15

3.3 Data architecture 16

3.4 Functional architecture 17

3.5 Technical architecture 18

4 Smart contract architecture and business sequence **19**

4.1 Architecture of the currently deployed contracts 20

4.2 Contract responsibilities and deployment boundaries 21

4.3 Cross-contract sequence (issuance and settlement) 23

4.4 Cross-contract sequence (billing, distributions and exit) 25

5 Wallets and test accounts **27**

5.1 Role accounts 1 28

5.2 Role accounts 2 29

5.3 Role accounts 3 30

6 Website, network and blockchain explorer **31**

6.1 Unified public entry 31

6.2 Public homepage 31

6.3 Business application 31

6.4 Blockchain explorer . . . 31
6.5 Wallet RPC . . . 31
6.6 Network selection and session isolation . . . 31

7 Full lifecycle testing on the private demonstration 33

7.1 Purpose and prerequisites . . . 33
7.2 Entry and network confirmation . . . 33
7.3 Direct investor testing . . . 33
7.4 Lifecycle and illustrated-step index . . . 34
7.5 Clock advancement, shared state and evidence . . . 35

8 Base Sepolia public-chain testing 36

8.1 Test objectives and the separate Oracle environment . . . 36
8.2 Entry and network configuration . . . 36
8.3 Environment components and data flow . . . 37
8.4 Funding and subscription steps . . . 37
8.5 Public-chain checks and completion criteria . . . 38
8.6 Real time and testing scope . . . 38
8.7 Transaction checks and recovery . . . 39

9 Local Oracle on Sepolia testing 42

9.1 Entry point and contracts . . . 42
9.2 Data and transaction flow . . . 43
9.3 Completed checks and the later billing period . . . 43
9.4 Operation, reconciliation and limits . . . 44

10 Local deployment and independent testing 45

10.1 Purpose and default environment . . . 45
10.2 Installation, startup and health checks . . . 46
10.3 Local entry points and wallets . . . 46
10.4 Empty scenes, data sources and workflow reproduction . . . 47
10.5 Development regression, shutdown and recovery . . . 48

11 Dynamic guidance and handoffs **49**

11.1 Product overview and task entry 50

12 Application entry **51**

12.1 A0 Application access 51

12.2 A1 Role selection and wallet sign-in 52

12.3 A2 Creation of an issuance draft 53

12.4 A3 Test funds for the selected product 54

13 Application, valuation and agreement **55**

13.1 B1 Station identity and evidence source 55

13.2 B2 Station dashboard review 56

13.3 B3 Revenue-right application 57

13.4 B4 Application queue 58

13.5 B5 Application assessment 59

13.6 B6 Revenue-right valuation and offer 60

13.7 B7 Independent offer approval 61

13.8 B8 Owner cashflow and electricity discount 62

13.9 B9 Acceptance of the approved offer 63

13.10 B10 Revenue-right agreement signature 64

14 Credentials, admission and publication **65**

14.1 C1 Station credential minting 65

14.2 C2 Pool custody authorization 66

14.3 C3 Revenue-right admission 67

14.4 C4 Pool valuation submission 68

14.5 C5 Pool valuation approval 69

14.6 C6 Fundraising publication 70

15 Qualification, subscriptions and settlement **71**

15.1 D1 Product selection 71

15.2 D2 Investor cashflow timeline 72

15.3 D3 Investment qualification application 73
15.4 D4 Qualification approval 74
15.5 D5 USDC approval and subscription 75
15.6 D6 Cancellation before settlement 76
15.7 D7 Cumulative subscription review 77
15.8 D8 Fundraising settlement and share issuance 78
15.9 D9 Confirmed investment position 79

16 Monthly collections, valuation and distributions 80

16.1 E1 Demonstration financial cutoff 80
16.2 E2 Publish the monthly payment bill 81
16.3 E3 Approve and pay the monthly bill 82
16.4 E3a Record the separate grid-export receipt 83
16.5 E3b Reconcile and confirm receipt into the pool 84
16.6 E4 Monthly pool expense accrual 85
16.7 E5 Monthly valuation approval 86
16.8 E6 Monthly distribution declaration 87
16.9 E7 Distribution claim 88

17 Peer transfers and quarterly redemption 89

17.1 F1 Peer transfer proposal 89
17.2 F2 Buyer consent and payment approval 90
17.3 F3 Transfer price review 91
17.4 F4 Atomic peer settlement 92
17.5 F5 Quarterly redemption request 93
17.6 F6 Proportional redemption processing 94
17.7 F7 Cancellation of the unfilled request 95

18 Servicing cases and trustee intervention 96

18.1 G1 Servicing case creation 96
18.2 G2 Snapshot-weighted support 97

18.3 G3 First trustee approval 98
18.4 G4 Second trustee approval and execution 99

19 Expiry, closure and credential release 100

19.1 H1 Product closure at rights expiry 100
19.2 H2 Final and outstanding distribution claims 101
19.3 H3 Station credential release 102

20 Cashflow calculations and valuation 103

20.1 Discount-rate meaning and pricing basis 103
20.2 Currency convention and current system limitation 104
20.3 Comparables and approval record 104
20.4 Comparison of the three valuation baselines 105
20.5 PVGIS irradiation-based climate valuation 106
20.6 Provisional Shanghai station inputs 106
20.7 Owner cashflow timeline for the first three periods 108
20.8 Investor cashflow timeline for the first three months 109

21 On-chain record reconciliation 110

22 Troubleshooting and test boundaries 112

22.1 Wallet or role mismatch 112
22.2 Insufficient settlement funding 112
22.3 Missing valuation or fee prerequisites 112
22.4 Distribution or balance not updated 112
22.5 Rejection, pending transactions and recovery 112
22.6 Remote environment unavailable 112

1 Testing approaches and their purposes

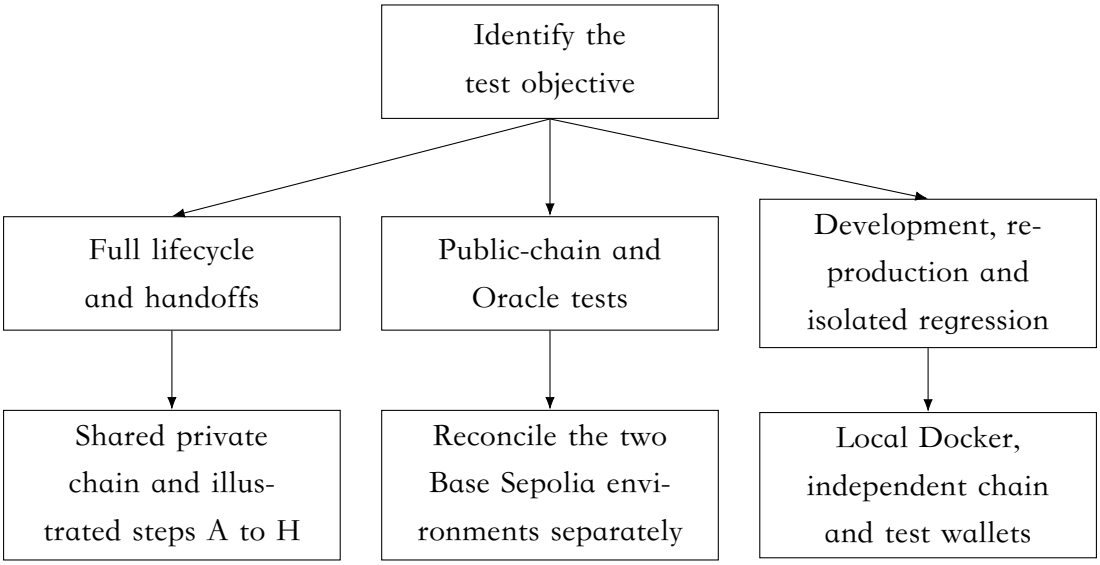
The unified site offers three business environments: the private demonstration, the original Base Sepolia product, and Local Oracle on Sepolia. A separate local deployment is also available. The private chain supports the complete lifecycle under a controlled clock. The original Base Sepolia product tests public-chain compatibility. The Oracle environment tests Deye API retrieval, signed report registration, and mandatory verification by the new billing contract. A local deployment supports development and regression testing.

Local deployment is a hosting approach, not a third public blockchain. The default local profile creates its own private Anvil chain. The shared private chain and a new local chain run the same business logic but have different Chain IDs, contracts, account configuration and storage. Conversely, a backend running on your computer but connected to Base Sepolia is still a public-chain test and cannot advance or reset that chain.

Approach	Main purpose	Recommended checks	Time and funding
Shared private demo	Complete lifecycle demonstrations and multi-role business acceptance.	Applications, issuance, monthly payments, distributions, transfers, redemptions, servicing and closure.	20261008; test mUSDC; intermediary-controlled demo time.
Base Sepolia	Public-chain compatibility and a baseline for later oracle integration.	Wallets, gas, Circle USDC, approvals, subscriptions, refunds, receipts and recovery.	84532; test ETH and USDC; real chain time.
Local Oracle on Sepolia	Validate the independent self-hosted Oracle product and enforced report checks.	Direct Deye API reads, EIP-712 signing, on-chain registration and trustee policy approval.	Also 84532; wallet test-token balances are shared with the original Base product.
Independent local deployment	Code changes, issue reproduction and repeatable isolated regression.	Build a product from an empty scene and check the UI, model, permissions and workflow.	Default 20260922; independent wallets and local test tokens.

The three business environments reuse the role addresses. Token balances are separate between the private chain and Base Sepolia. The two Base Sepolia environments share the same chain, so test ETH and Circle USDC balances in a given wallet are shared. Product contracts, eligibility, holdings, records and sign-in sessions remain isolated by environment. A fresh local instance normally generates different accounts.

1.1 Selecting a test path



Use the private-chain illustrations for the full lifecycle. Choose the original Base Sepolia product for public-chain transactions and the third entry for Oracle reports. Both public-chain environments follow real chain time; demonstration controls cannot bring a monthly bill forward.

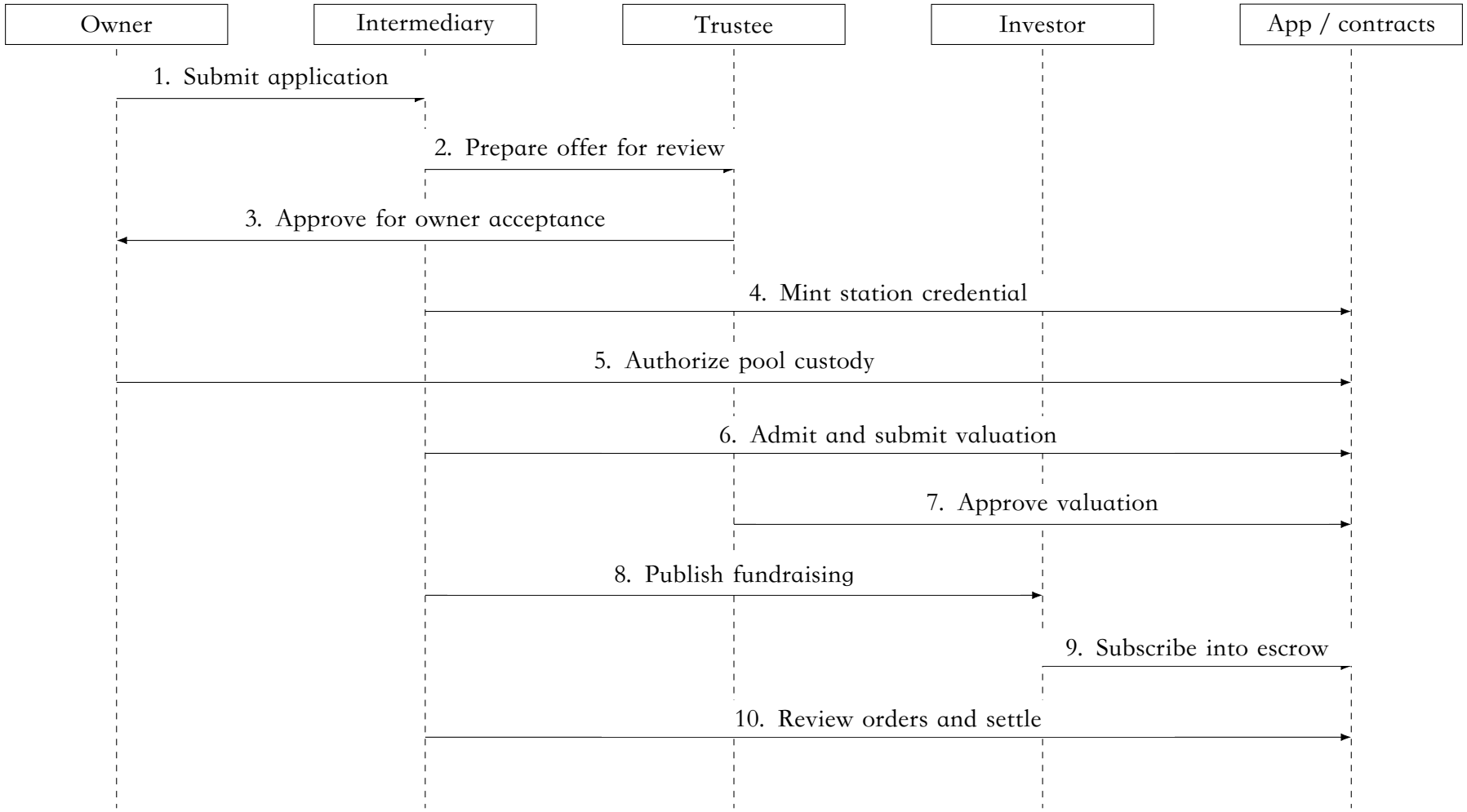
2 Product states and role handoffs

A product starts as a draft, opens fundraising after approval and issues DRS only on successful settlement. It then enters servicing. Failed fundraising allows refunds, and a live product is reconciled and closed when its rights expire. Contract deployment is distinct from publication, and escrowed subscriptions are distinct from issued shares.

The owner applies, signs the rights agreement and pays monthly bills using their own wallet. The intermediary assesses materials, prepares valuations, organizes issuance and operates the pool. Trustees independently approve offers and pool valuations. Investors complete qualification, subscriptions, claims and exits. Two separate trustees are required for the defined servicing intervention, rather than for every ordinary valuation approval.

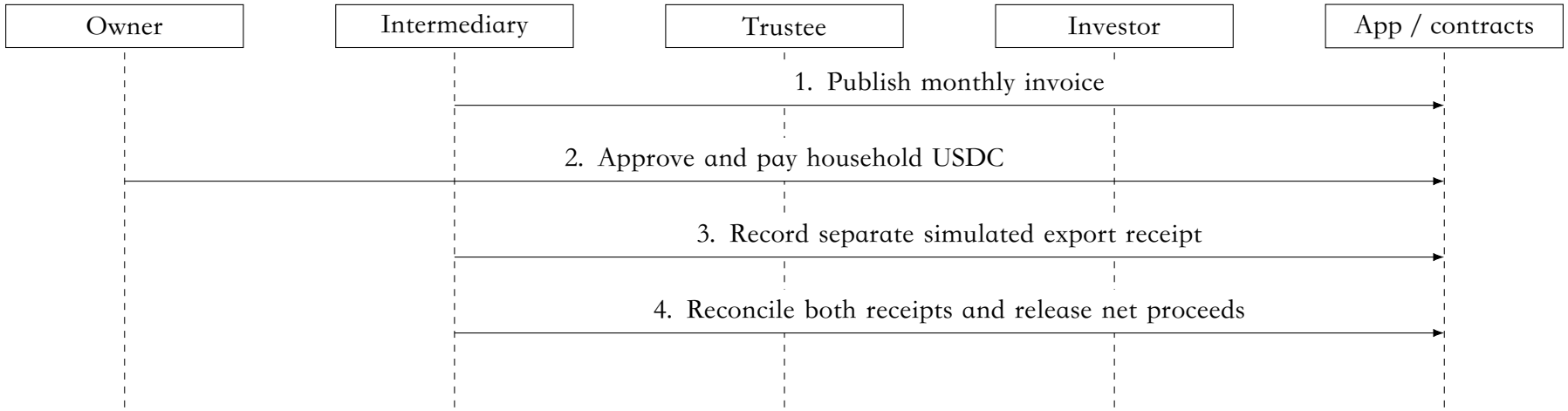
2.1 Application, admission and issuance sequence

Arrows show the order of handoffs. Message signatures and off-chain review are distinct from confirmed blockchain transactions.



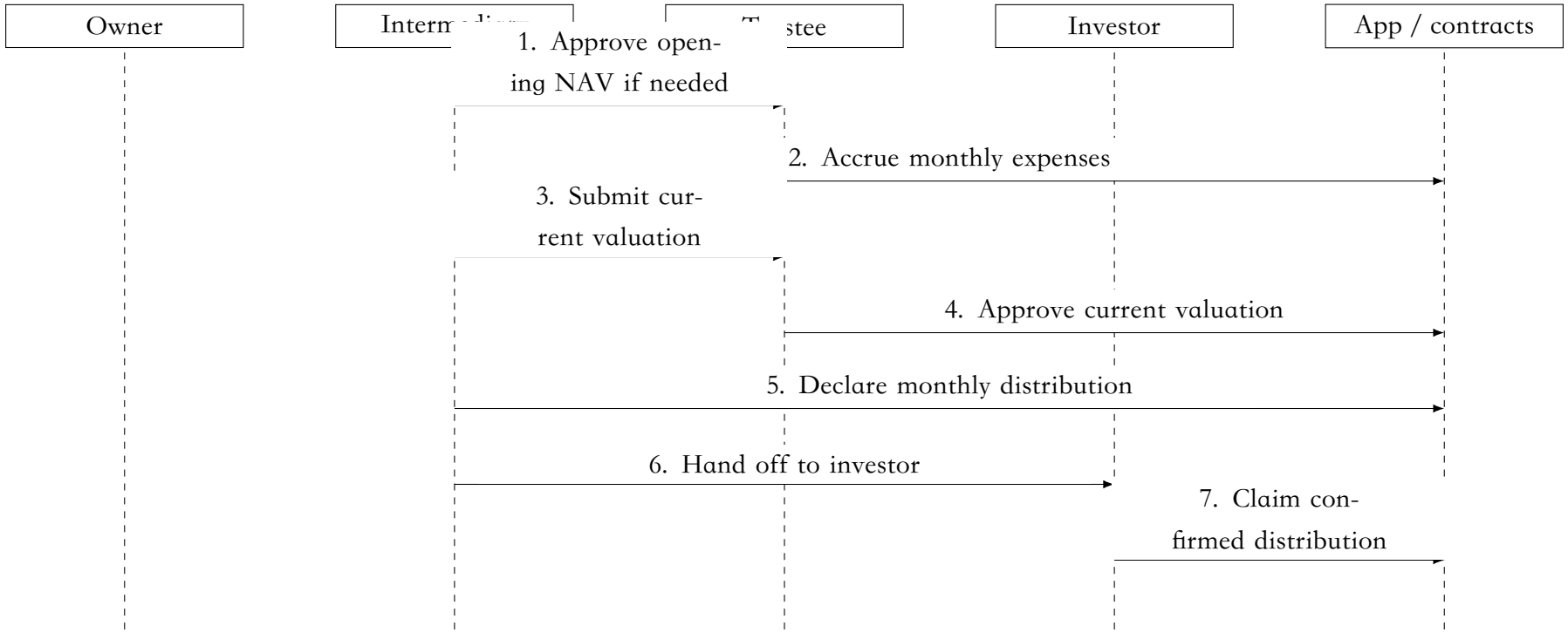
2.2 Monthly bills and household payment

Arrows show the order of handoffs. Message signatures and off-chain review are distinct from confirmed blockchain transactions.



2.3 Monthly valuation and investor handoff

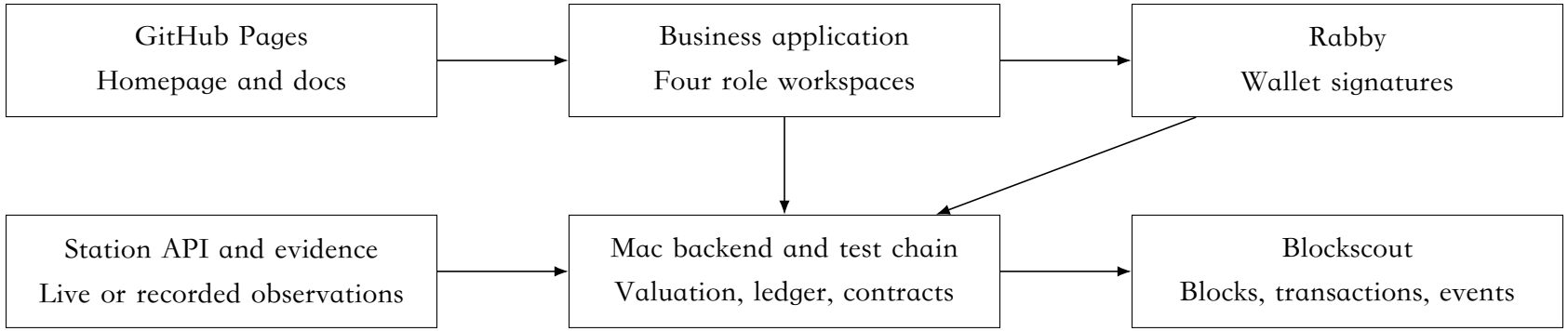
Arrows show the order of handoffs. Message signatures and off-chain review are distinct from confirmed blockchain transactions.



3 System composition and architecture

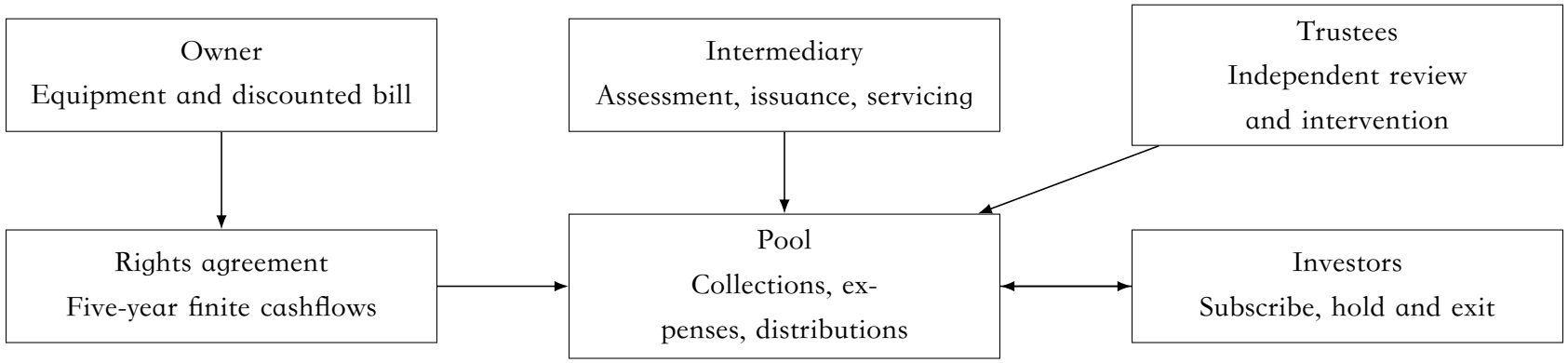
3.1 System composition

The homepage remains readable independently. The application, RPC and explorer run in an isolated Docker environment on the Mac and are exposed through HTTPS tunnels.



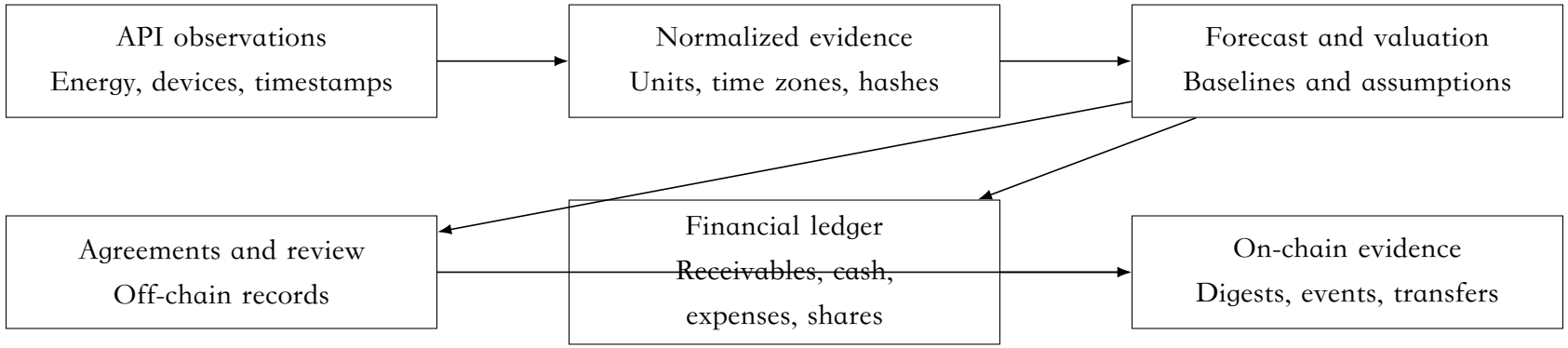
3.2 Business architecture

Discounted owner electricity payments and eligible export receipts fund the rights. The pool pays the agreed purchase price and handles expenses, distributions and exits without acquiring equipment ownership.



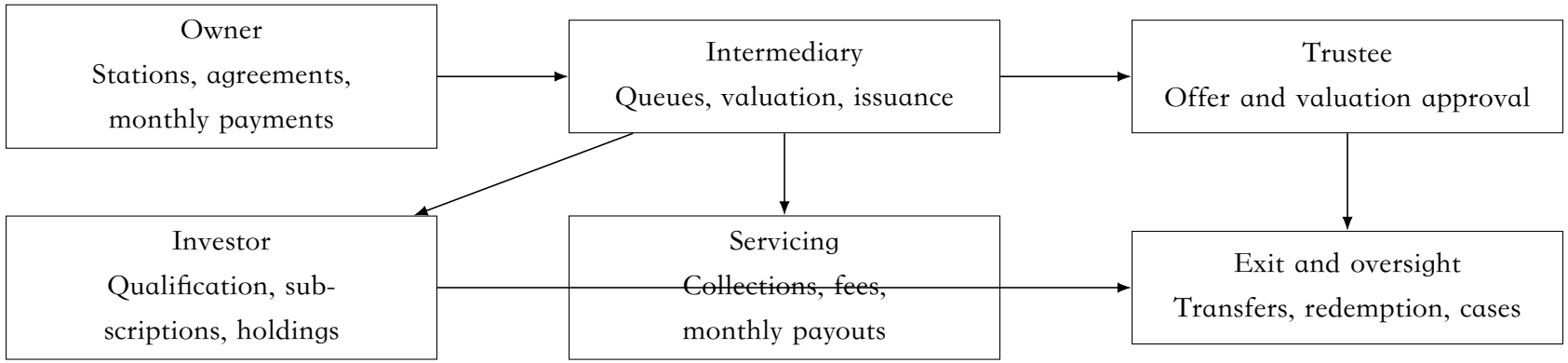
3.3 Data architecture

Observed energy, modeled values, review status and chain confirmation are recorded separately. Collection is not another revenue recognition, and anchoring a digest does not publish the underlying raw evidence.



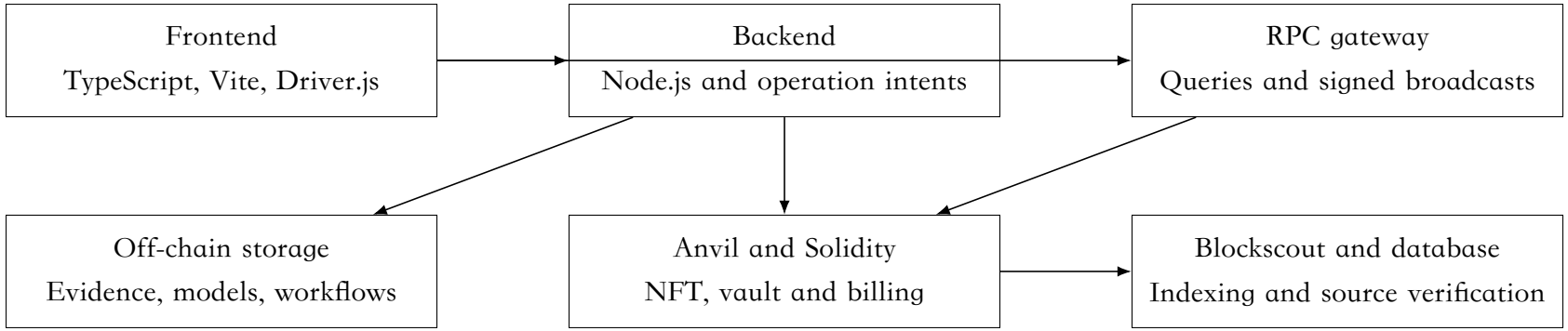
3.4 Functional architecture

Workspaces show the operations permitted for each role. Guidance derives from actual state, and choosing a role does not grant approval, payment or signing authority.



3.5 Technical architecture

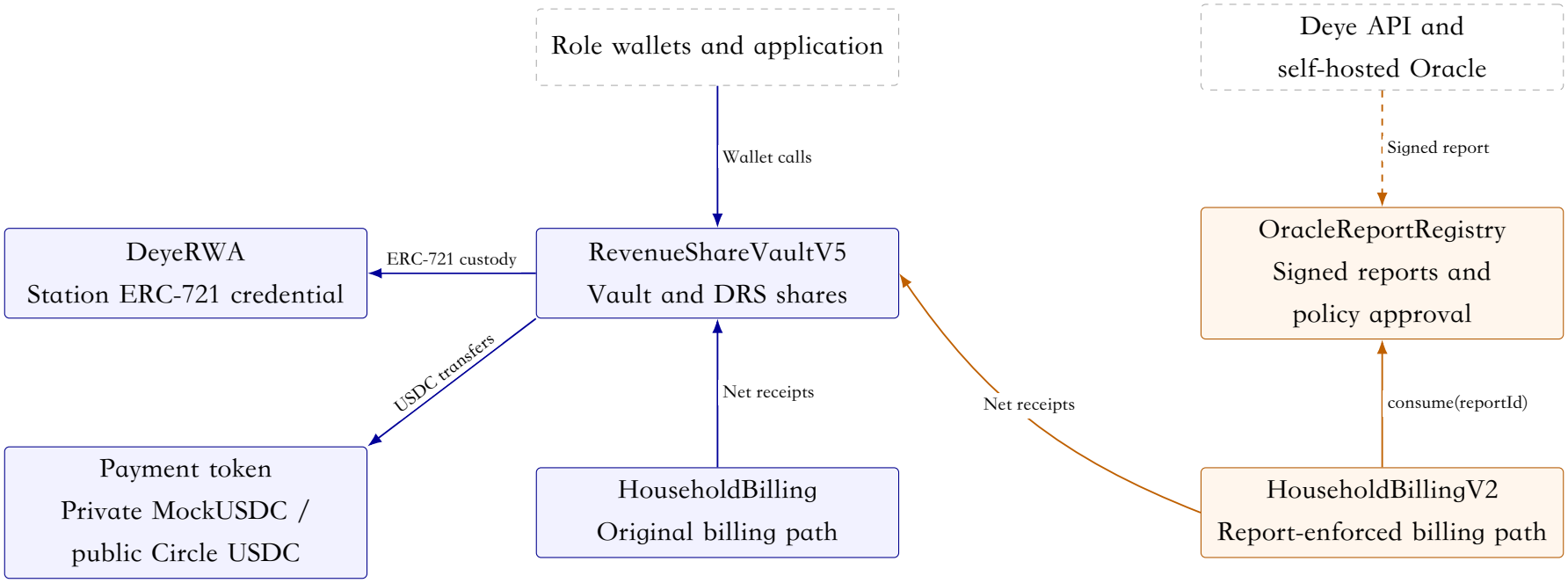
Wallets sign on the client. The backend checks sender, target, method, arguments, receipt and events before advancing workflow state. The public RPC rejects reset, impersonation and balance-editing methods.



4 Smart contract architecture and business sequence

This section describes the revenue-rights contracts used by the three current test environments. Historical prototypes in the source tree are not combined into one product. The private demonstration and original Base Sepolia product use HouseholdBilling. The independent Local Oracle on Sepolia product replaces that billing path with OracleReportRegistry and HouseholdBillingV2. Each environment has its own station credential and vault. The two Base Sepolia environments share only the public test chain and Circle test USDC.

4.1 Architecture of the currently deployed contracts



Blue denotes the shared contract path; orange is used only by the independent Oracle product. A product does not activate both billing contracts.

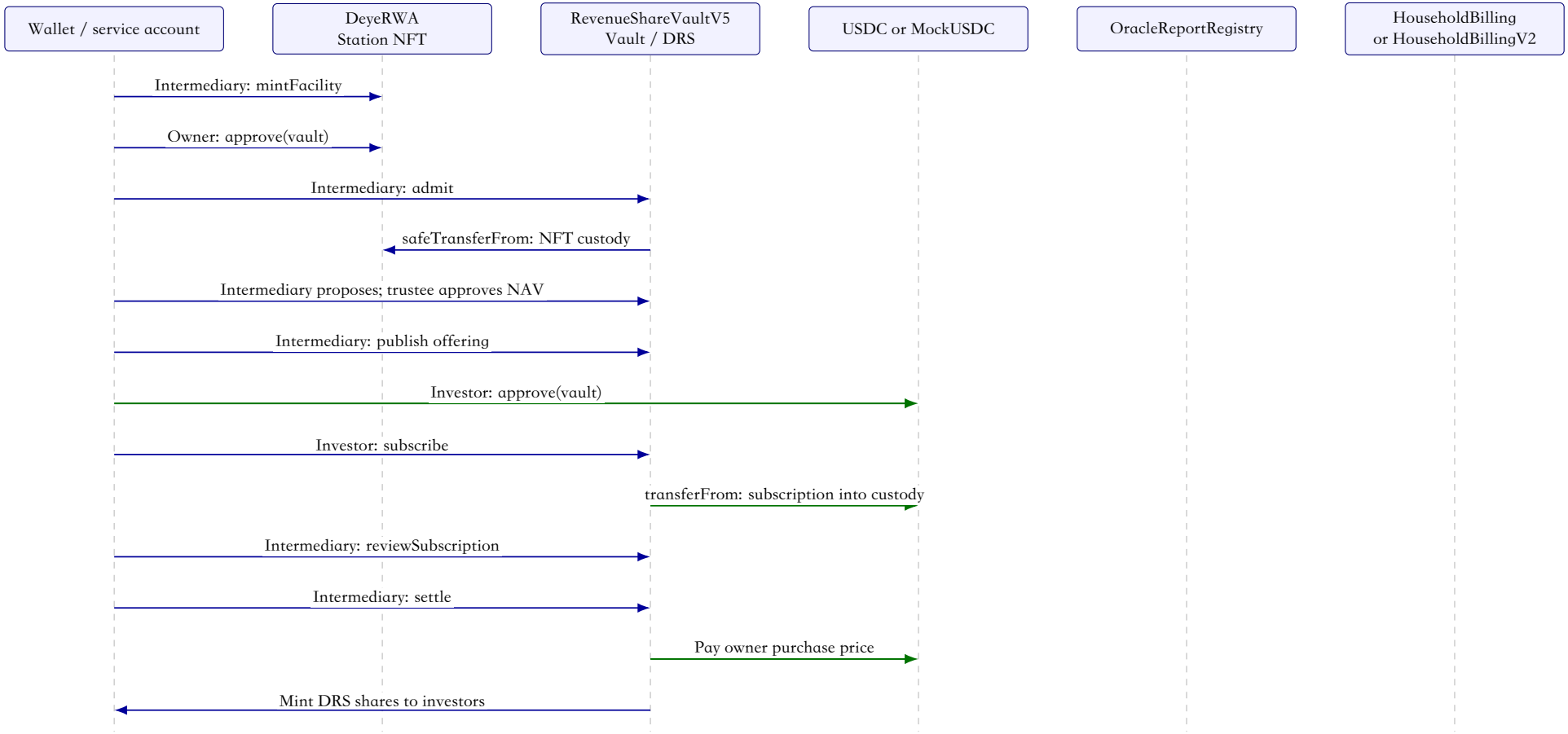
4.2 Contract responsibilities and deployment boundaries

Contract or component	Role in the current product
DeyeRWA	The intermediary mints the station ERC-721 credential. After owner approval, the vault receives it on admission and returns it after failure or closure. Its hashes do not transfer ownership of the physical equipment.
RevenueShareVaultV5	The vault is itself the six-decimal DRS ERC-20 share contract. It manages eligibility, subscription custody, settlement, valuation, receipts, fees, monthly distributions, transfers, redemption, servicing cases and closure. There is no separately deployed DRS token.
HouseholdBilling	The private and original Base Sepolia products use this bill contract. The intermediary publishes off-chain reviewed amounts. The owner pays USDC, a separate test treasury provides simulated export funding, and the intermediary confirms the net receipt into the vault. It does not verify an Oracle report.
OracleReportRegistry	Only the Oracle product uses this registry. Two distinct trustees approve billing parameters. The registry verifies the Oracle EIP-712 signature, product, calendar month, service period, expiry and revision state. It does not hold owner payments.
HouseholdBillingV2	Only the Oracle product uses this version. Bill publication calls consume on the registry in the same transaction and uses the amounts in the report. Owner payment, simulated export funding and intermediary confirmation remain separate transactions.
USDC / MockUSDC	The private chain deploys MockUSDC. Both Base Sepolia products use the same Circle test USDC contract. The payment token differs from DRS shares; a shared on-chain USDC balance does not merge product orders or holdings.

Calendar is an internal calendar library compiled into the vault, not a separate interactive deployment. The source tree also retains earlier prototypes such as DBondMarket, RWAVault and EnergyRevenue, and older vault versions for historical scenarios or compatibility checks; they are

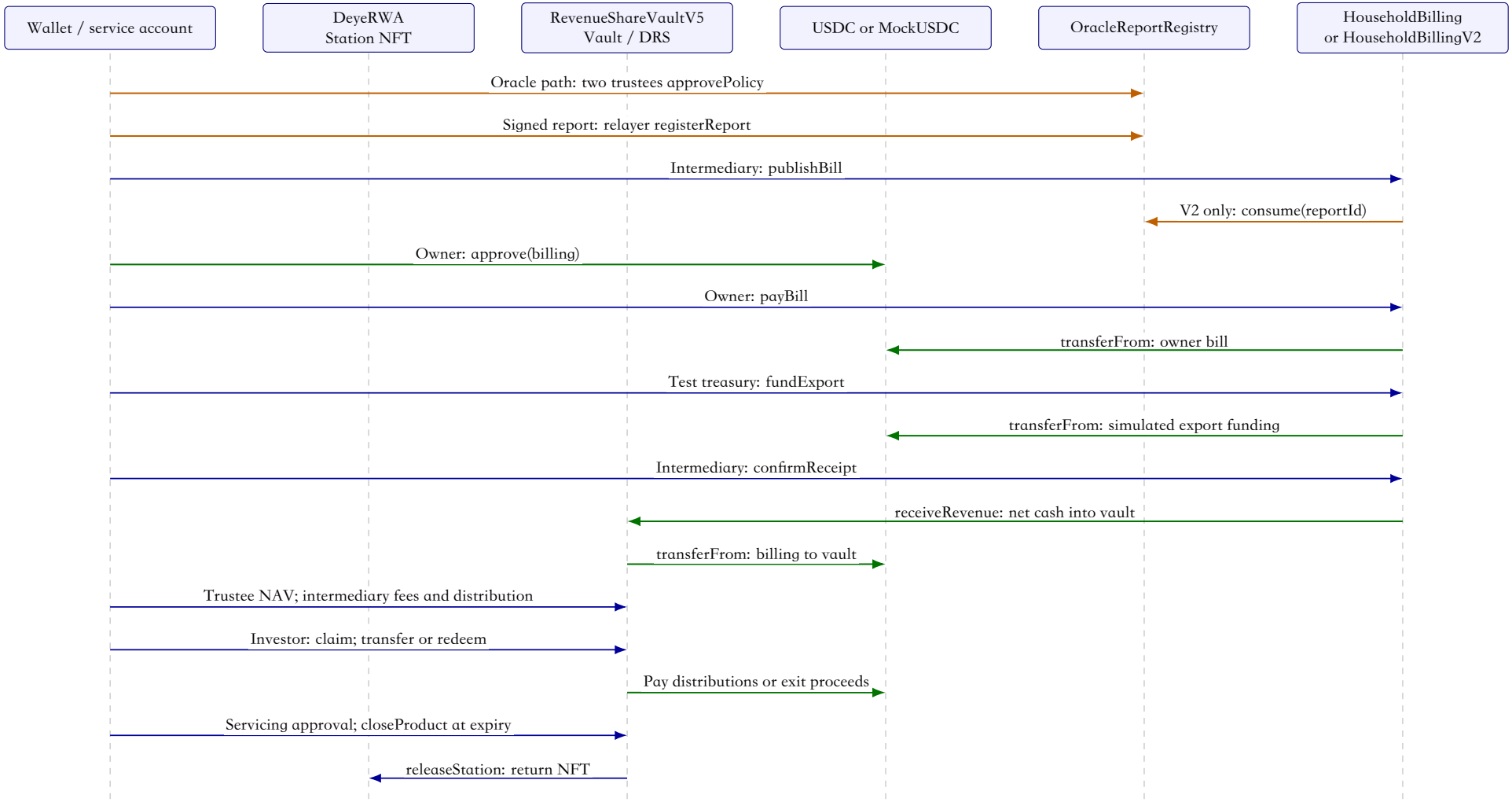
outside the current revenue-rights path shown here. Check contract addresses against the selected environment manifest. The same Chain ID alone does not identify the product.

4.3 Cross-contract sequence (issuance and settlement)



The first sequence shows the issuance path in transaction order. Application review, agreement text, model calculations and ordinary wallet sign-in are off-chain steps. Only the corresponding mint, approval, custody, valuation approval, subscription and settlement calls change on-chain state. Subscription USDC first remains in vault custody. At settlement, the vault pays the owner and mints DRS shares within the same vault contract.

4.4 Cross-contract sequence (billing, distributions and exit)

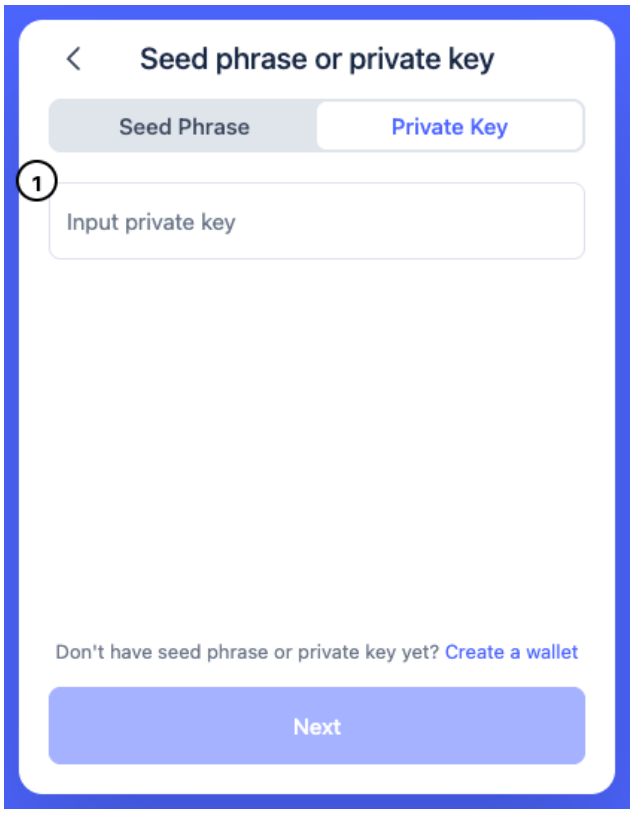


The second sequence continues the same product timeline. The original HouseholdBilling publishes off-chain reviewed amounts without calling OracleReportRegistry. Only the Oracle product has HouseholdBillingV2 consume a registered report during bill publication; a validation-only report cannot become a payable bill. Owner payments and simulated export funding enter the billing contract separately. Net cash reaches the vault only after intermediary confirmation. The vault handles distributions, claims, transfers, redemptions, servicing cases and closure, while actual USDC transfers occur through the payment-token contract. Returning the station credential after failure or closure is a separate transaction and does not transfer physical equipment title.

5 Wallets and test accounts

Install the browser extension through Rabby's official website. Choose an existing address and private-key import, then verify the resulting full address. Enter private keys only inside the wallet extension. The business website never needs them.

https://rabby.io/



Marker 1 identifies the private-key field inside the wallet extension. The illustrated field is empty.

This release preserves the public owner, intermediary and two trustee addresses. Role bindings are not generated by clicking a role button. An ordinary wallet may apply as an investor. Investor A/B and trustee A/B use separate accounts for a full walkthrough.

The two trustees use the same login role but different wallets. A servicing case requires 30% support under its fixed snapshot and approval from two distinct trustees before changing the demo collection recipient. Repeating approval with one address does not replace the second trustee.

5.1 Role accounts 1

Owner

0xadC4920e9036dadC146ba1aF84fb1475C612E15b

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

Intermediary

0xcee71e610f50435B8C9B0B25d3D322a5E184423a

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

5.2 Role accounts 2

Investor A

0x34cAACf0E56129F461afA460cDa3B220a3bfe2F2

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

Investor B

0xcC245B1140F25b91BD8C13B05d9516A85da75605

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

5.3 Role accounts 3

Trustee A

0x2641feCc09b01C7C482F51deDB28372daa41326a

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

Trustee B

0xfcf63f6E1100C4dD1bd0630C358848d990E33e64

The demo operator supplies the private key through an internal channel. It is excluded from this source document.

6 Website, network and blockchain explorer

6.1 Unified public entry

https://sell-dec-sir-org.trycloudflare.com/

The unified entry currently shows the private demonstration, the original Base Sepolia product and Local Oracle on Sepolia. Cloudflare addresses are temporary and may change; use the public homepage to find the current entry. If the private service is under maintenance, its unavailability does not imply an Oracle-environment failure.

6.2 Public homepage

https://mengzhon001.github.io/deye-rwa-demo

6.3 Business application

https://sell-dec-sir-org.trycloudflare.com/networks/private/app/

6.4 Blockchain explorer

https://cope-jim-itunes-emacs.trycloudflare.com

6.5 Wallet RPC

https://affiliate-respond-shortly-technological.trycloudflare.com

The test network Chain ID is 20261008 and its gas token is DETH. Check the current RPC when adding the network. A tunnel change requires a new RPC URL but does not change the chain, accounts or existing balances. The address 127.0.0.1 points to the reader's own computer and is not a remote access address.

Click a confirmed transaction hash to open its Blockscout page. Alternatively, paste a full hash, account or contract into the explorer search field. Check status, sender, target and block, followed by token transfers and events. Login and agreement message signatures are off-chain and do not create transaction records.

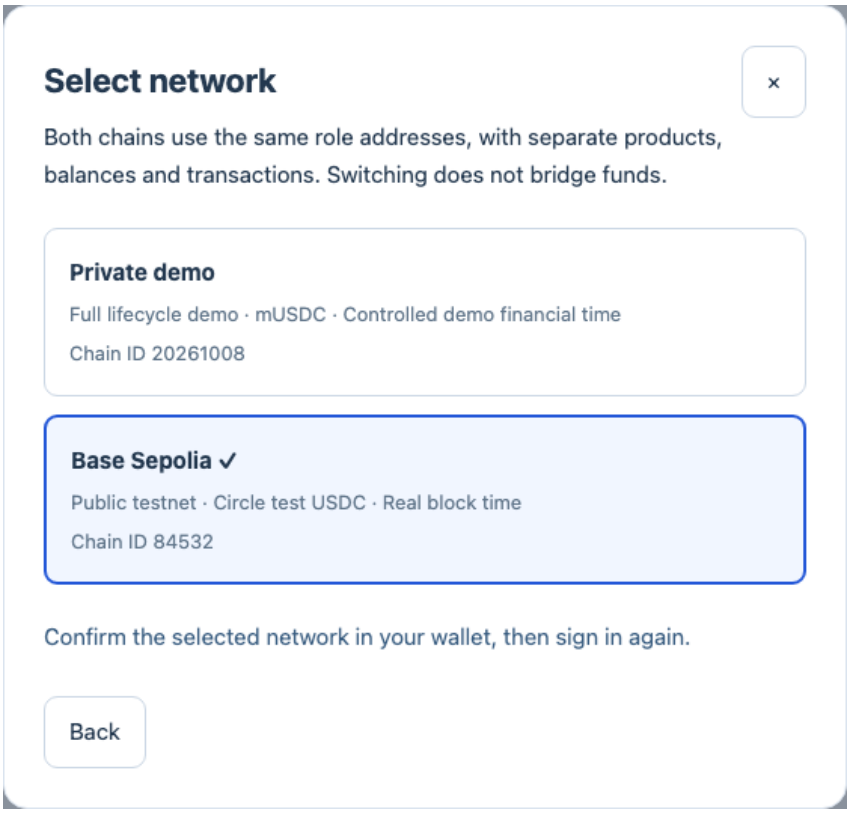
6.6 Network selection and session isolation

Select an environment on the GitHub homepage and click Launch App. The network button in the application lists three environments and their Chain IDs. Both Base Sepolia entries use 84532. A switch between them does not require changing the wallet chain, but requires a new

sign-in signature; records and sessions are not merged. The sign-in dialog also offers environment selection.

After the wallet signature is verified, the role workspace and navigation appear first. Product metadata, wallet balances and saved records load independently. Transaction controls remain unavailable until the required business state is loaded, and a pending amount does not mean zero. An unavailable section can be retried separately. If business-state loading fails, sign-in is retained without repeatedly connecting or signing again.

To switch, confirm the destination network in Rabby and sign in again with the intended role. Role addresses remain unchanged, while sessions, products, pending operations and page selections are isolated by chain. Switching does not convert or bridge private-chain mUSDC into Circle USDC. Reconcile any pending operation on its original page or in the transaction log before switching. A rejected wallet request or unavailable destination leaves the application on its original network.



Network selection in the unified application. Select the chain, verify it in the wallet and sign in again.

7 Full lifecycle testing on the private demonstration

7.1 Purpose and prerequisites

The shared private demonstration checks the complete handoff of the revenue-rights business between roles. It uses Anvil, demo mUSDC and its own Blockscout. Authorized intermediary time advancement allows monthly operations, the six-month lock, quarterly redemption and expiry to be exercised without waiting through the real calendar. Advancing time creates no station telemetry and does not replace invoices, payments, approvals or distribution transactions.

Obtain the role test accounts from the operator and agree which product the group will use. A full walkthrough requires an owner, intermediary, investors A/B and trustees A/B. An investor-only check can use a prepared product that is published and still fundraising, without repeating the application and issuance preparation. If the product is already active, primary fundraising subscriptions are closed; select another fundraising product or ask the operator to prepare a new scene.

7.2 Entry and network confirmation

https://sell-dec-sir-org.trycloudflare.com/networks/private/app/

Click Launch App on the public homepage and select Private demo in the upper-right network menu. Verify Chain ID 20261008 in Rabby, select the intended role and sign in. Check the product name and vault address. Wallet funds, escrowed orders and issued DRS are separate records. Claiming demo funds increases the wallet balance but does not submit a subscription.

7.3 Direct investor testing

Choose a fundraising product, inspect its cashflow timeline and risks, and submit the eligibility declarations. After intermediary approval, obtain demo funds, enter an amount within the remaining capacity, approve mUSDC and confirm the subscription transaction. The intermediary reviews the current cumulative order. Additional subscriptions are allowed while fundraising and capacity permit, and each addition requires another review.

To check refunds, the original investor can cancel before settlement and recover the entire cumulative escrow. To continue to holdings and distributions, cover the displayed settlement shortfall and have the intermediary review and settle. The contract retains the agreed 10% cash automatically; there is no separate reserve-transfer step. Until settlement completes, escrow must not be treated as issued shares.

7.4 Lifecycle and illustrated-step index

The table organizes the full walkthrough by business state. Confirm both the page result and corresponding chain record before handing off to the next role. The identifiers link to the annotated illustrations later in the manual. Investor testing on a prepared product starts with group D, while an empty scene starts with group B.

Stage	Participant and action	Result to confirm	Figures
Application and agreement	Owner applies; intermediary reviews and prices; trustee approves; owner accepts and signs.	Agreement, valuation and station evidence match.	B3 / B10
Admission and publication	Intermediary mints; owner authorizes; intermediary admits and prepares NAV; trustee approves before publication.	Fundraising opens with reconciled custody and issuance terms.	C1 / C6
Subscription and settlement	Investors obtain eligibility and subscribe; intermediary reviews and settles.	Escrow pays for rights, cash is retained and investors receive DRS.	D3 / D8
Monthly servicing	Intermediary bills; owner pays; intermediary reconciles export and receipts; trustee approves NAV.	Both receipt sources, expenses and approved NAV are recorded separately.	E1 / E5
Declaration and claim	Intermediary declares the monthly distribution; investors switch to their own wallets and claim.	Declared, claimable and paid amounts reconcile with USDC transfers.	E6 / E7
Transfer and redemption	Participants and intermediary act after lock and notice conditions are met.	Shares, payments, partial fills and remaining requests agree.	F1

Stage	Participant and action	Result to confirm	Figures
Servicing and closure	Obtain snapshot support and two trustee approvals where required, then reconcile expiry.	Receivables and liabilities are reconciled, with final distribution and credential release.	G1 / H1

7.5 Clock advancement, shared state and evidence

When waiting for a cutoff, the intermediary opens Product operations, expands Demo tools: financial time, selects the next monthly fee cutoff and confirms. Where supported, the guide locates the same shortcut. Complete each month's billing, receipts, expenses and valuation approval before moving to the next month. Skipping dates does not complete the missing accounting.

Products in the shared environment use the same chain clock. Advancing it can expire another product's fundraising or valuation, so coordinate through one operator. The operator should also control resets and preparation of new sample products. Record the product and role addresses, key hashes, business identifiers, expected outcomes and actual results. Do not record off-chain review as an on-chain transaction.

8 Base Sepolia public-chain testing

The original public demonstration remains the environment for the full accelerated lifecycle. Base Sepolia is a separate environment for public-chain contracts and wallet transactions. Role addresses are shared, but contracts, balances and workflow records are independent. The complete lifecycle screenshots later in this manual illustrate the private-chain demonstration. Public-chain testing follows the real-time and funding restrictions in this section.

8.1 Test objectives and the separate Oracle environment

Base Sepolia checks whether the business code handles wallet signatures, gas, Circle USDC approvals and transfers, contract calls, event reads and recovery on a real public chain. It also provides the foundation for later oracle integration, where report receivers, report digests, billing-period binding, permissions and failure handling can be tested on the same public test chain without disturbing the private-chain demonstration.

The original Base Sepolia product in this chapter still uses its earlier billing contract and does not verify Oracle reports. The new feature belongs to another isolated product with a new billing contract, described in the next chapter. Both use Base Sepolia, but their business records must not be mixed.

8.2 Entry and network configuration

Public-chain application

https://sell-dec-sir-org.trycloudflare.com/networks/base-sepolia/app/

Public-chain explorer

https://base-sepolia.blockscout.com

Wallet RPC

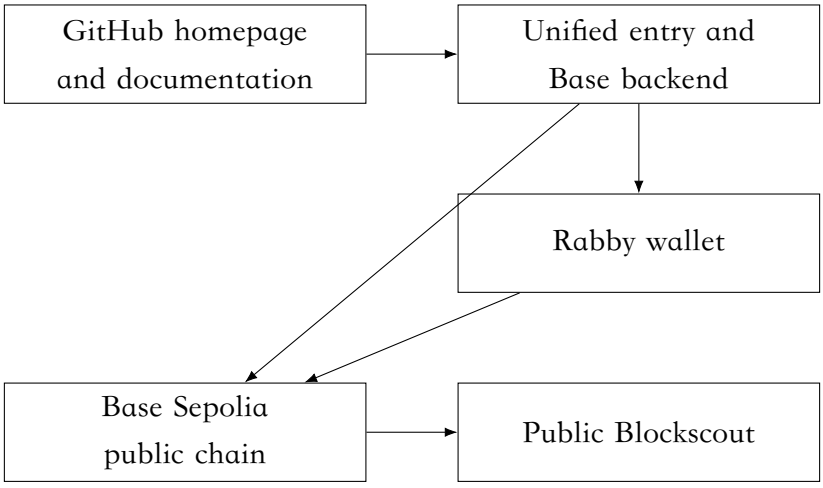
https://sepolia.base.org

The Chain ID is 84532, gas is paid in test ETH, and payments use Circle test USDC. The homepage and application share one network-selection entry. Select Base Sepolia and verify the wallet network before signing in.

0x036CbD53842c5426634e7929541eC2318f3dCF7e

The address above identifies the official Circle test USDC used here, with six decimals. It is separate from private-chain mUSDC. Test tokens have no monetary value, and balances on another chain cannot pay for subscriptions here.

8.3 Environment components and data flow



The shared HTTPS application routes network-specific paths to the corresponding independent backend. The wallet signs transactions for Base Sepolia, the backend checks receipts and events, and the public explorer indexes the results. This environment does not run Anvil or depend on the private-chain RPC tunnel.

8.4 Funding and subscription steps

Open the official faucets from the test-funds entry, select Base Sepolia and request test ETH and Circle test USDC for the current role address. Providers may require login or limit claims. The application cannot mint Circle USDC. After confirmation, return and refresh the balance. Enter private keys only inside the wallet extension, never in a faucet or the application.

https://portal.cdp.coinbase.com/products/faucet

https://faucet.circle.com/

Select a fundraising product and apply for eligibility. After intermediary approval, enter an amount, approve the corresponding USDC allowance and confirm the subscription transaction. Additional subscriptions are allowed while fundraising remains open, and the intermediary reviews the new accumulated amount. Cancellation before settlement refunds the full accumulated escrow for that address. Escrow does not mean DRS has been issued.

Settlement still requires a current valuation, order approvals, the fundraising threshold and sufficient purchase funding and reserve. The contract automatically retains 10%; no separate reserve transfer is needed. Small test balances support approvals, subscriptions, review and refunds, but do not establish that a larger funded settlement has completed.

8.5 Public-chain checks and completion criteria

Start with one investor account and test sign-in, eligibility, USDC approval, a small subscription and its refund. Then check the intermediary and trustee permissions using their own accounts. Verify the network and balance before each transaction, and reconcile its receipt, events and token transfers in the public explorer. While fundraising is open, add a subscription and check that the intermediary reviews the latest cumulative amount. Retain and reconfirm the original hash when a submitted transaction is slow to appear rather than paying again.

This round is complete when actual transaction amounts agree with business records, wrong wallets and networks are rejected, and records recover after refresh or restart. If settlement funding or the real monthly cutoff has not been reached, record settlement, billing and distribution as not executed in this round and explain why. Small test balances and a short observation window do not amount to lifecycle acceptance; use the private demonstration or your own local instance for that path.

8.6 Real time and testing scope

The application cannot advance public-chain time. Monthly billing requires a settled product and the end of the relevant service period. The six-month transfer lock, redemption notice period and expiry remain unchanged. Reset and time-advance controls are therefore unavailable. Use the original public private-chain environment for the complete accelerated lifecycle.

The original Base Sepolia product uses Circle test USDC for household payments. Grid-export receipts remain a disclosed economic simulation funded from a separate account. The original product has no mandatory Oracle verification in billing. The separate Oracle product and its data retrieval, signing and enforcement are described next.

8.7 Transaction checks and recovery

After submission, use the hash to check the sender, destination, method, events and USDC transfers in the public explorer. The wallet and backend RPC may see the same transaction at different times. Retain the original hash and retry confirmation rather than paying again. Explorer indexing can lag without a transaction having failed.

Station credential

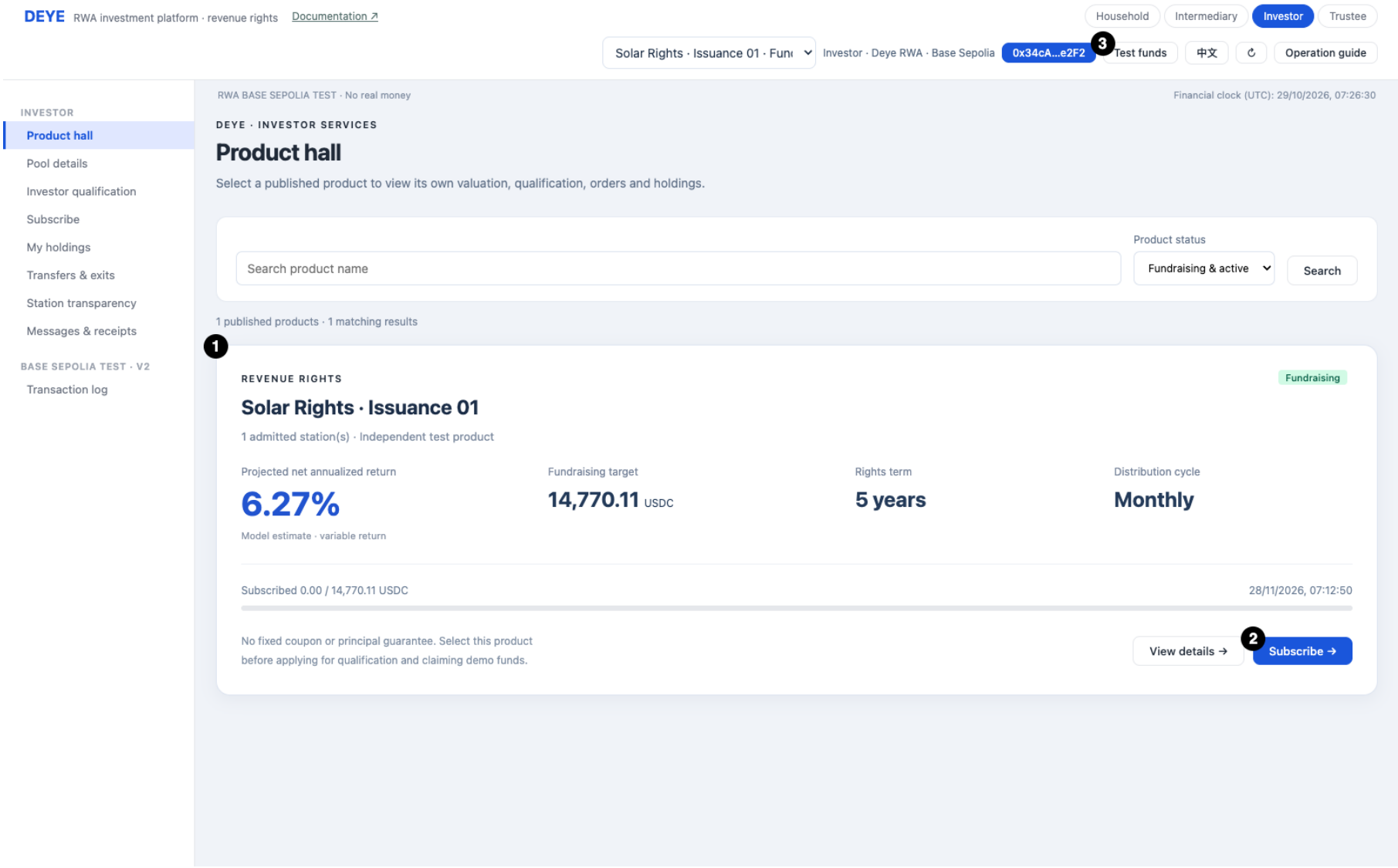
0xf17FFEAb245CebCd260a2E6aFAe90A43F2749176

Revenue-rights vault

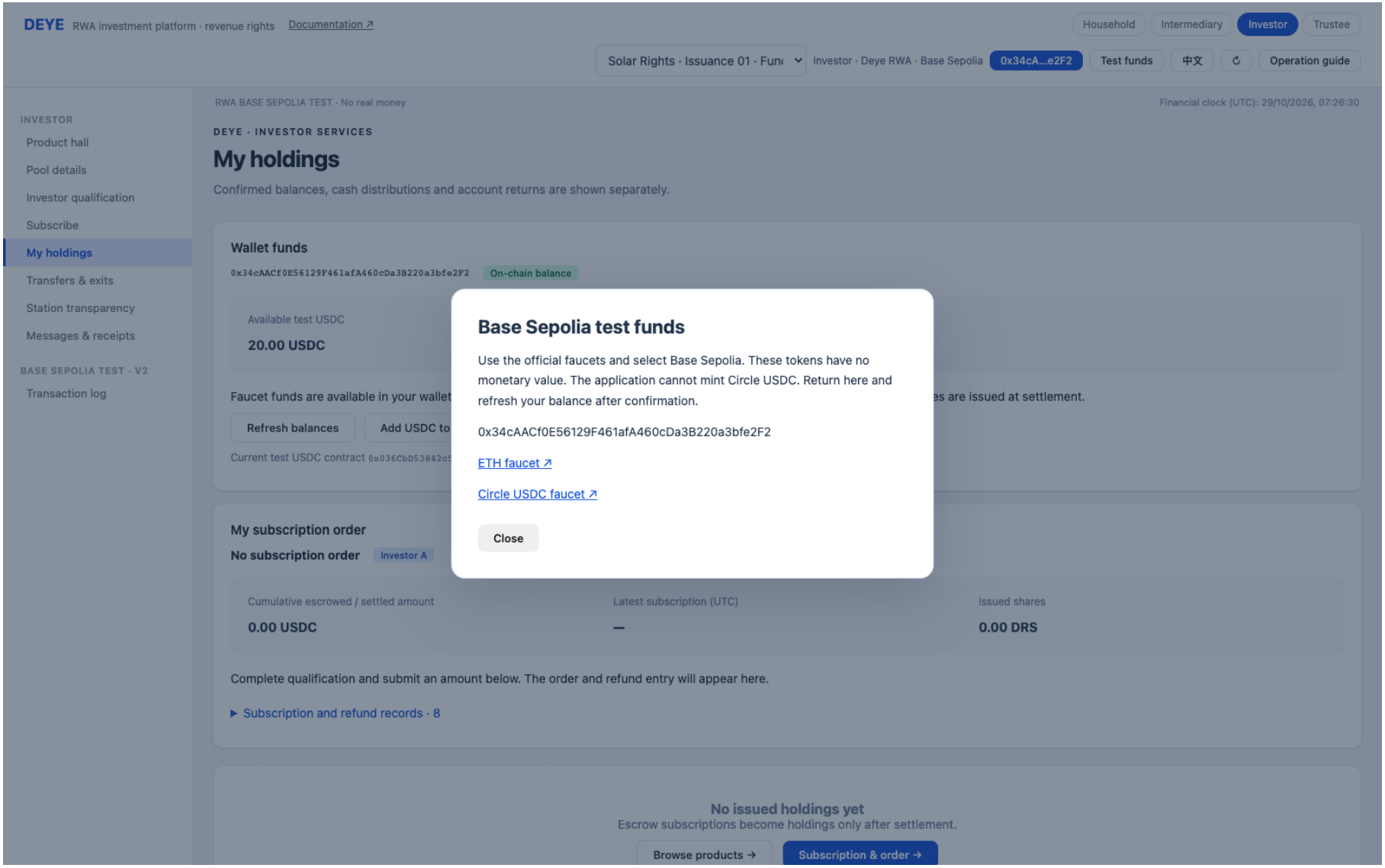
0xB84F6E2EB3b9e63bBE447190dF268c49D617A445

Monthly billing contract

0xa2748E730069A38913b5E2e09a45C5aB546a0908



Public-chain product and network. Markers identify the product, subscription navigation and test-funds entry.



Wallet balances and official-faucet instructions. The backend does not mint Circle USDC.

9 Local Oracle on Sepolia testing

The third network-menu entry is Local Oracle on Sepolia. It runs on the same public Base Sepolia chain as the original product, with Chain ID 84532; it is not a third chain. Role wallets and their test ETH and Circle USDC balances are shared, while contracts, products, sign-in sessions and business records are isolated. Switching between the two environments does not require a chain switch, but it does require a new sign-in signature for the target backend.

9.1 Entry point and contracts

https://sell-dec-sir-org.trycloudflare.com/networks/oracle-base-sepolia/app/

https://base-sepolia.blockscout.com

Station credential

0x18a88350eb54FD2A98902509DF110851097c6a9b

Isolated vault

0x0479eB8a2bb68D23bAa88de32B27Af63bFCf0C32

Report registry

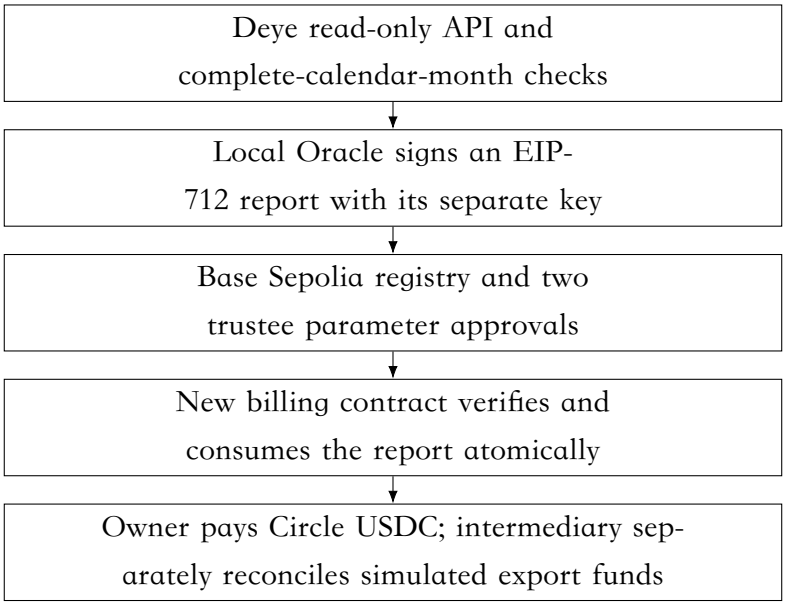
0x49a3b54c2972786987B24dC44490332bcb5D41c6

Enforced billing contract

0xFa650D2619452E1763695c780F478bDeA7027D28

This environment uses the same Circle test USDC contract as the original Base Sepolia product. Check the Chain ID, role account, product and transaction destination in the wallet. The read-only Oracle reports page shows the observation month, source and parameter digests, reporter and registration transaction. It does not reveal Deye API credentials or raw observations.

9.2 Data and transaction flow



The Oracle service reads the authorized Deye API directly, checks each day and reconciles daily and monthly totals, then calculates six-decimal USDC amounts using the approved parameters. Missing data, an API failure or a parameter mismatch stops submission; there is no replay or synthetic fallback. A labeled recorded capture may support initial product preparation but cannot replace a monthly Oracle report. Deye has not digitally signed these observations, so the report proves checking and signing by our self-hosted Oracle, not independent certification by Deye or Chainlink.

9.3 Completed checks and the later billing period

As of 9 November 2026, the Oracle has actually read the complete August 2026 calendar month from the Deye API, signed a historical report and registered it on Base Sepolia. It is explicitly validation-only; the new billing contract rejects it as a source of a household bill. The registration transaction and report ID below can be checked independently in the Base Sepolia explorer.

Report ID

0x27a8e1073e45d570c03dcd07c48efe5f655ad23f37b045ef277359c323f7a965

Registration transaction

0x3df71ae9d301c9a5736316fac0857e9f1507549492f5c39e6a79be41cf51e411

The new product still requires issuance, fundraising conditions and settlement. After settlement, the first complete calendar month is set, and trustees A and B separately approve the station, first observation month and parameter digest in their review queue. Only after both the service period and matching observation month end may the Oracle submit a billable report. The intermediary then generates the bill from that report, the owner pays USDC, and the intermediary confirms pool

receipt before expenses and investor distributions follow actual cash. The full billable-report-to-distribution sequence has not yet passed on-chain acceptance. Historical report registration is not evidence of actual monthly collection.

9.4 Operation, reconciliation and limits

npm run oracle:check
npm run oracle:dry-run
npm run oracle:validate-report
npm run oracle:billable

The operator runs these commands using restricted local configuration. oracle:dry-run reads and calculates without submitting a transaction. oracle:validate-report registers a validation-only report; check the prior report and receipt before repeating it. The isolated Compose environment also starts a persistent Oracle worker. It checks the two trustee approvals, product settlement and real monthly cutoff each hour. Only after those conditions pass does it read the API and submit a billable report. oracle:billable lets the operator inspect the same flow manually. Reporter and gas-payer keys are stored separately and never enter the browser, GitHub or this manual. Export receipts remain simulated funding from an independent test account; an Oracle report does not prove payment by the grid.

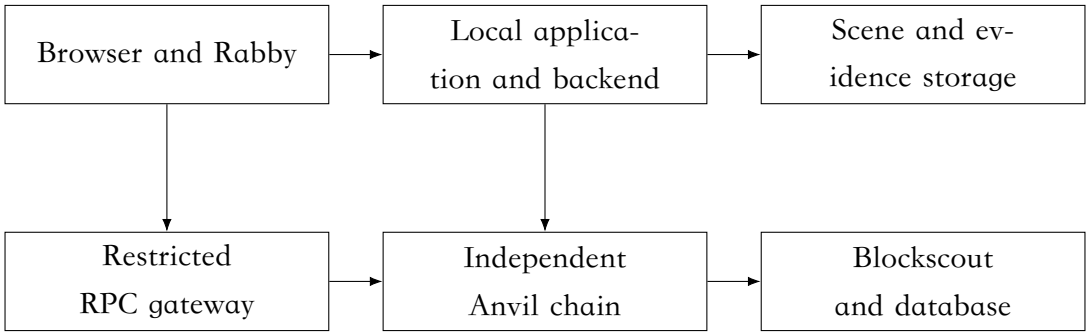
10 Local deployment and independent testing

10.1 Purpose and default environment

A local deployment lets developers reproduce issues, change the model and contracts, validate the interface and repeat a walkthrough from an empty scene. It needs no public tunnel and does not depend on the demonstration operator's Mac staying online. The commands below assume a fresh clone in an independent directory, not a working directory already hosting the shared demonstration.

The default Compose project is deye-rwa-platform and its Chain ID is 20260922. The application, Blockscout and wallet RPC use local ports 4174, 4000 and 18545. Check that the ports are free before starting. Resolve a conflict without stopping someone else's active demonstration. The address 127.0.0.1 refers only to the current computer and does not give a colleague access from another machine.

The existing maintenance profile on port 4191 and Chain ID 20261014 is a separate guidance regression environment with separately prepared recorded evidence. A fresh deployment does not copy that profile, shared public wallets, prepared products or real station credentials. These addresses differ because the environments are isolated; use the network information returned by your own deployment.



10.2 Installation, startup and health checks

Prepare Node.js 22.19 or a newer 22.x release, npm, and running Docker with Compose. The scripts call docker-compose, so installations offering only docker compose need a compatible command first. Full deployment has primarily been validated on Apple Silicon Macs. Check image architecture on other platforms; the source-verification compiler uses amd64 and needs emulation on Apple Silicon. The first installation downloads npm dependencies, container images and explorer dependencies and therefore needs an internet connection.

Run these commands in a new directory. web3:init creates local configuration, web3:wallets exports this instance's test accounts, and web3:up builds and starts the application, node and explorer. Initial Blockscout migrations may take time. Wait for the health checks before connecting a wallet.

```
git clone https://github.com/Mengzhon001/deye-rwa-demo.git deye-rwa-local
cd deye-rwa-local
npm ci
DEYE_LOCAL_SOLC=1 npm run build
npm run web3:init
npm run web3:wallets
npm run web3:up
npm run web3:health
```

Health checks cover the RPC Chain ID, application API, homepage, business page, documentation and Blockscout interface and API. If a check fails, use web3:status to inspect container status and web3:logs to inspect startup. Reinitializing wallets or deleting volumes is not a remedy for indexing delay. A healthy service does not itself prove source verification; inspect the relevant contract page in Blockscout.

```
npm run web3:status
npm run web3:logs
```

10.3 Local entry points and wallets

Homepage

http://127.0.0.1:4174/

Role sign-in and application

http://127.0.0.1:4174/app/

Product documentation

http://127.0.0.1:4174/docs/

Explorer

http://127.0.0.1:4000/

Wallet RPC

http://127.0.0.1:18545

Import the six accounts into Rabby from .data/local-wallets/wallet-import.private.json on your own machine. The file has mode 600 and the terminal prints addresses only. Verify Chain ID 20260922, choose the role and sign in. The generated accounts differ from the shared-environment addresses listed earlier in the manual; importing a shared role key does not automatically grant its role in a new instance.

10.4 Empty scenes, data sources and workflow reproduction

The public source excludes raw station records, API credentials and pre-issued products. An empty product hall on first startup is therefore expected. The intermediary opens Monitoring and evidence, explicitly selects Synthetic fixture as the source and confirms. The owner then submits an application for the sample station. The synthetic label must remain visible; the fixture tests workflow and does not prove a real asset's value.

Follow group B for application, quotation, review and signing, group C for credential approval, admission, valuation and publication, and group D for eligibility, subscription and settlement. Drafts do not appear in the investor hall. Group E covers actual household wallet payments, intermediary confirmation and distributions. Follow groups F to H for exits and expiry. Settlement still requires sufficient funds, and advancing time does not approve orders.

The standard local Docker profile exposes time controls under Demo tools: financial time at the bottom of the intermediary's Product operations page. Complete billing, payment, confirmation, expenses and valuation month by month before advancing, without relying on a maintenance-profile-specific guide shortcut. A bill does not mean cash has arrived; household payment and simulated grid-export funding still require separate reconciliation.

For real-data integration, obtain authorization and configure the station API on the server, or import validated recorded evidence. A container needs a read-only mount for the replay file, and DEYE_REPLAY_FILE must identify its container path rather than only a host filename. An API failure never silently selects synthetic data. Field mappings, units and suitability for valuation must still be checked; displaying numbers is not data certification.

10.5 Development regression, shutdown and recovery

After a code change, run the following checks from the repository root. They test types, contracts and guidance rules and do not replace real-wallet and browser acceptance. Perform the interface walkthrough in your own instance and retain hashes and business records.

```
npm run typecheck:ui
DEYE_LOCAL_SOLC=1 npm run test:onchain
npm run test:guidance
```

For automated lifecycle regression, web3:e2e operates on your own local Docker instance. It backs up the scene, executes signed transactions, advances time and restores the scene, stopping services in the process. It is not a read-only check for the shared public environment. Use web3:down to stop and web3:up to restart; ordinary shutdown preserves volumes. Remove volumes only for an intentional complete rebuild after preserving the local wallets and deployment records.

```
npm run web3:e2e
npm run web3:down
npm run web3:up
npm run web3:health
```

These commands follow the current repository scripts. This documentation revision does not restart or reset the existing public, Base Sepolia or maintenance environments, and successful document compilation is not a claim that a fresh machine has been deployed and tested. A new machine still needs its health checks and validation of wallets, container platform and data source.

11 Dynamic guidance and handoffs

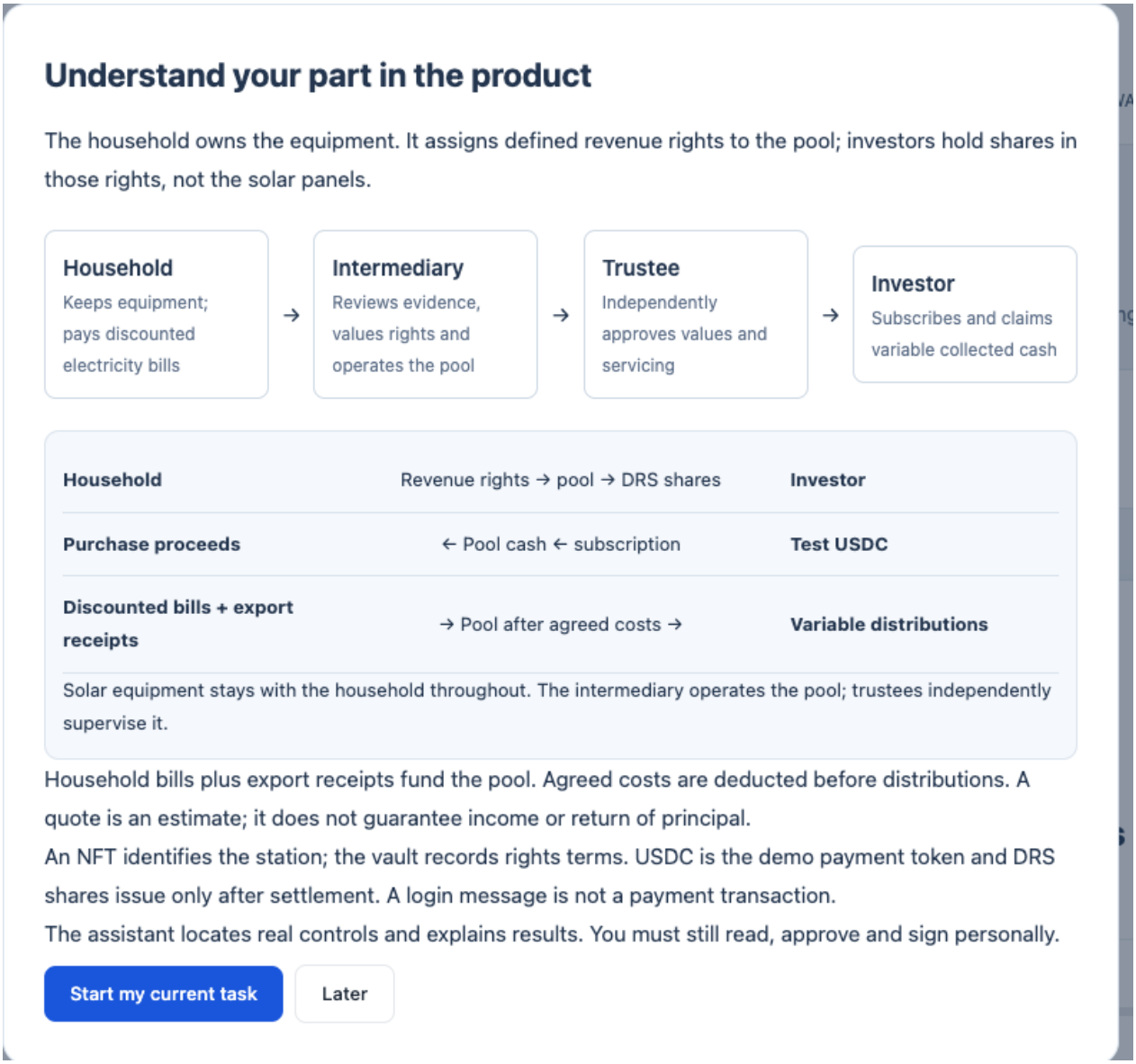
The first-entry introduction explains equipment, rights, money and participant responsibilities. The guide panel identifies the product, object, prerequisites, handler and expected result. Pause or collapse it and reopen Operation guide from the top bar. Completion comes from business and chain state rather than a Next button.

Participant mode focuses on the connected participant's tasks and handoffs. Four-role demonstration mode can inspect other roles'tasks and open their login entry. The correct wallet and a new signature are still required. It does not sign on behalf of another role or grant intermediary authority to an investor.

Locate operation opens the relevant page and identifies a control, field or prerequisite. If a condition is missing, use the specific remedy shown. Switch to the named reviewer, prepare the missing report or let the intermediary confirm the controlled time advancement. A read-only inspection is not presented as a blocked payment.

In the monthly cycle, a pending trustee approval must be completed first. The guide hands off to investors only after a new declaration creates an actual claimable amount. Fully paid earlier distributions do not remain the current task. Investors can distinguish historical payments, an undeclared current period and remaining NAV.

11.1 Product overview and task entry



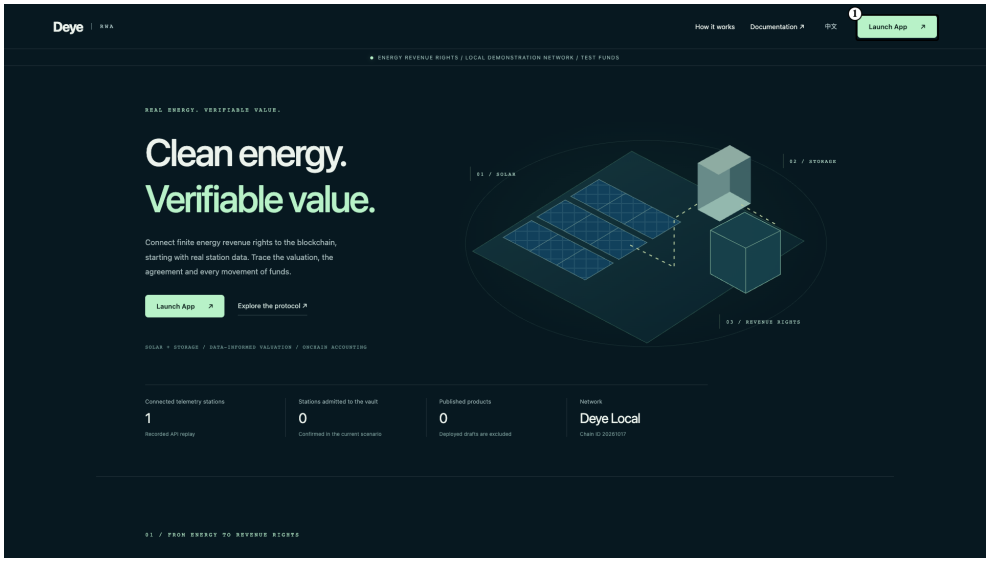
Read the overview and start the current task, or choose Later. The guide does not sign for a participant.

12 Application entry

12.1 A0 Application access

This step is performed by All participants. Open the public homepage and click Launch App. The browser should enter the current HTTPS application address and show the role selection dialog. If the environment is unavailable, ask the demonstration operator to check the Mac and tunnels.

After this step, select the intended role before connecting the wallet.



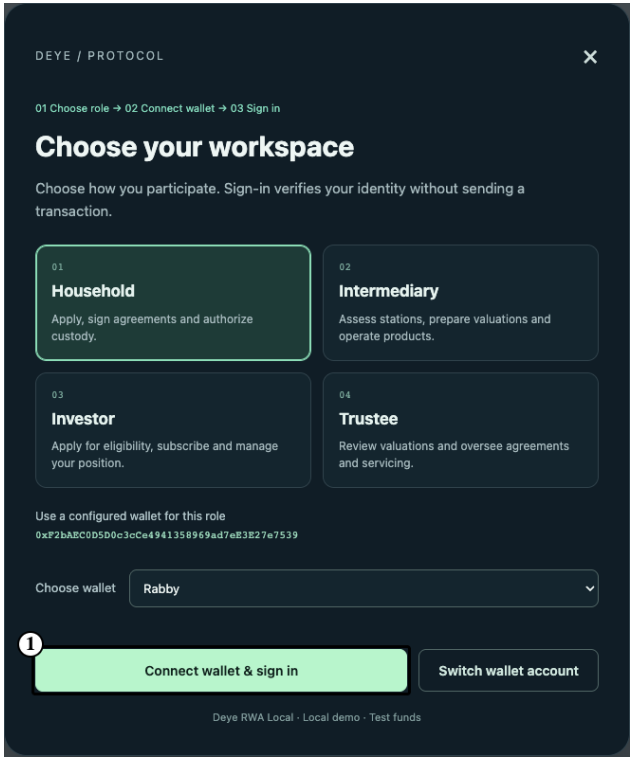
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



12.2 A1 Role selection and wallet sign-in

This step is performed by The participant performing the next task. Select the role, choose Rabby and click Connect wallet and sign in. In the extension, check the account, application domain, role and Chain ID before signing. A successful login opens the corresponding workspace without transferring funds.

After this step, continue in the selected role's workspace.

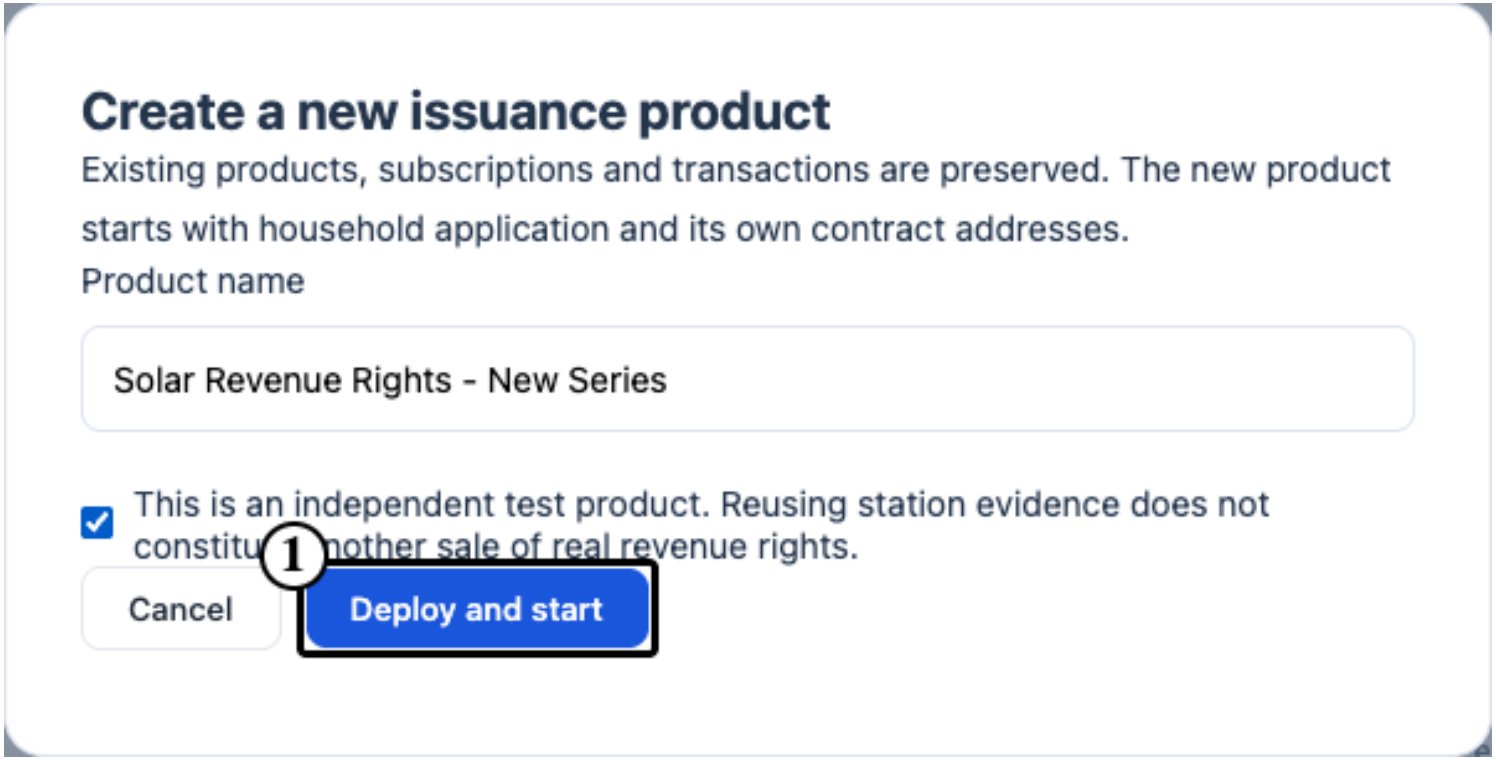


Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

12.3 A2 Creation of an issuance draft

This step is performed by Intermediary. Click Create new series in the top bar. Enter the product name, acknowledge that this is a test and confirm deployment. The new draft has separate contracts and does not appear in the investor hall until fundraising is published.

After this step, ask the owner to select this same product before applying.

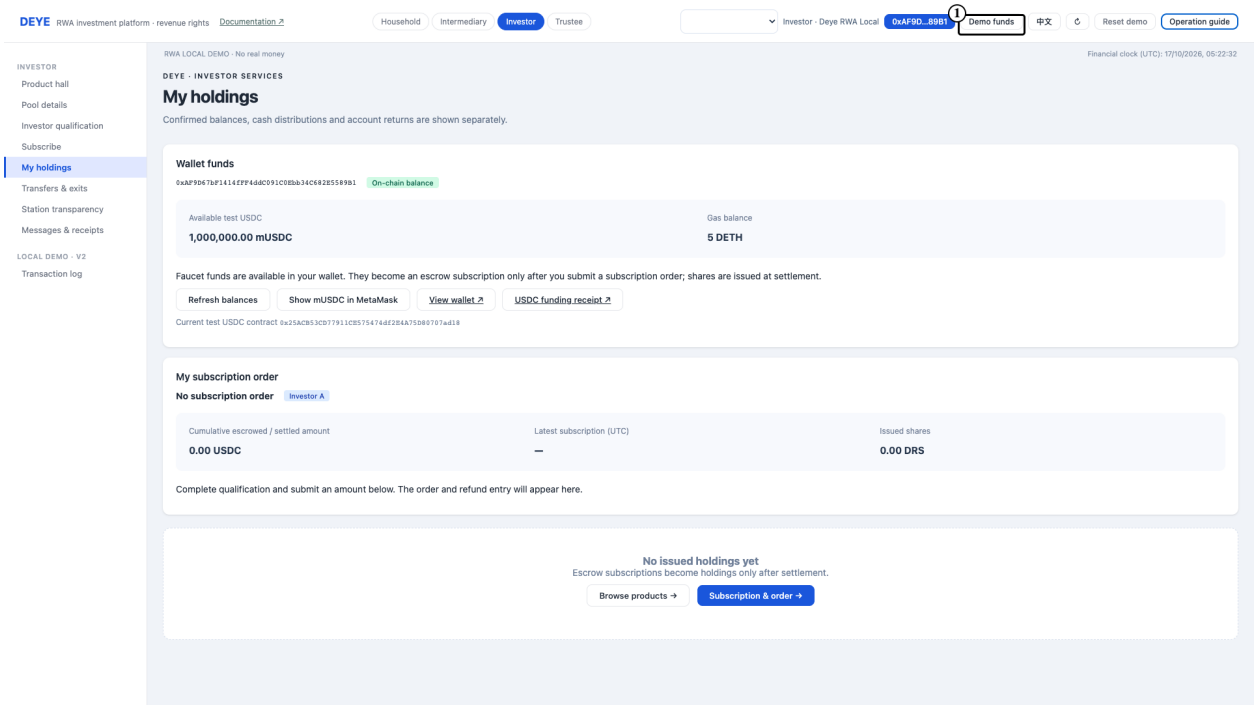


Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

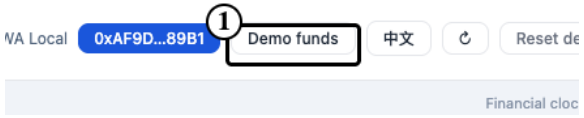
12.4 A3 Test funds for the selected product

This step is performed by The connected participant. Click Receive test funds in the top bar and check the wallet balances. The faucet supplies demo gas and the selected product's test USDC. It does not create an investment or issue DRS, and another product may have a different test USDC contract.

After this step, the investor can subscribe after qualification is approved.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

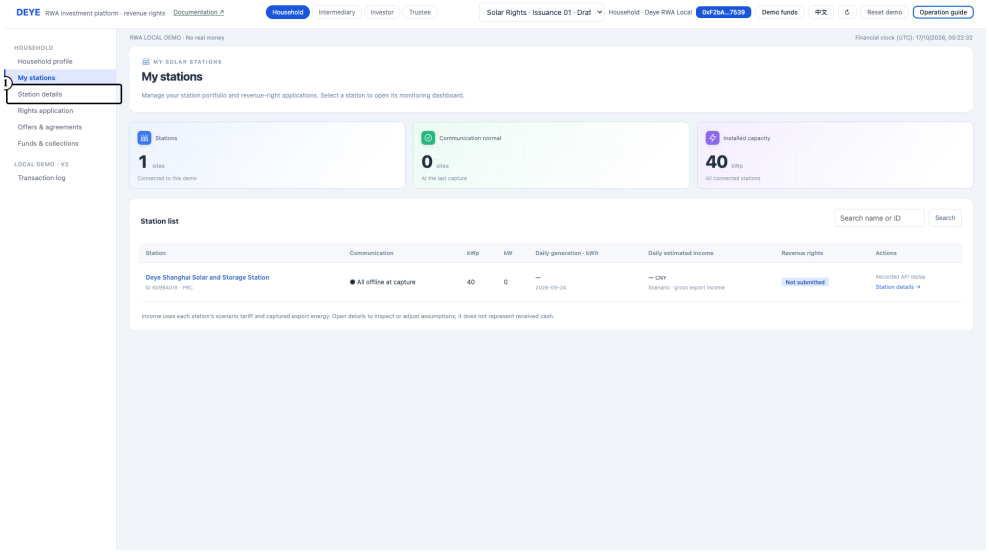


13 Application, valuation and agreement

13.1 B1 Station identity and evidence source

This step is performed by Owner. Open My stations and select the station details entry. Check the station identity, capacity and evidence timestamp before starting an application. The station list and the single-station dashboard serve different purposes.

After this step, review the detailed evidence and then submit an application.



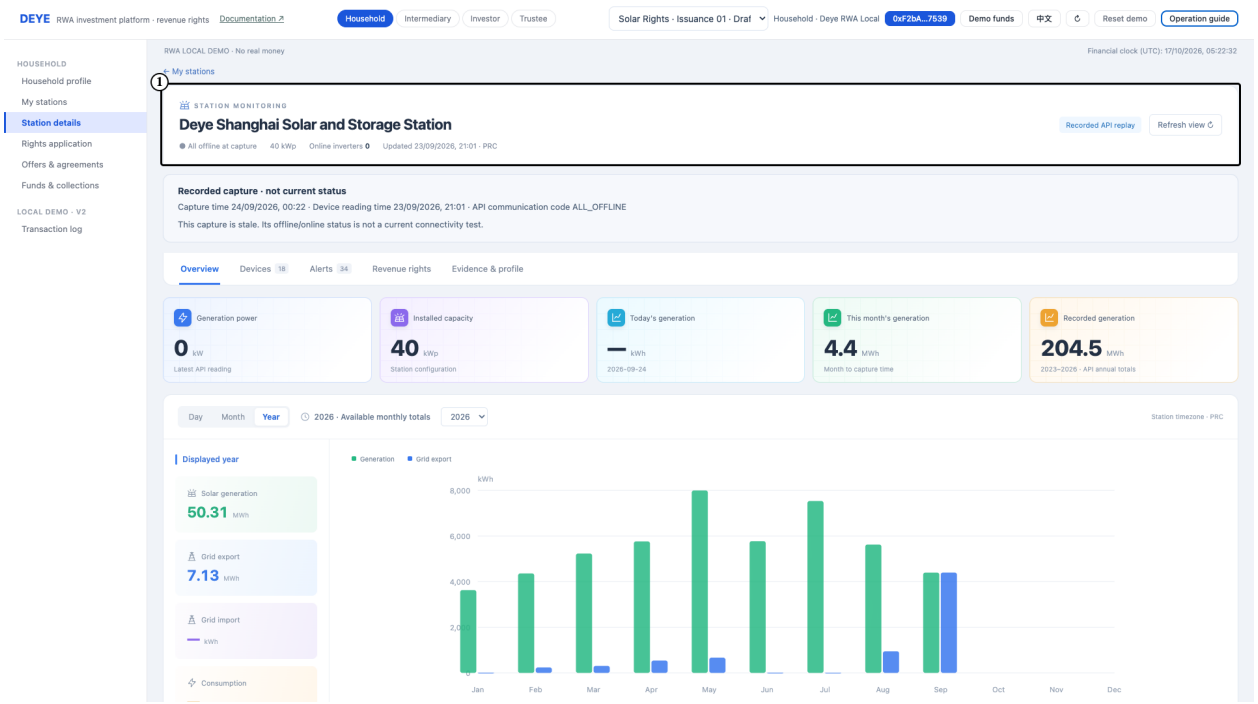
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



13.2 B2 Station dashboard review

This step is performed by Owner. Review generation, export, devices, alerts and the source badge in Station details. Recorded replay describes an earlier collection and is not a live reading. ALL_OFFLINE is the reported telemetry status and does not by itself invalidate a blockchain product.

After this step, submit the revenue-right application for the selected station.



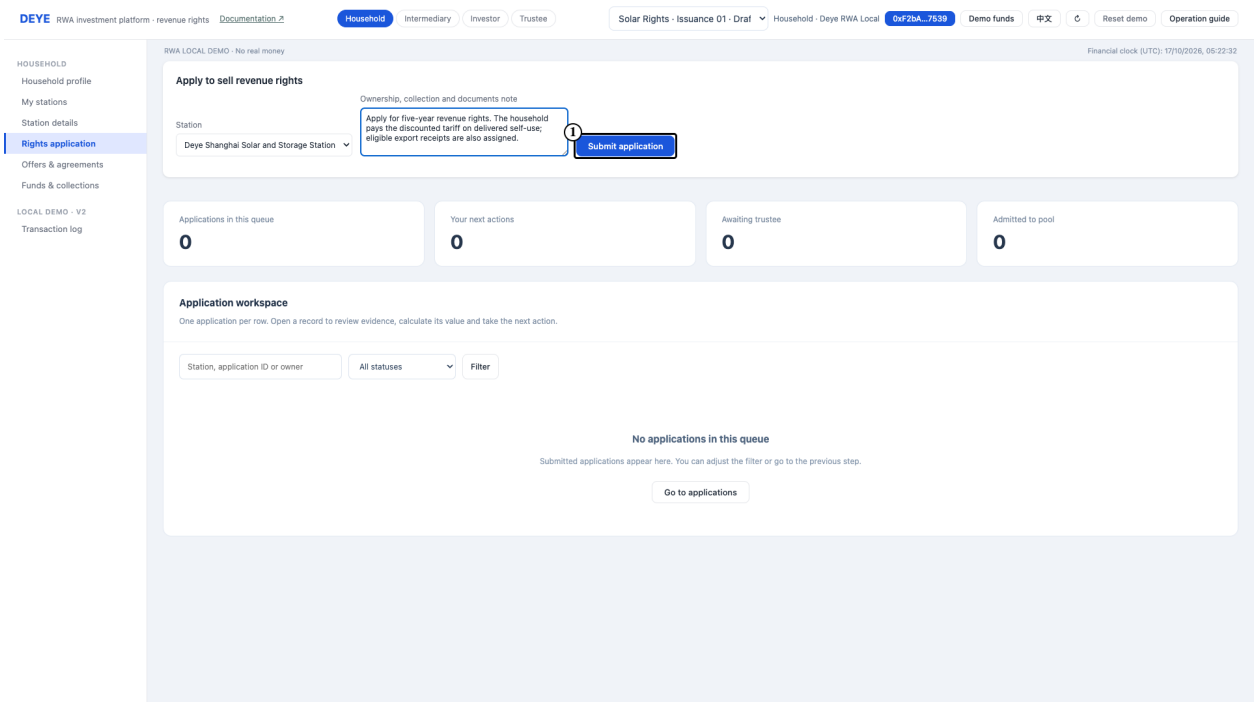
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

, 21:01 · PRC

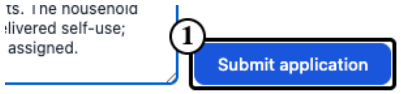
13.3 B3 Revenue-right application

This step is performed by Owner. Select the station and enter the requested information in the application form. Submit the record and retain its application identifier. This step stores the application and evidence references without minting an NFT or issuing investor shares.

After this step, the intermediary reviews the application queue.



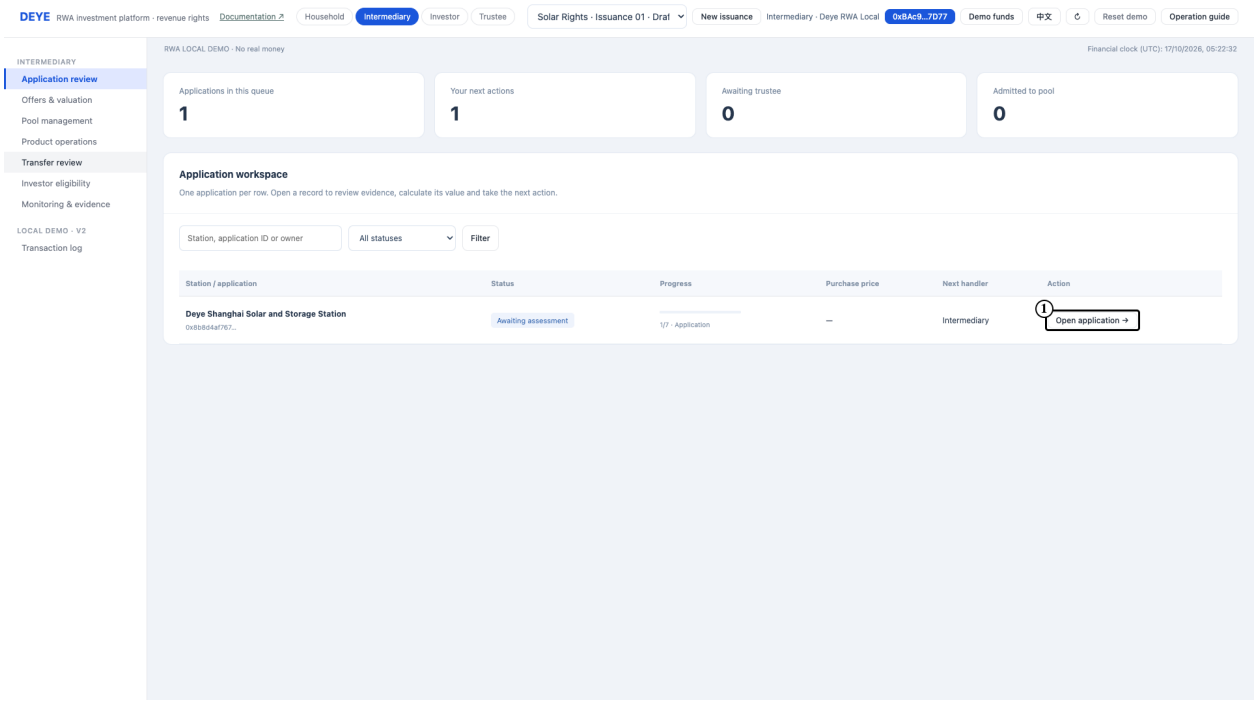
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



13.4 B4 Application queue

This step is performed by Intermediary. Open Application review and find the row for the relevant product and station. Check its status and next handler, then open the detailed record. Use the application identifier to distinguish separate requests for the same station.

After this step, review the evidence and terms in the application detail.



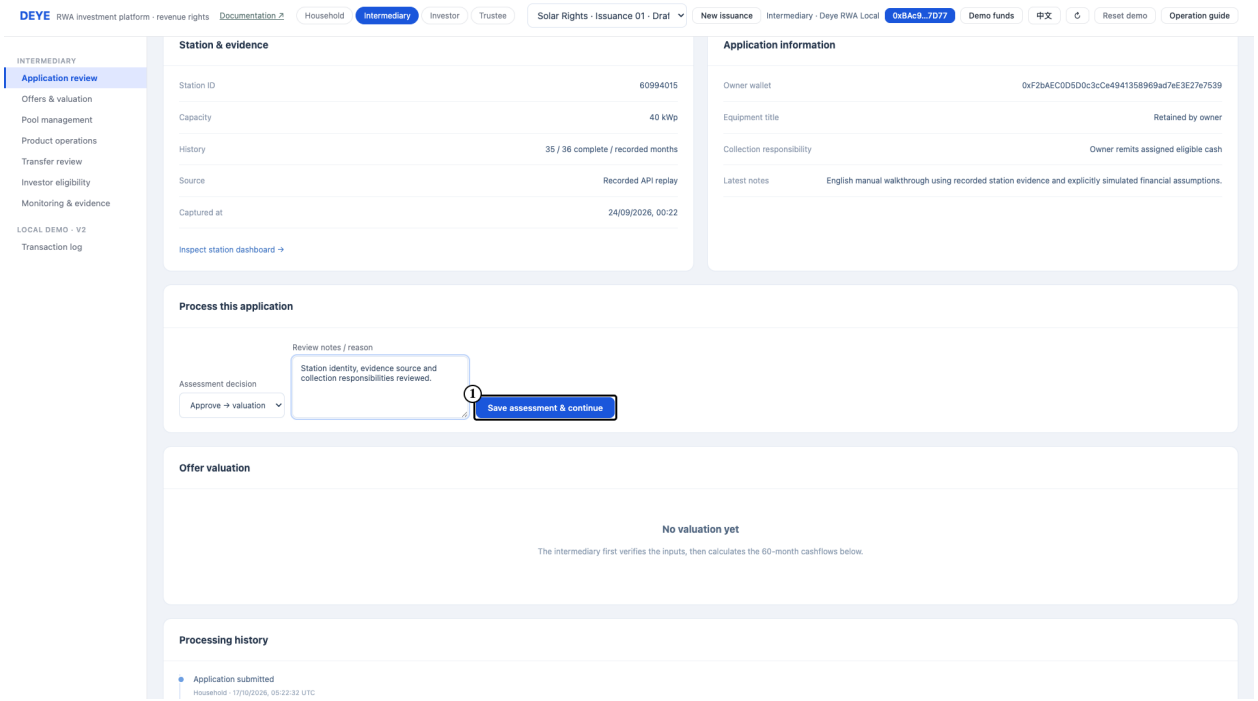
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



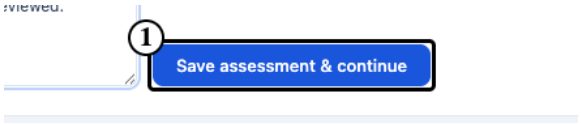
13.5 B5 Application assessment

This step is performed by Intermediary. Check station identity, evidence, the proposed rights scope and collection responsibilities. Approve complete materials or request supplementation with a clear explanation. A rejected or incomplete application must be resolved before pricing continues.

After this step, prepare the revenue-right offer for an approved application.



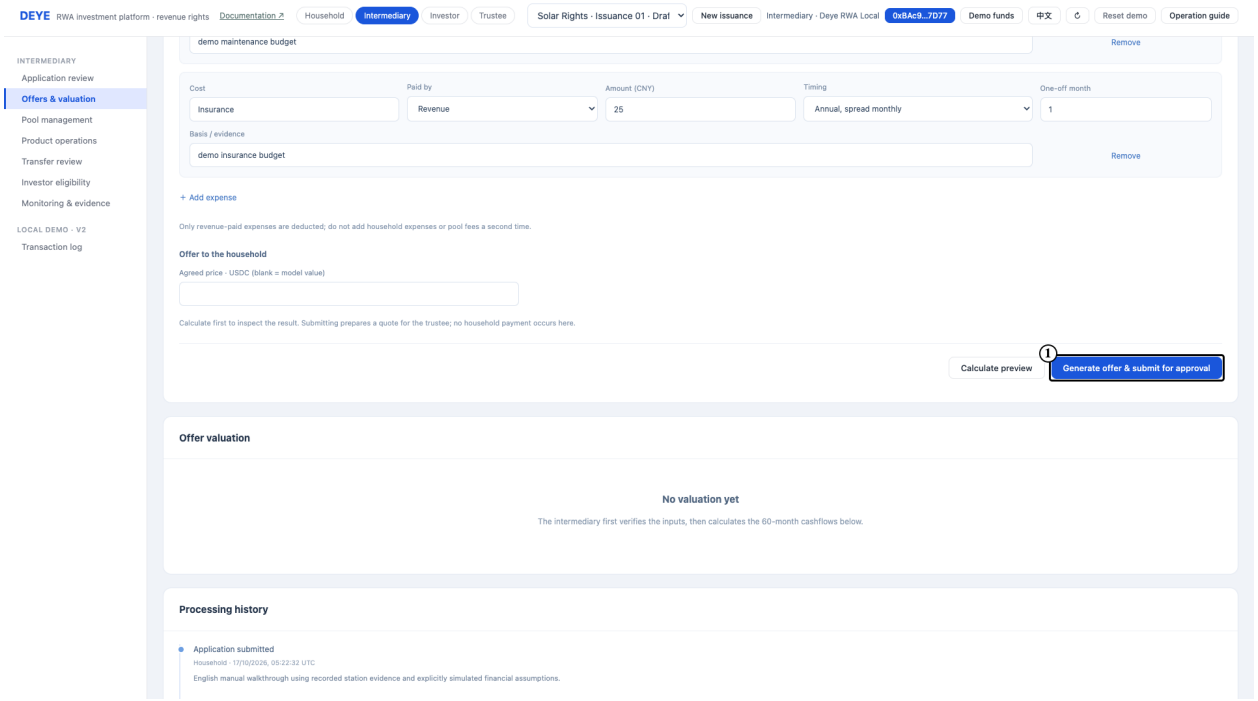
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



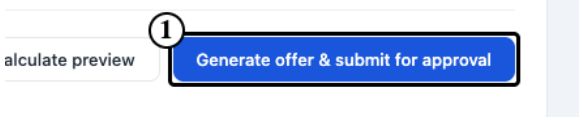
13.6 B6 Revenue-right valuation and offer

This step is performed by Intermediary. Open Quotes and valuation for the approved application. Review historical sample quality, seasonal generation and export estimates, delivered self-use, tariffs, owner discount, costs, FX and discounting. Record the reason for any provisional baseline before generating the offer.

After this step, the trustee independently reviews the offer and assumptions.



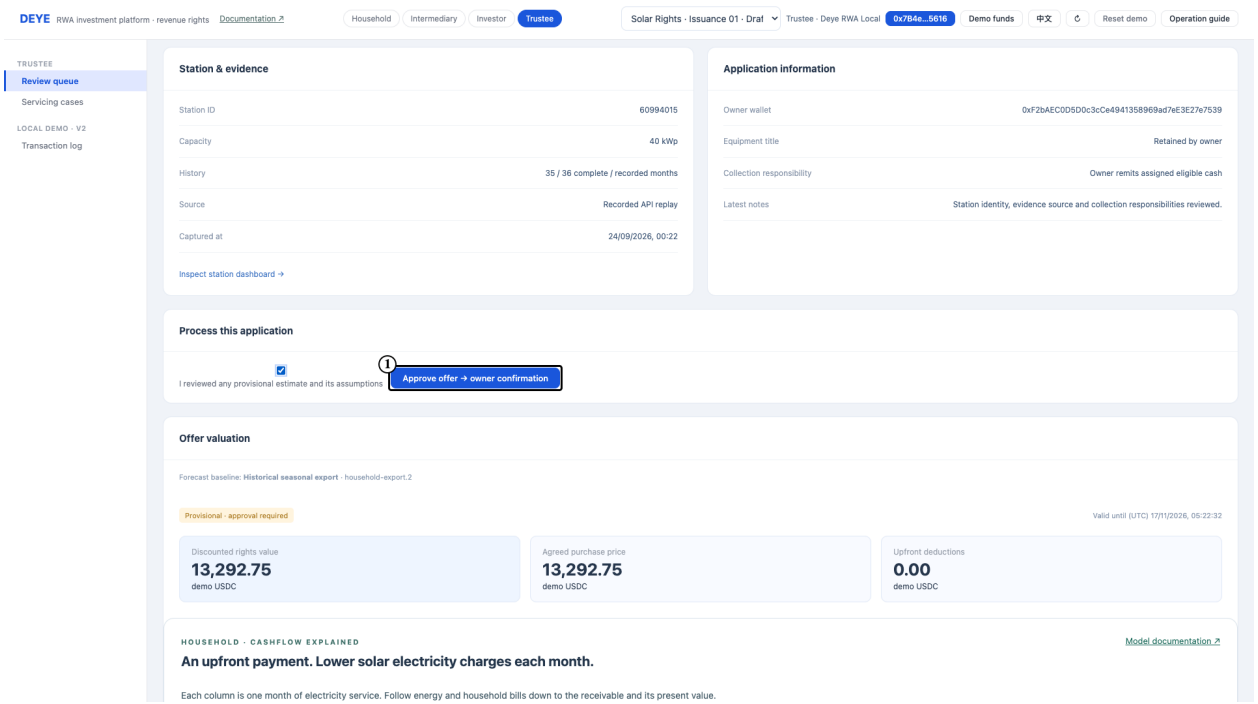
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



13.7 B7 Independent offer approval

This step is performed by Trustee. Open the application in the review queue and compare the forecast with its evidence and assumptions. Explicitly acknowledge provisional data when justified, then approve the offer. Approval does not sign the owner's agreement or transfer funds.

After this step, the owner reviews and accepts the approved offer.



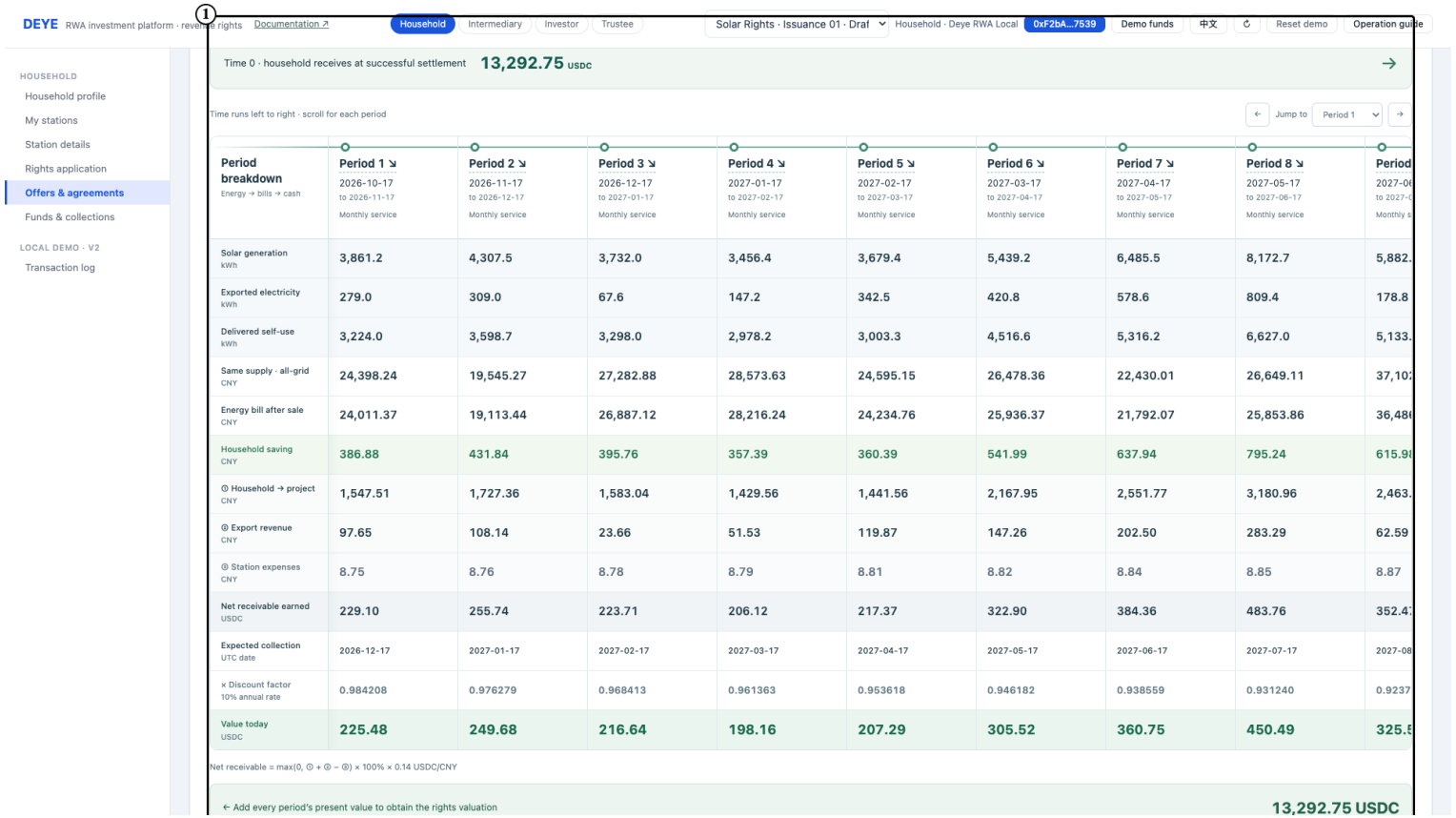
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



13.8 B8 Owner cashflow and electricity discount

This step is performed by Owner. Open the approved quote and inspect the monthly timeline from left to right. For each period, compare generation, export, delivered self-use, the grid-price comparison, the owner's discounted payment, export income, costs and discounted present value. These are forecasts rather than recorded receipts.

After this step, accept only after the price and future payment obligation are understood.

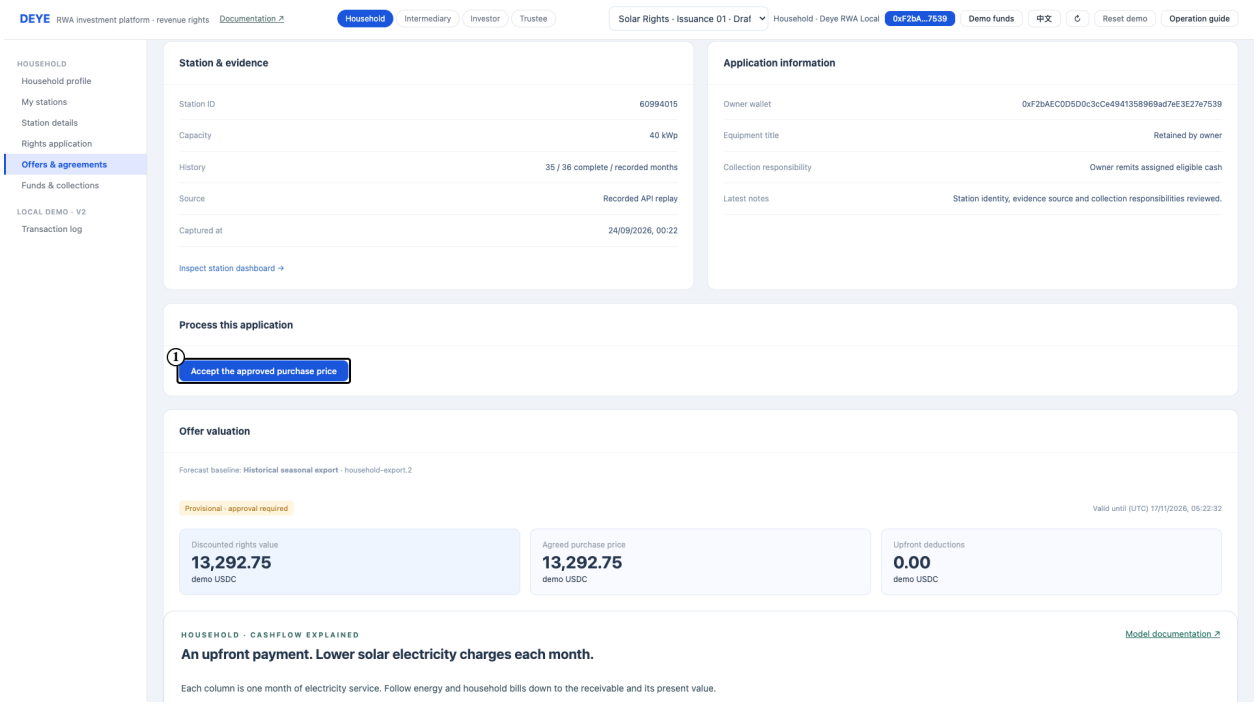


Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

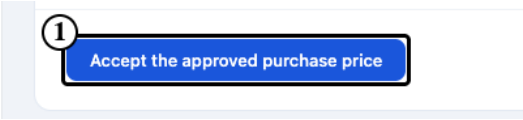
13.9 B9 Acceptance of the approved offer

This step is performed by Owner. Check the rights percentage, five-year term, purchase price, deductions and discounted electricity payment rule. Click Accept for the approved offer. Acceptance records consent to these terms and does not yet mint the station credential.

After this step, sign the corresponding agreement with the owner wallet.



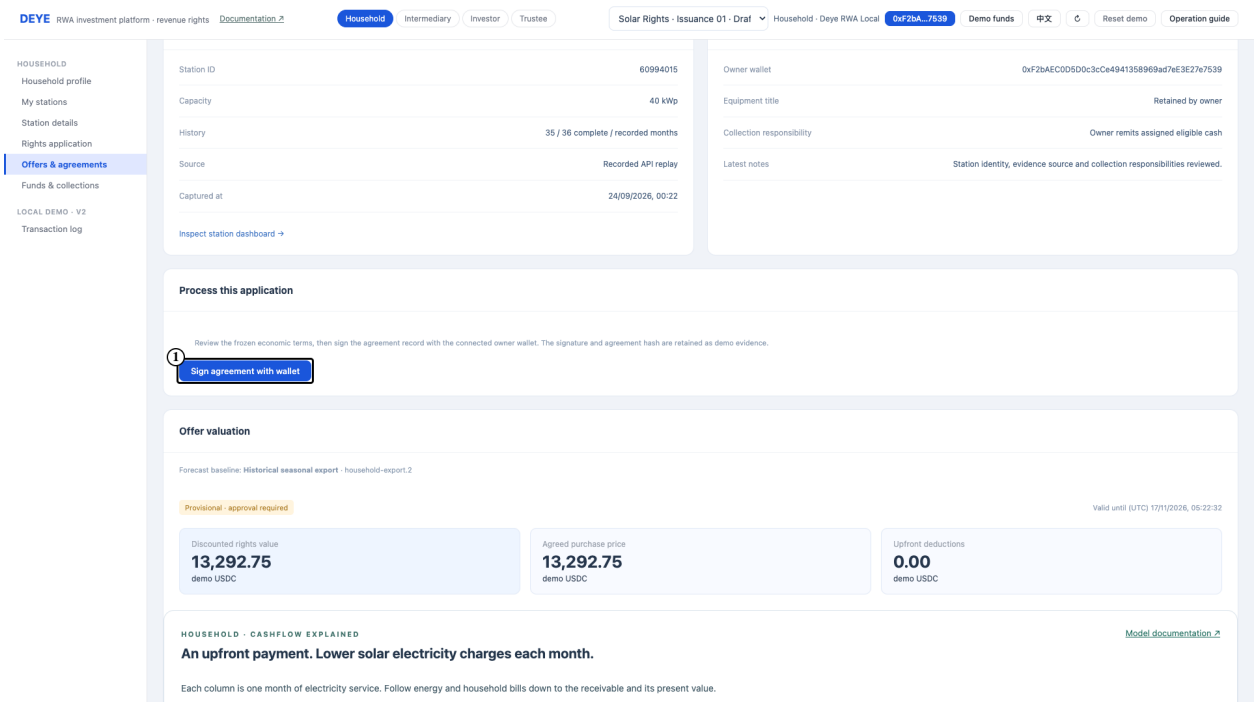
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



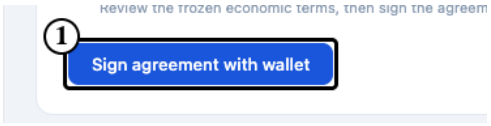
13.10 B10 Revenue-right agreement signature

This step is performed by Owner. Read the agreement and request the wallet signature. Verify the station, product, Chain ID, price, term and content digest in the message. The signature records agreement acceptance off-chain without transferring equipment ownership.

After this step, the intermediary mints the station identity credential.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

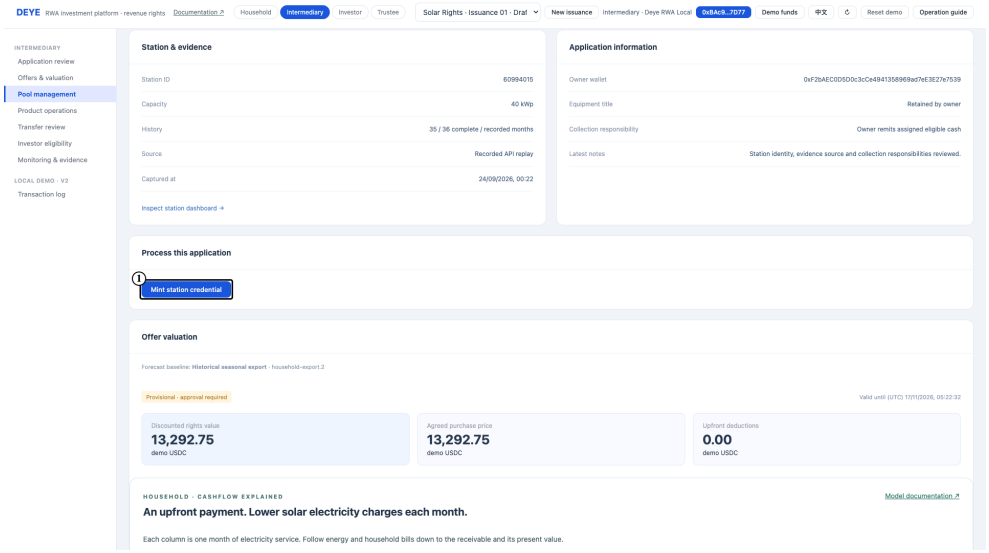


14 Credentials, admission and publication

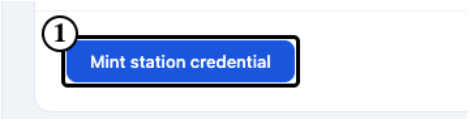
14.1 C1 Station credential minting

This step is performed by Intermediary. In Pool management, open the signed application and click Mint station credential. Confirm the transaction with the intermediary wallet. After confirmation, the station NFT belongs to the owner and the transaction can be checked in Blockscout.

After this step, the owner authorizes custody by this pool.



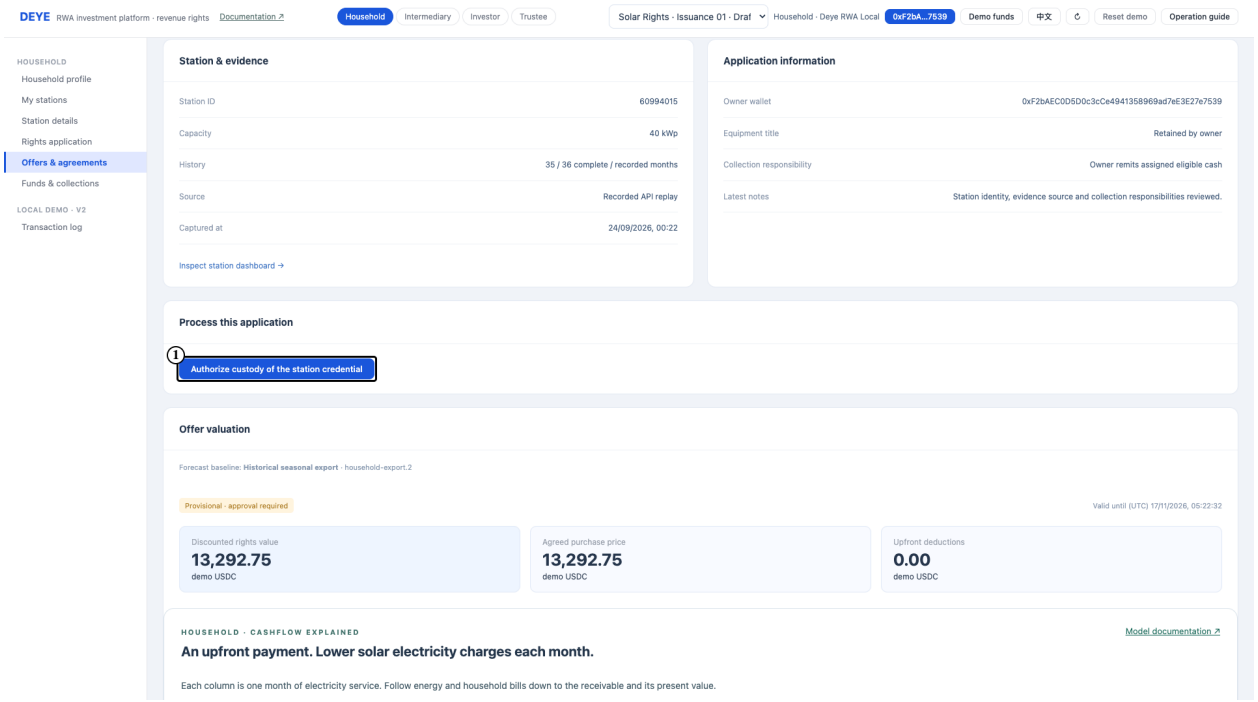
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



14.2 C2 Pool custody authorization

This step is performed by Owner. Open the agreement's custody task and check the station NFT and destination vault. Authorize this pool in the wallet and wait for chain confirmation. Approval permits custody but is not itself admission into the pool.

After this step, the intermediary executes admission after authorization is confirmed.



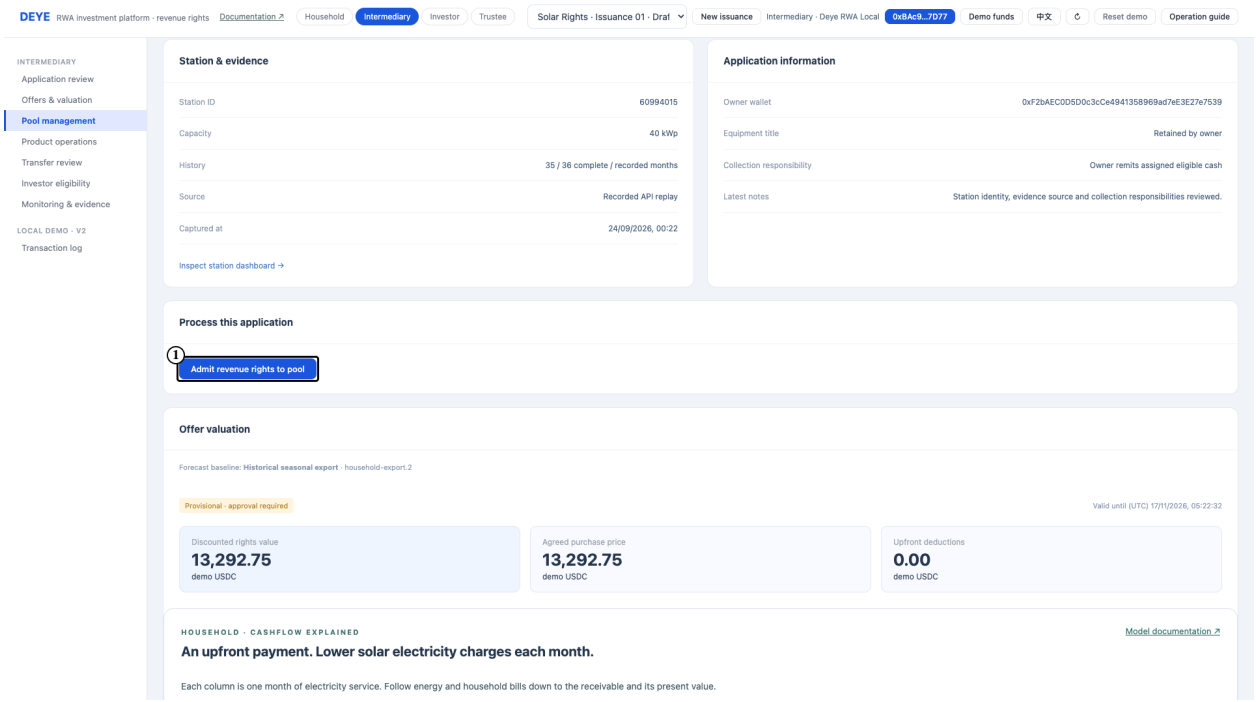
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



14.3 C3 Revenue-right admission

This step is performed by Intermediary. Check the signed agreement, owner authorization, purchase price and rights term, then execute admission. Confirm that the station NFT is held by the vault and the agreement digest is recorded. Admission does not mint investor shares.

After this step, prepare the pool valuation.



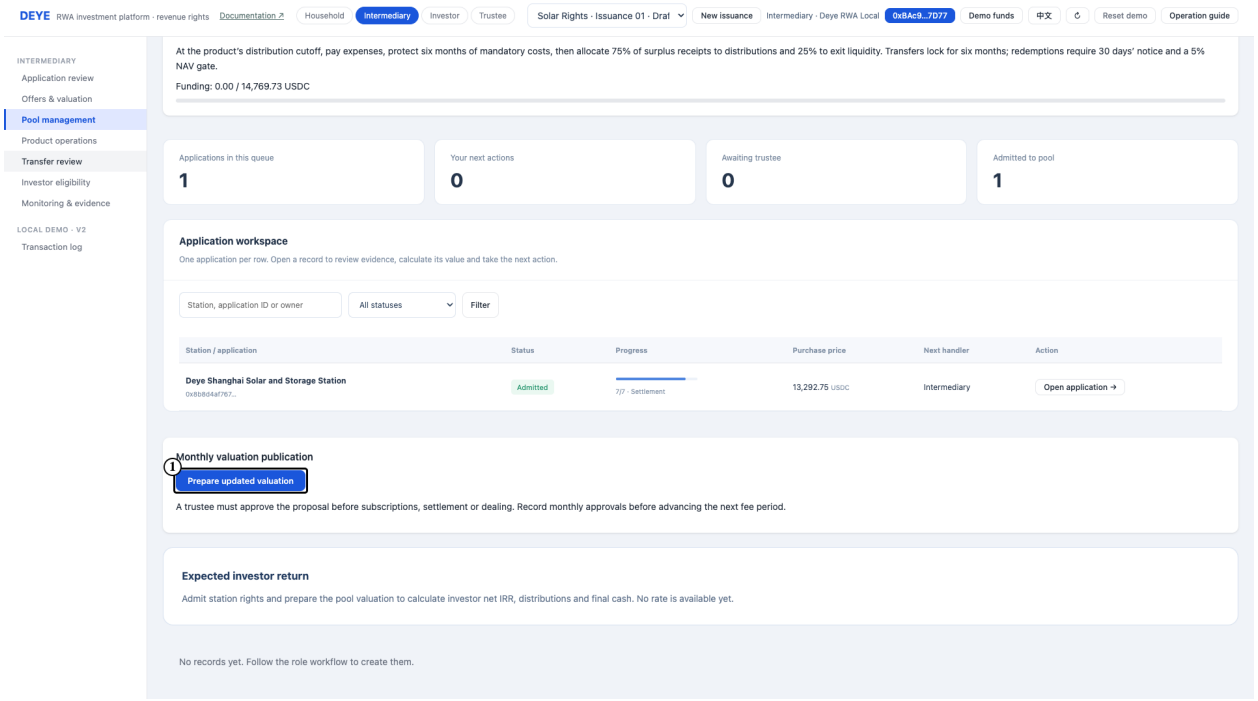
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



14.4 C4 Pool valuation submission

This step is performed by Intermediary. Prepare the pool report after reconciling rights value, cash, receivables, liabilities and future pool expenses. Confirm the submission transaction and retain its report digest and validity period. A material cashflow change may require a new report.

After this step, the trustee approves the submitted report.



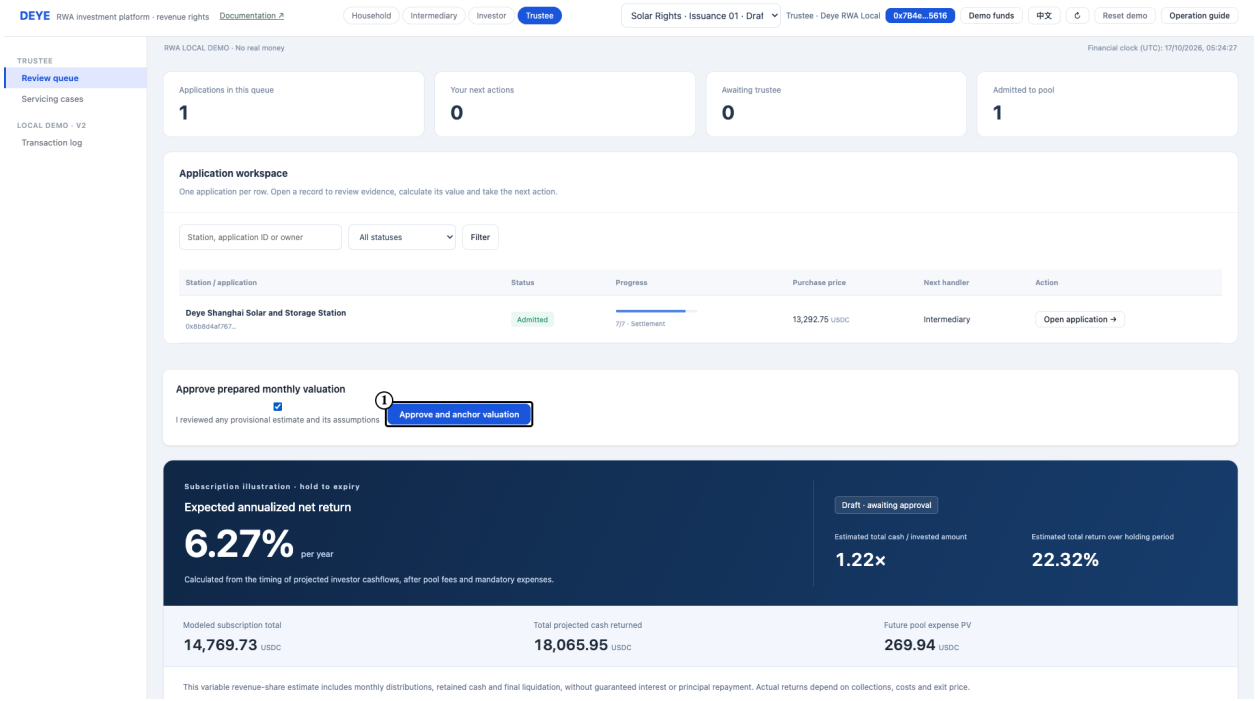
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



14.5 C5 Pool valuation approval

This step is performed by Trustee. Review the current report and explicitly check the assumptions acknowledgement before clicking Approve and anchor valuation. Confirm with the trustee wallet. An expired or changed report must return to the intermediary for preparation rather than being forced through approval.

After this step, the intermediary publishes fundraising after approval.



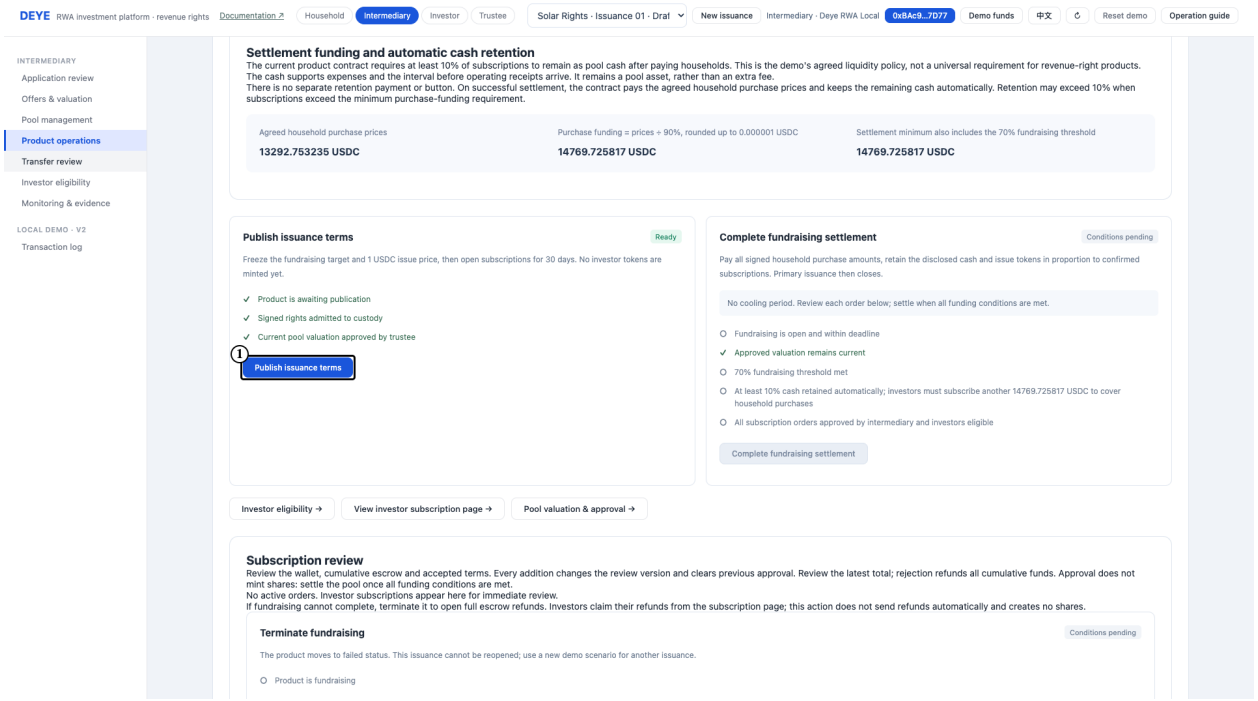
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



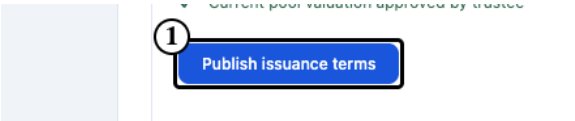
14.6 C6 Fundraising publication

This step is performed by Intermediary. Open Product operations and review projected net IRR, expected distributions, reserve policy, target, price and deadline. Publish the issuance terms with the wallet. Investors can then see the product as Fundraising, although no investor shares have been issued yet.

After this step, investors review the published product and apply for qualification.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

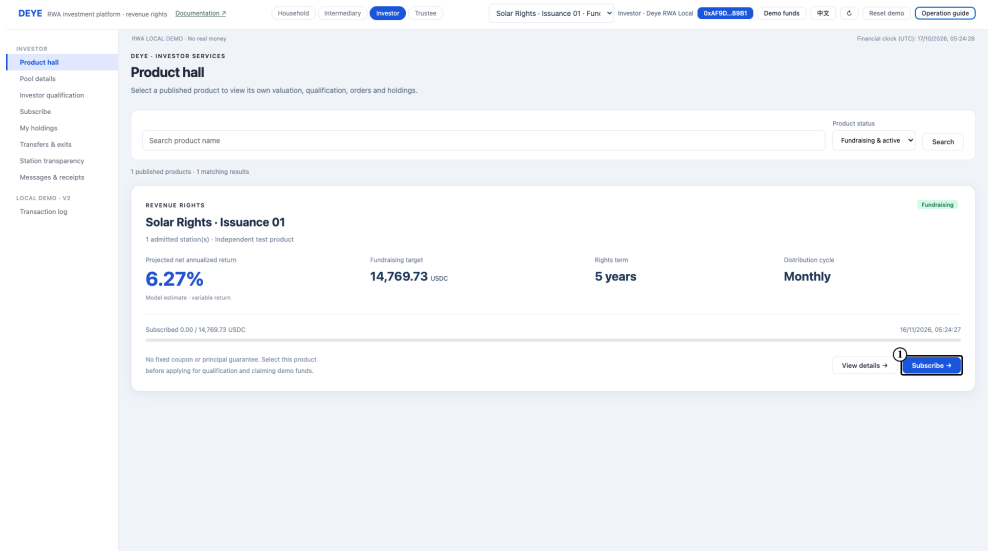


15 Qualification, subscriptions and settlement

15.1 D1 Product selection

This step is performed by Investor. Open Product hall and select a published Fundraising or Active product. Check its status, cadence, target and contract before entering details. Draft contracts are management work and are excluded from the investor hall.

After this step, review the product timeline and eligibility requirements.



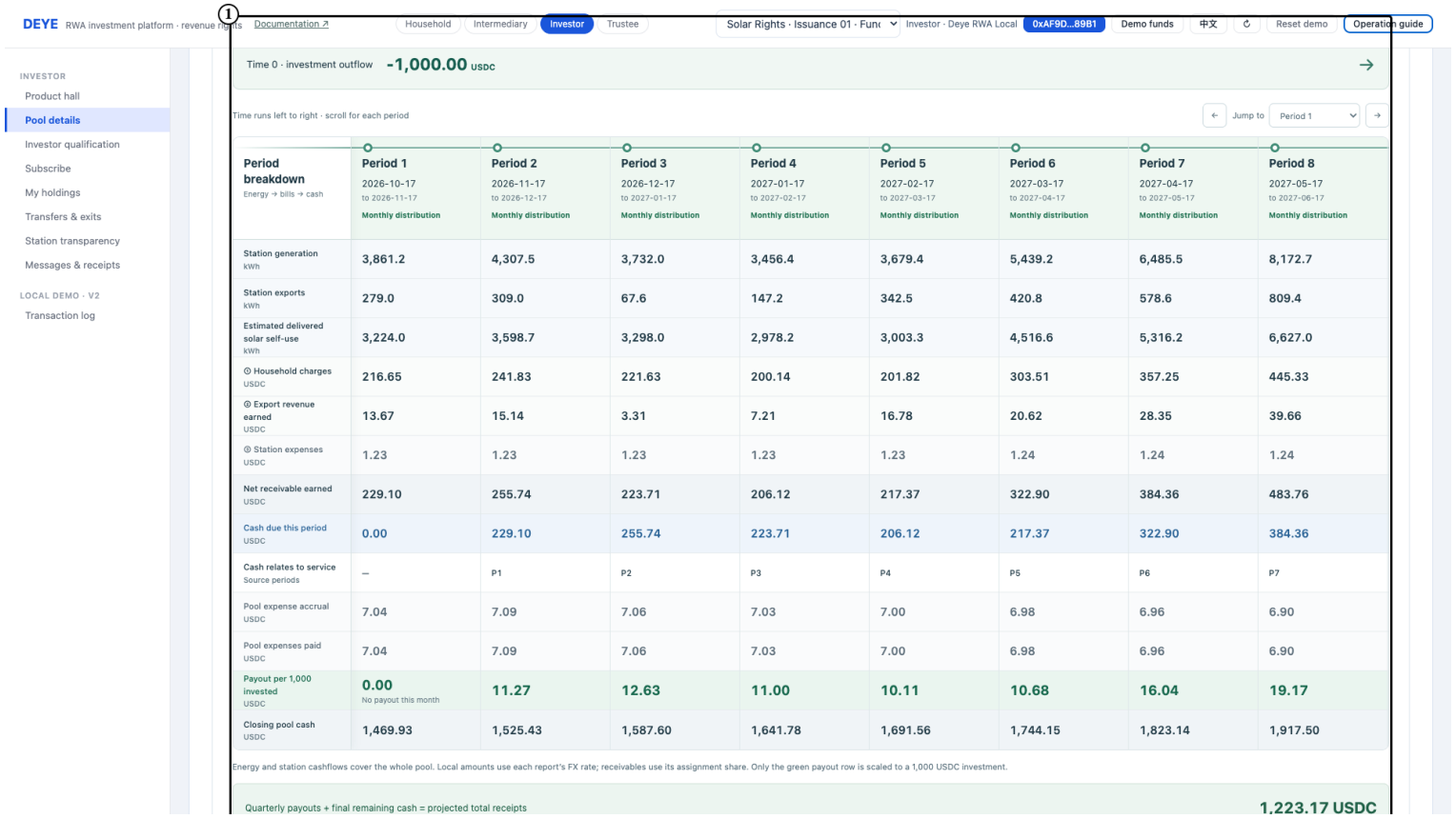
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



15.2 D2 Investor cashflow timeline

This step is performed by Investor. Read the monthly timeline in Pool details. Distinguish the initial investment, monthly collections, accrued expenses, declared distributions, retained cash and remaining NAV. The projected net IRR is a model result and is not the valuation discount rate or a guaranteed coupon.

After this step, apply for qualification before subscribing.

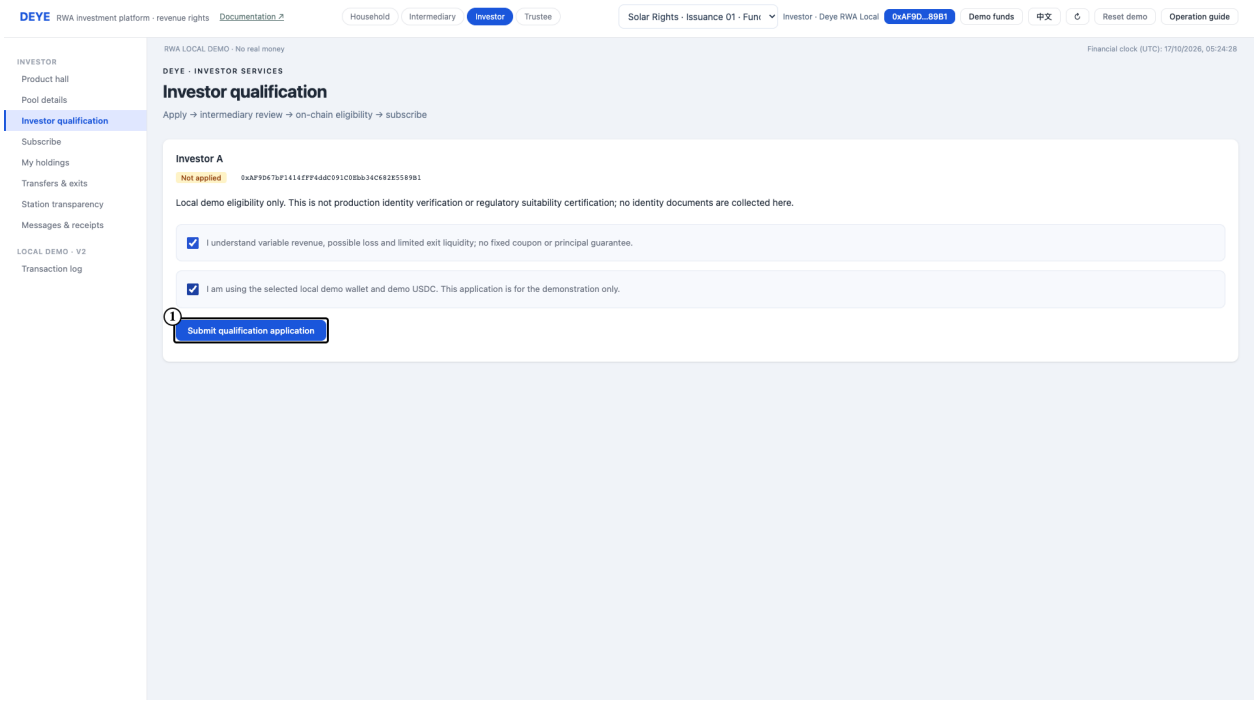


Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

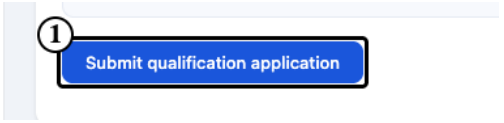
15.3 D3 Investment qualification application

This step is performed by Investor. Open Investment qualification, read the risk statement and acknowledge the test-only setting. Submit the application for the connected wallet and selected product. A staff member must approve it before the contract accepts a subscription.

After this step, the intermediary reviews this wallet's qualification.



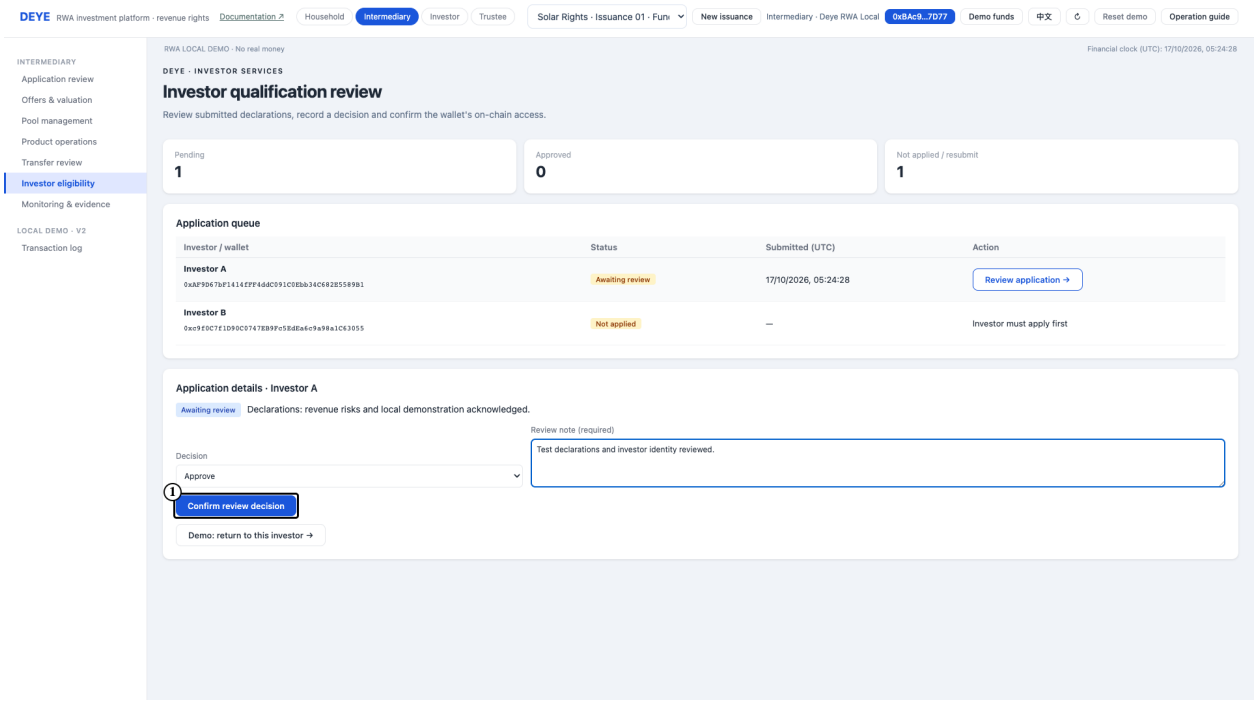
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



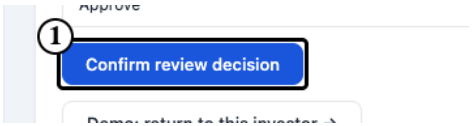
15.4 D4 Qualification approval

This step is performed by Intermediary. Open the qualification queue, select the investor and review the application. Approve the correct wallet for the selected product and confirm the on-chain transaction. Choosing an investor role in the UI does not grant eligibility.

After this step, the investor receives test funds and submits a subscription.



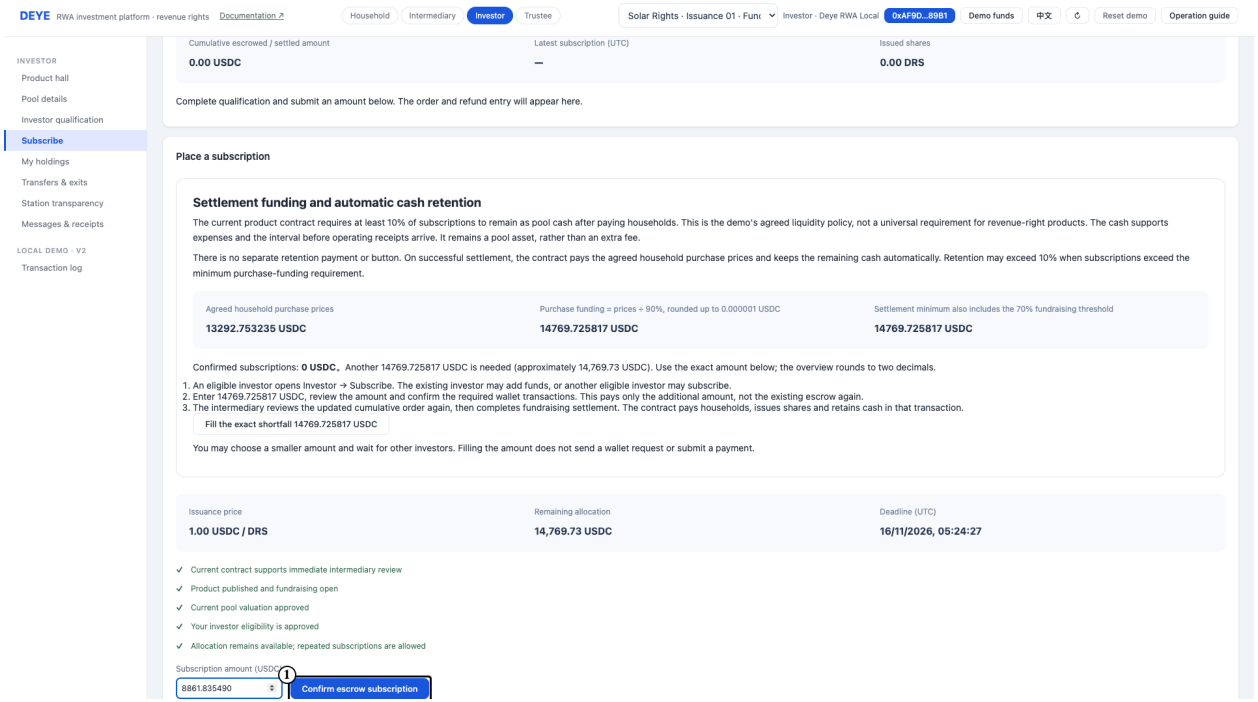
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



15.5 D5 USDC approval and subscription

This step is performed by Investor. Enter the additional amount in Subscription and check the remaining capacity and wallet balance. Confirm USDC approval if requested, then confirm the subscription transaction. Each addition increases cumulative escrow and invalidates the earlier order review. Repeat subscriptions remain available while fundraising permits them.

After this step, the intermediary reviews the latest cumulative amount before settlement.



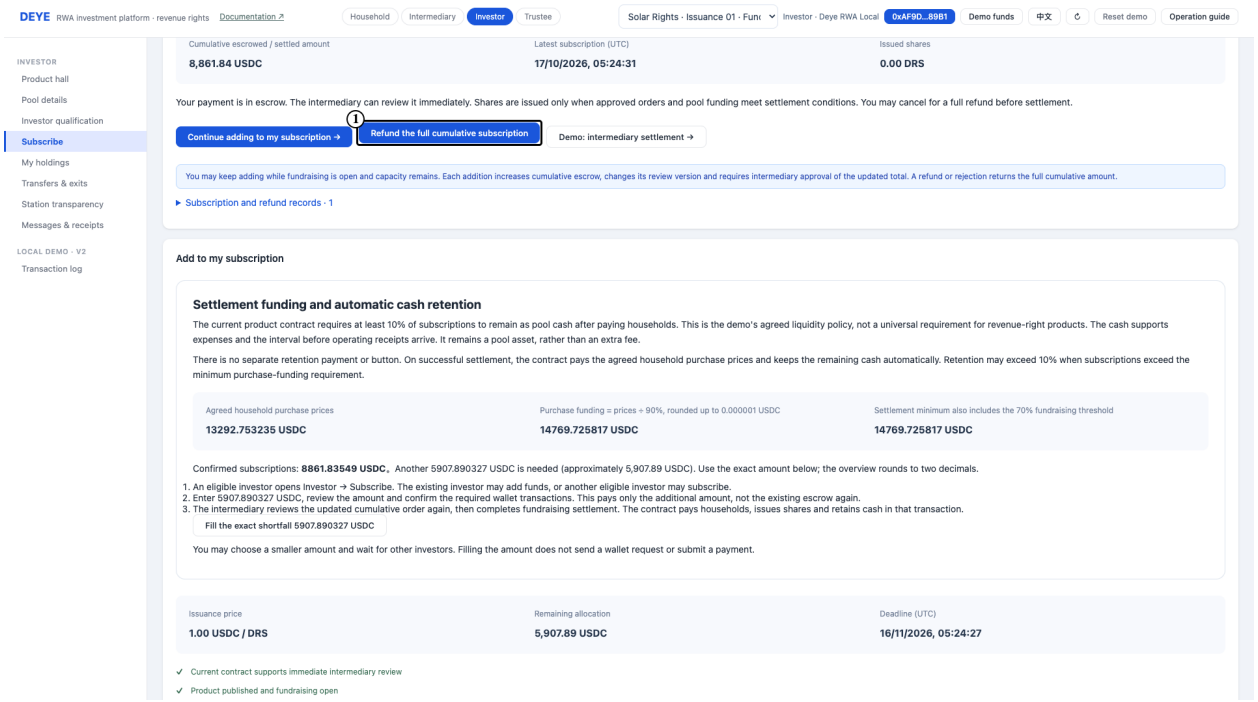
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



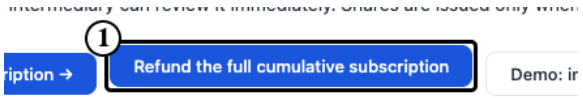
15.6 D6 Cancellation before settlement

This step is performed by Investor. Open the order and click the refund action before issuance settlement. Confirm the wallet transaction and check that the full cumulative escrow returns to the wallet. This cancels the pending investment without issuing DRS.

After this step, the investor may subscribe again while fundraising remains open.



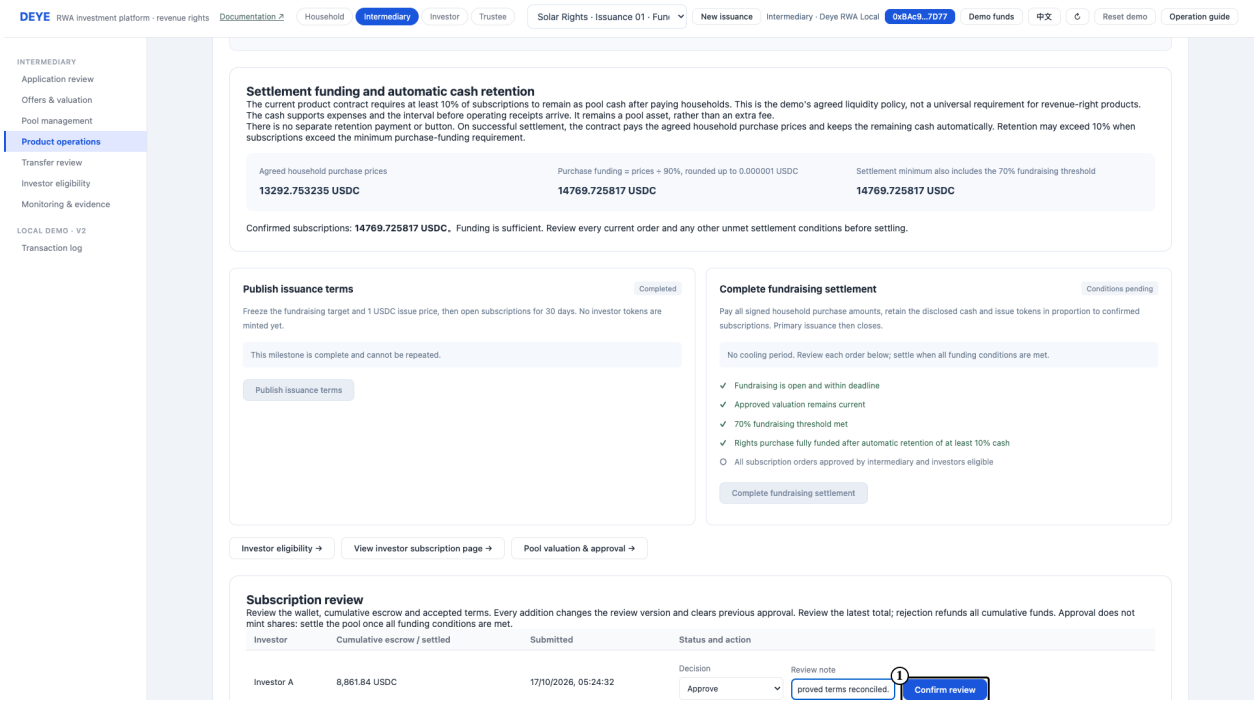
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



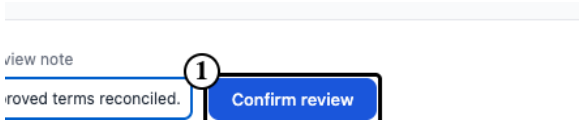
15.7 D7 Cumulative subscription review

This step is performed by Intermediary. In Product operations, locate the investor's order and compare wallet, cumulative escrow and issuance terms. Approve the latest review version with the intermediary wallet. Approval does not issue shares, and a later addition requires another review.

After this step, settle only when every order and all funding conditions are satisfied.



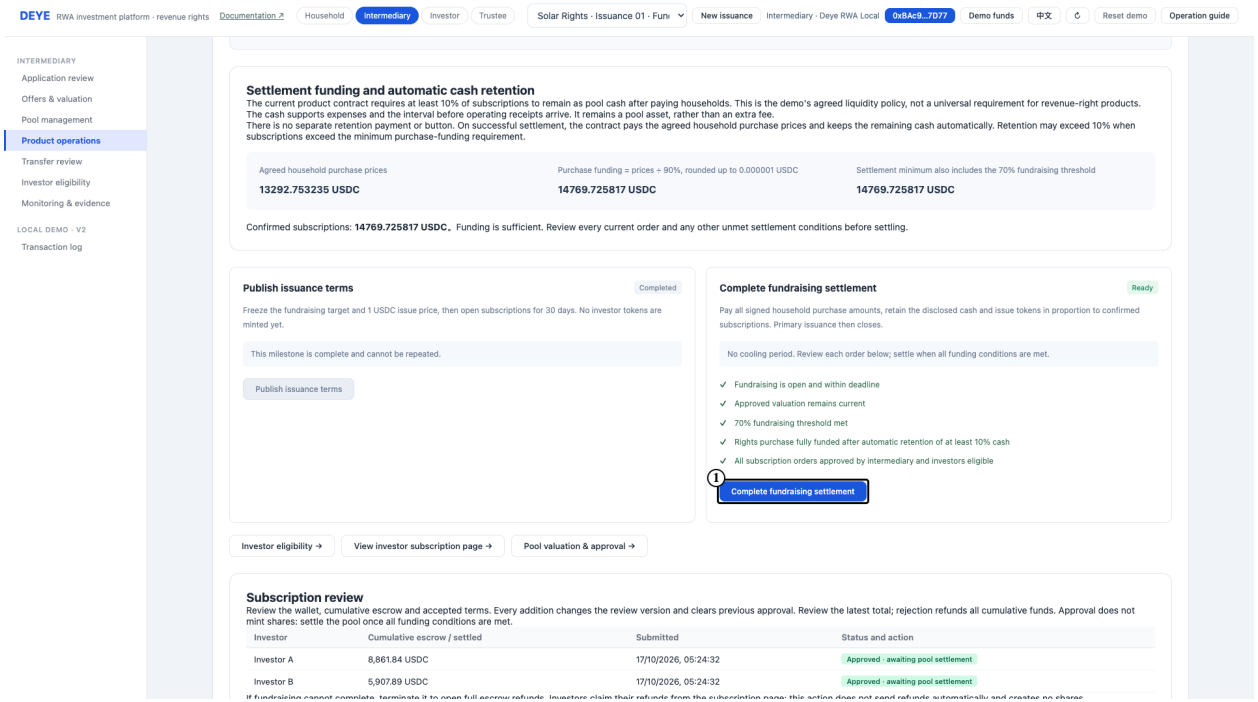
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



15.8 D8 Fundraising settlement and share issuance

This step is performed by Intermediary. Check that all orders are approved and the subscription total covers signed purchases plus the 10% initial cash retention. The contract retains the cash automatically during settlement. If a shortfall is shown, direct the investor to add exactly that amount and review the updated order before settling.

After this step, the owner receives the purchase payment and investors inspect their issued shares.



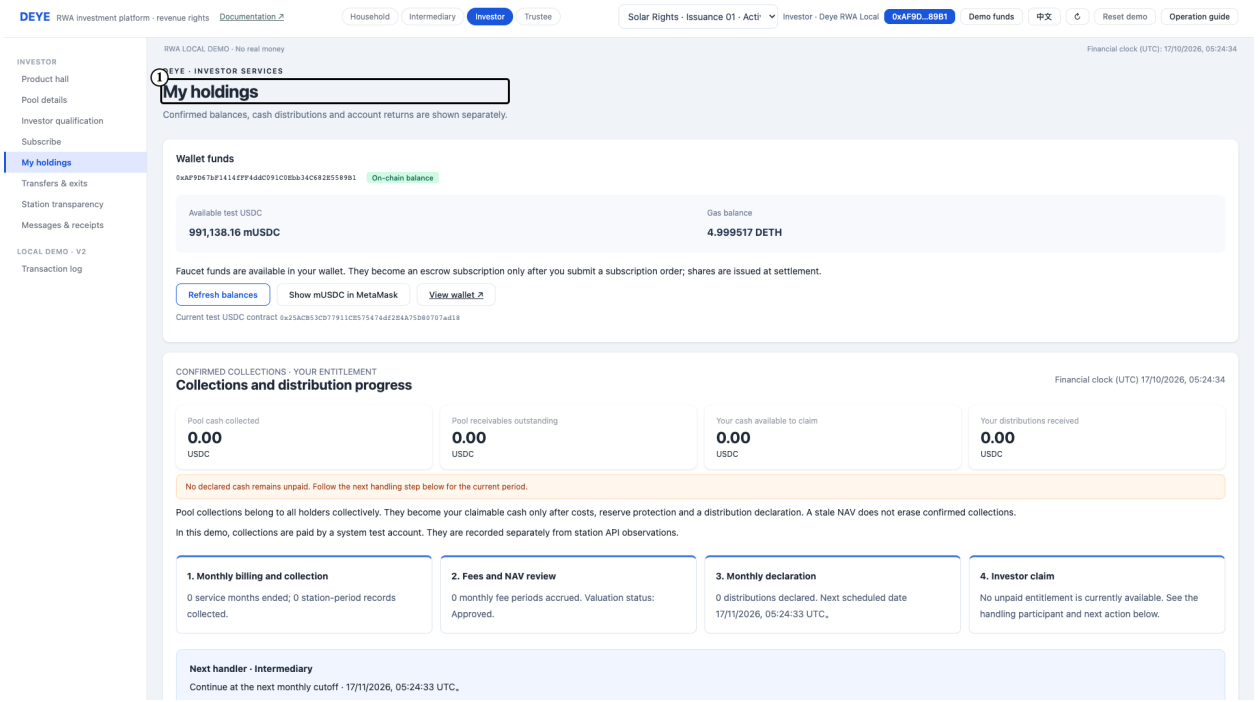
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



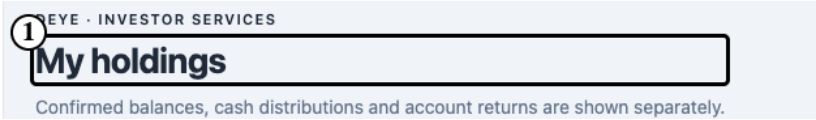
15.9 D9 Confirmed investment position

This step is performed by Investor. Open My holdings after settlement and refresh the confirmed state. Check DRS, the settlement record, the latest approved NAV and any declared distribution. Wallet USDC, subscription escrow and investment NAV are different amounts.

After this step, the intermediary begins monthly collection and accounting.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

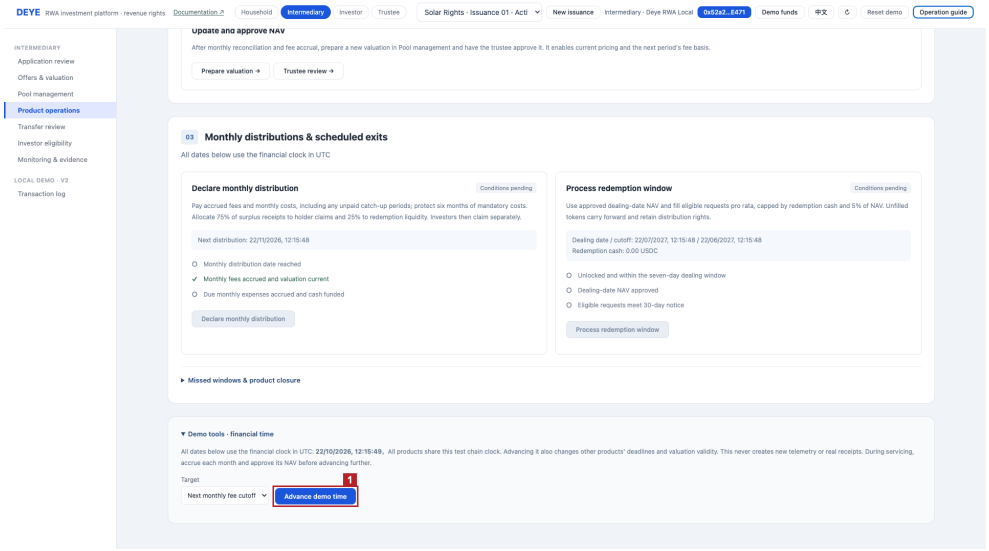


16 Monthly collections, valuation and distributions

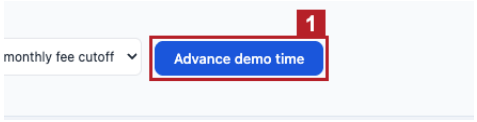
16.1 E1 Demonstration financial cutoff

This step is performed by Intermediary. Use Locate operation for the monthly fee task. If its cutoff has not arrived, choose the offered financial-time shortcut and read the shared-clock confirmation. Advancing the test clock affects all products on the chain and does not create live telemetry or approve valuations.

After this step, record completed service periods and reconcile the opening NAV before accruing fees.



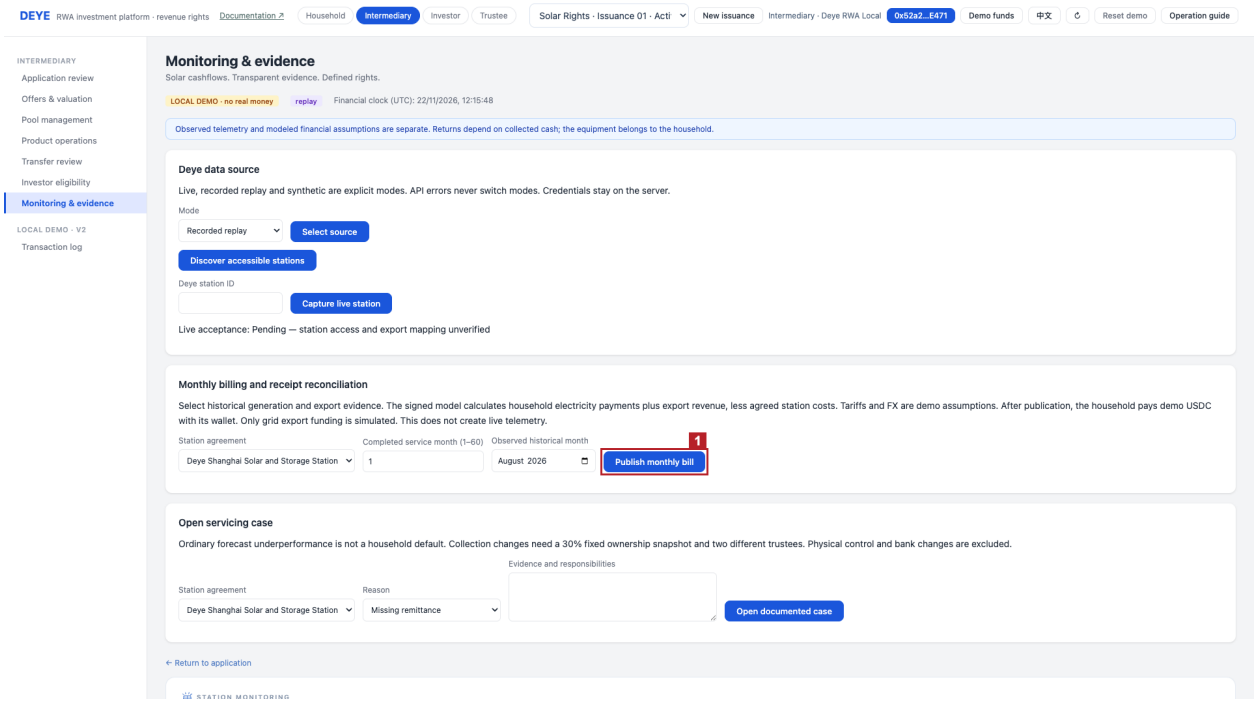
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



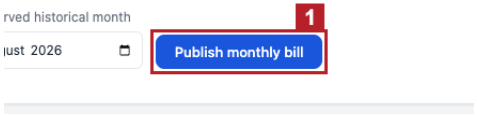
16.2 E2 Publish the monthly payment bill

This step is performed by Intermediary. Under Monitoring & evidence, check the service period and complete recorded history month. Review the household bill, export income, station costs and FX, then use marker 1 to publish and confirm the wallet transaction.

After this step, the invoice binds the designated household, exact test-USDC payment and evidence digest. No cash has been collected yet. The owner pays next.



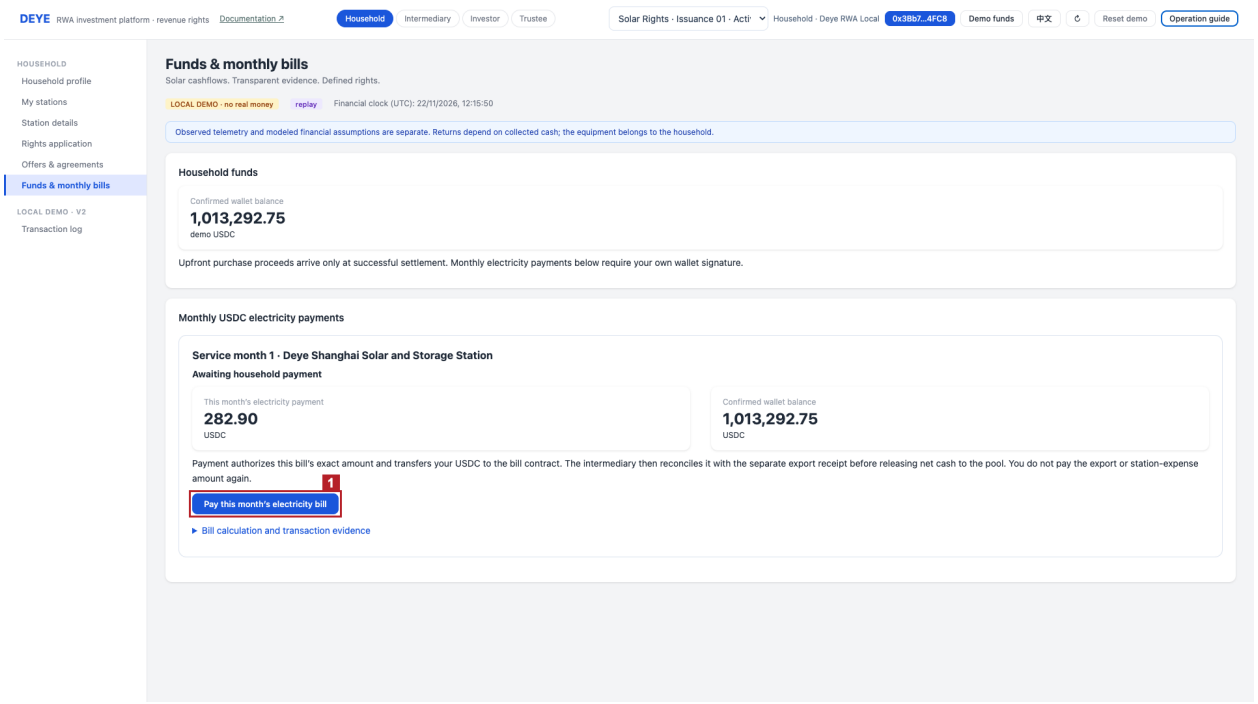
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



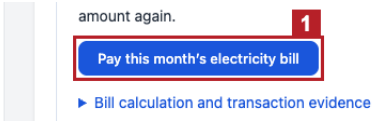
16.3 E3 Approve and pay the monthly bill

This step is performed by Owner. Under Funds & monthly bills, check the product, period, token and amount. Use marker 1, approve the exact invoice amount in the wallet and then confirm payment. Explicitly claim test funds first if the balance is insufficient.

After this step, the household wallet transfers the bill amount into the billing contract. The invoice awaits intermediary reconciliation. A direct token transfer does not settle a named bill.



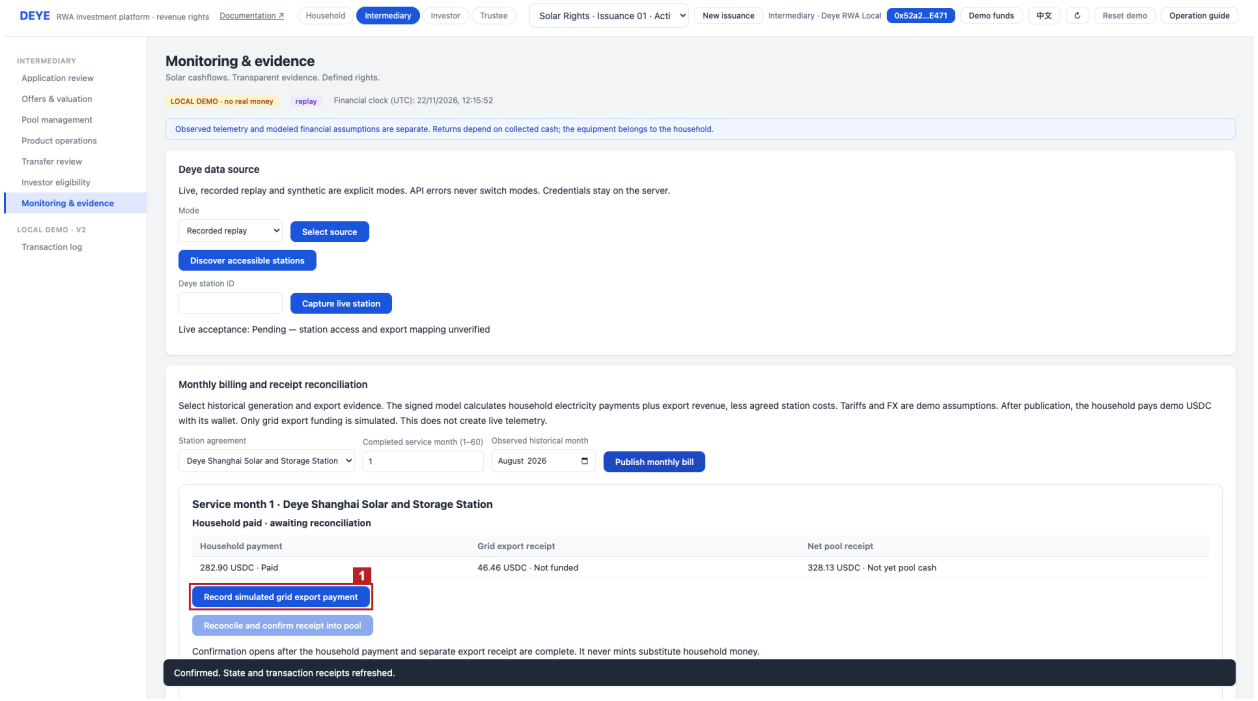
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



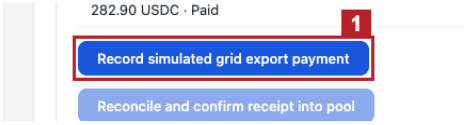
16.4 E3a Record the separate grid-export receipt

This step is performed by Intermediary. Under Monitoring & evidence, inspect the same invoice and household-payment status. Use marker 1 to record only the explicitly simulated grid-export receipt.

After this step, a separate system account funds the export component. It does not replace or top up the household payment. Both components await reconciliation.



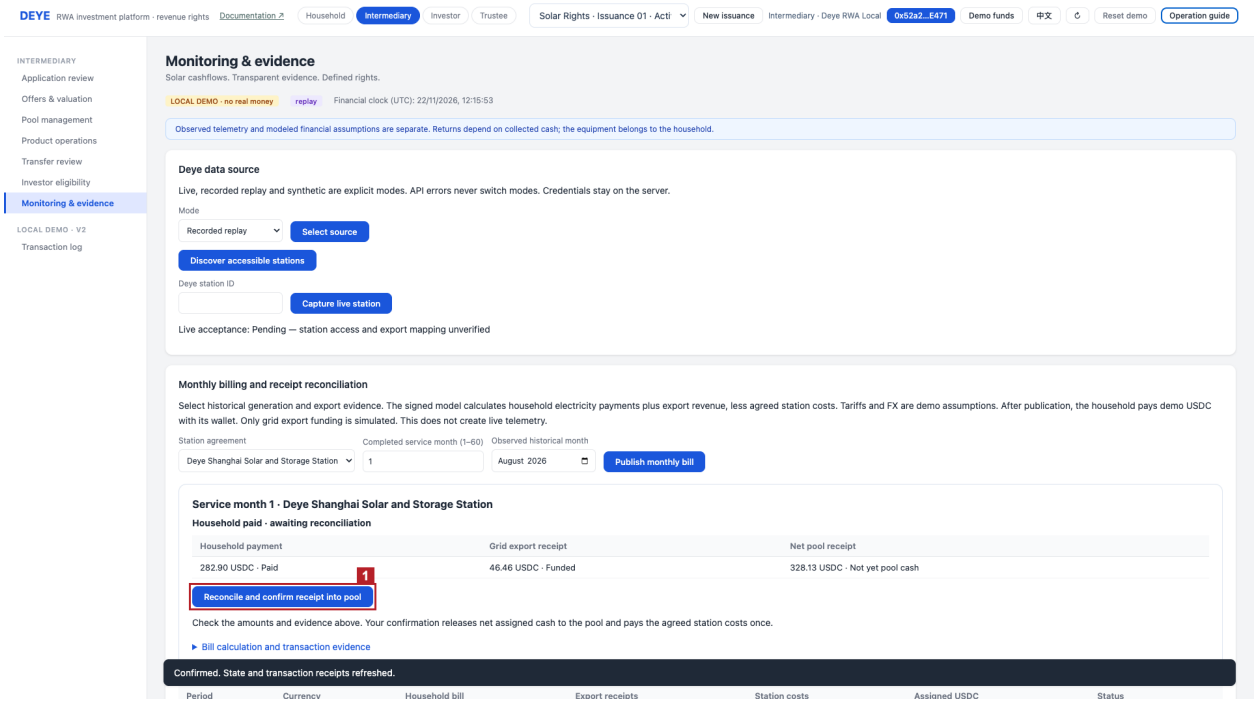
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



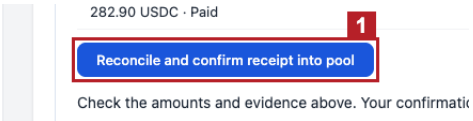
16.5 E3b Reconcile and confirm receipt into the pool

This step is performed by Intermediary. Inspect the household payment, export receipt, station expense and net amount on the same invoice. Use marker 1 to reconcile and confirm receipt into the pool, then confirm the intermediary wallet transaction.

After this step, the billing contract pays agreed station expenses once and transfers assigned net proceeds to the vault. The receivable clears once. Completed historical collections are not charged again. Continue with monthly fees and valuation approval.



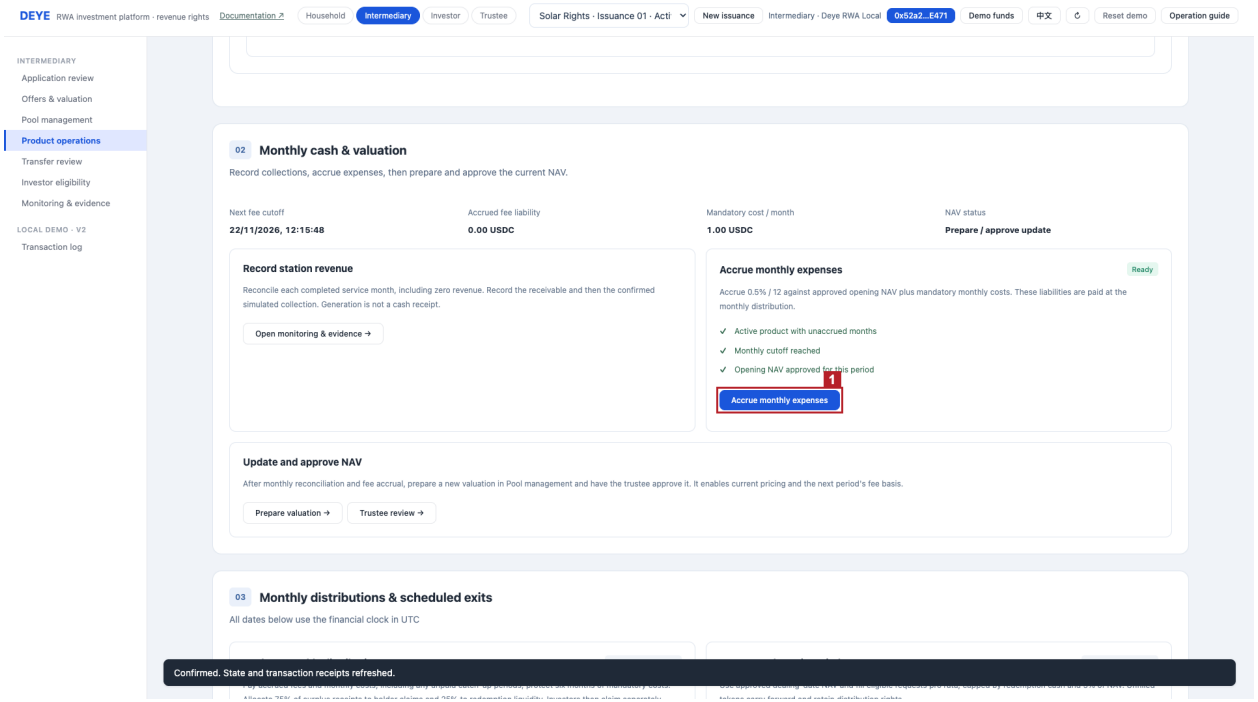
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



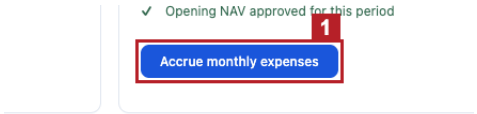
16.6 E4 Monthly pool expense accrual

This step is performed by Intermediary. Check the fee cutoff and the approved opening NAV for this fee period. If the basis is missing, follow the guide to prepare a report and obtain trustee approval. Accrue the monthly annual-fee fraction and mandatory pool costs once. Accrual records liabilities rather than another immediate cash payment.

After this step, prepare the post-accounting valuation for trustee approval.



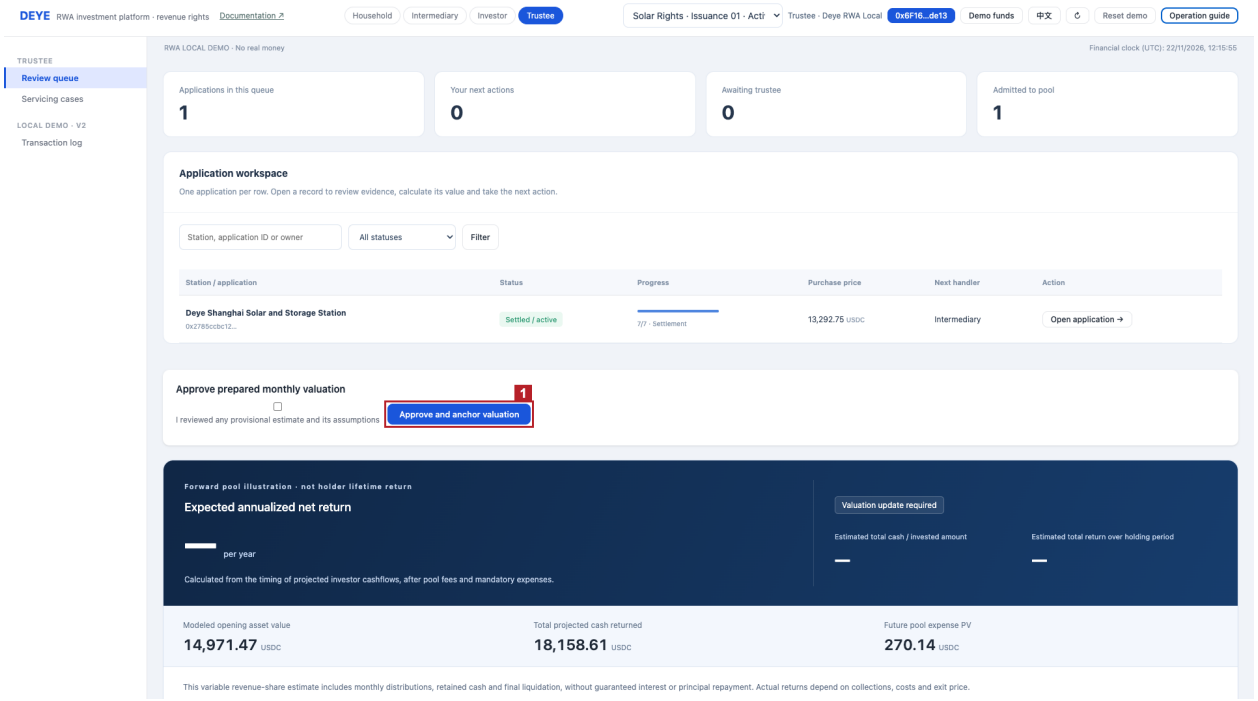
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



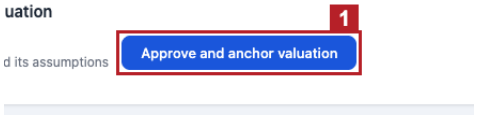
16.7 E5 Monthly valuation approval

This step is performed by Intermediary, followed by trustee. Prepare a report after receipts and fee accrual are reconciled. The trustee must check the current report and approve its anchor transaction. An older fully paid distribution must not replace this pending approval task. Reprepare a report whose inputs or validity have changed.

After this step, return to the intermediary for the monthly distribution declaration.



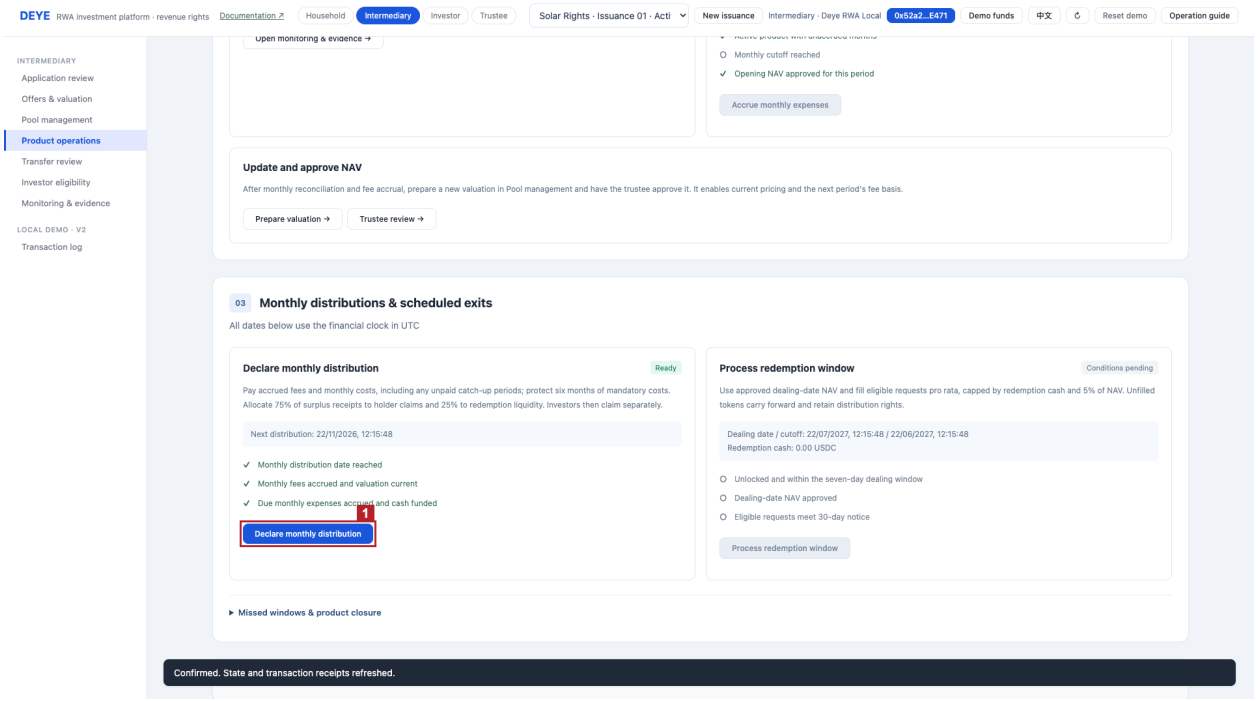
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



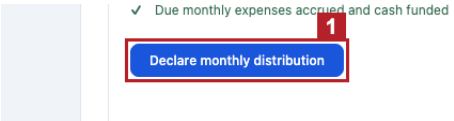
16.8 E6 Monthly distribution declaration

This step is performed by Intermediary. After the current report is approved, declare the due monthly distribution. The contract pays applicable accrued expenses, restores six months of mandatory-cost reserve and allocates eligible remaining receipts 75% to distribution and 25% to redemption liquidity. The initial retained cash is not treated as distributable revenue.

After this step, follow the explicit handoff to the investor to inspect and claim this new distribution.



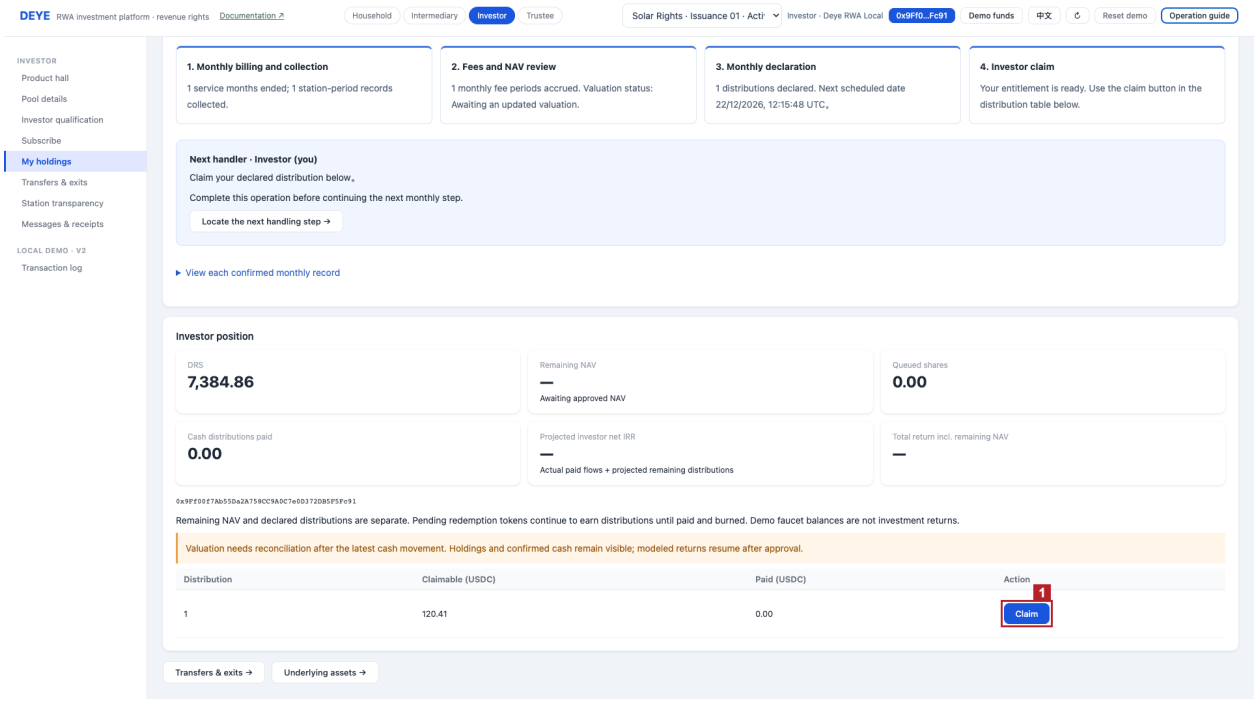
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



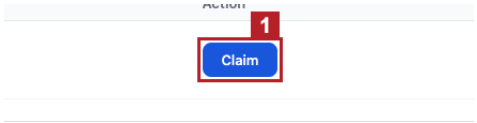
16.9 E7 Distribution claim

This step is performed by Investor. Open My holdings and locate the newly declared distribution identifier. Check the positive claimable amount, confirm Claim in Rabby and compare the wallet balance before and after confirmation. A paid record remains historical and must not continue to appear as payable.

After this step, staff can reconcile the next month after the claim or explicitly defer investor viewing.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

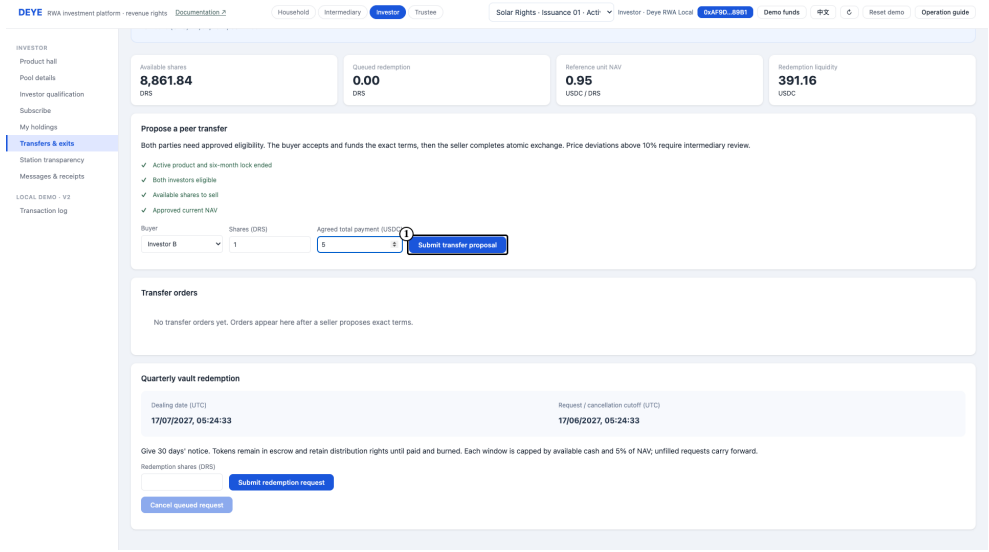


17 Peer transfers and quarterly redemption

17.1 F1 Peer transfer proposal

This step is performed by Investor A as seller. After the six-month lock, enter the eligible buyer, share quantity and negotiated USDC total in Transfers and exits. Submit the exact terms and inspect the NAV comparison. A proposal does not itself transfer shares or collect payment.

After this step, investor B accepts these terms and authorizes payment.



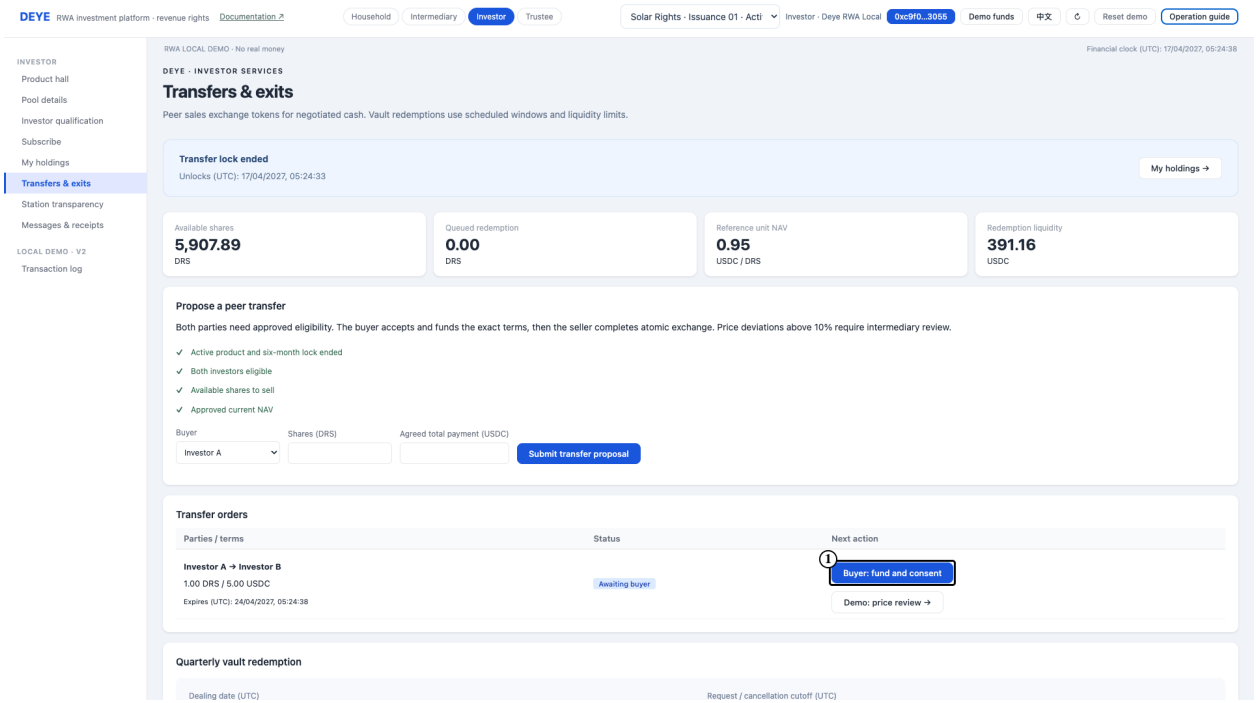
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



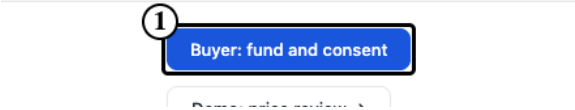
17.2 F2 Buyer consent and payment approval

This step is performed by Investor B. Open the proposed transfer and verify the seller, shares and total consideration. Complete the required USDC approval and buyer-consent transactions in the indicated order. Consent is bound to these exact terms and cannot be supplied by the seller.

After this step, obtain intermediary price review if the deviation exceeds 10%.



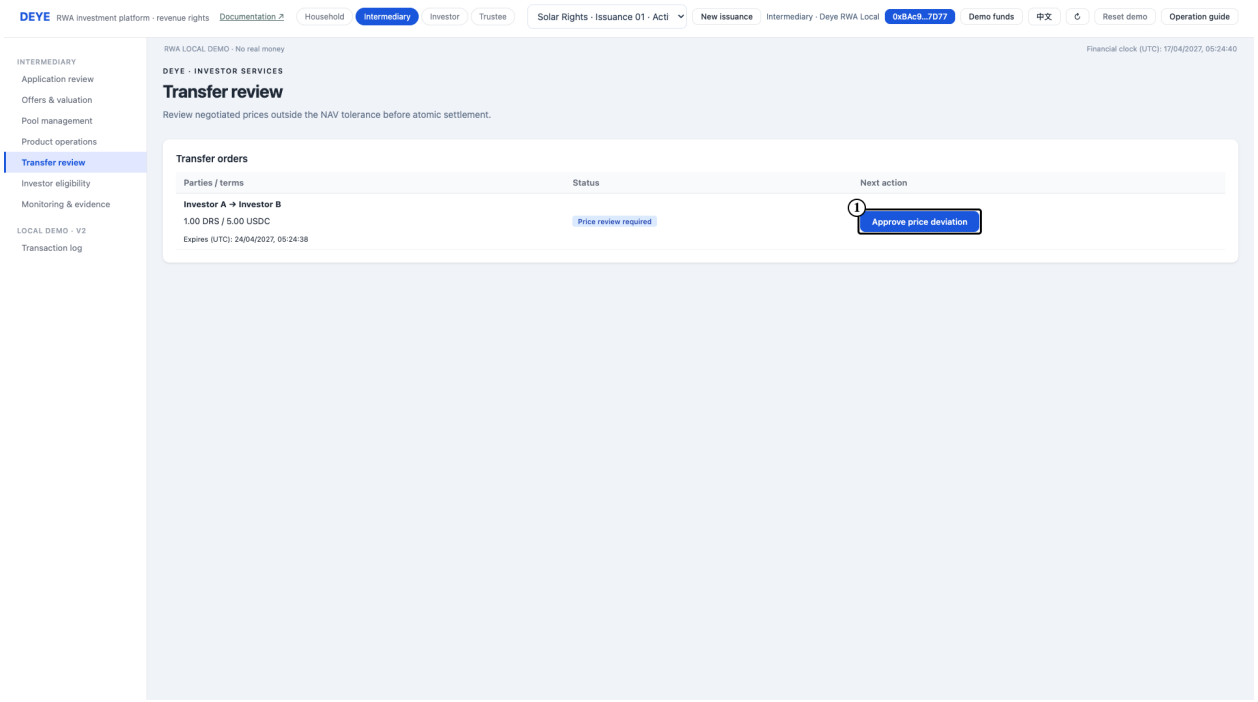
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



17.3 F3 Transfer price review

This step is performed by Intermediary. Open Transfer review and compare the negotiated price with the approved NAV reference. Record the required approval for a deviation above 10%. Review does not guarantee the price or complete either transfer leg.

After this step, the seller completes atomic settlement.



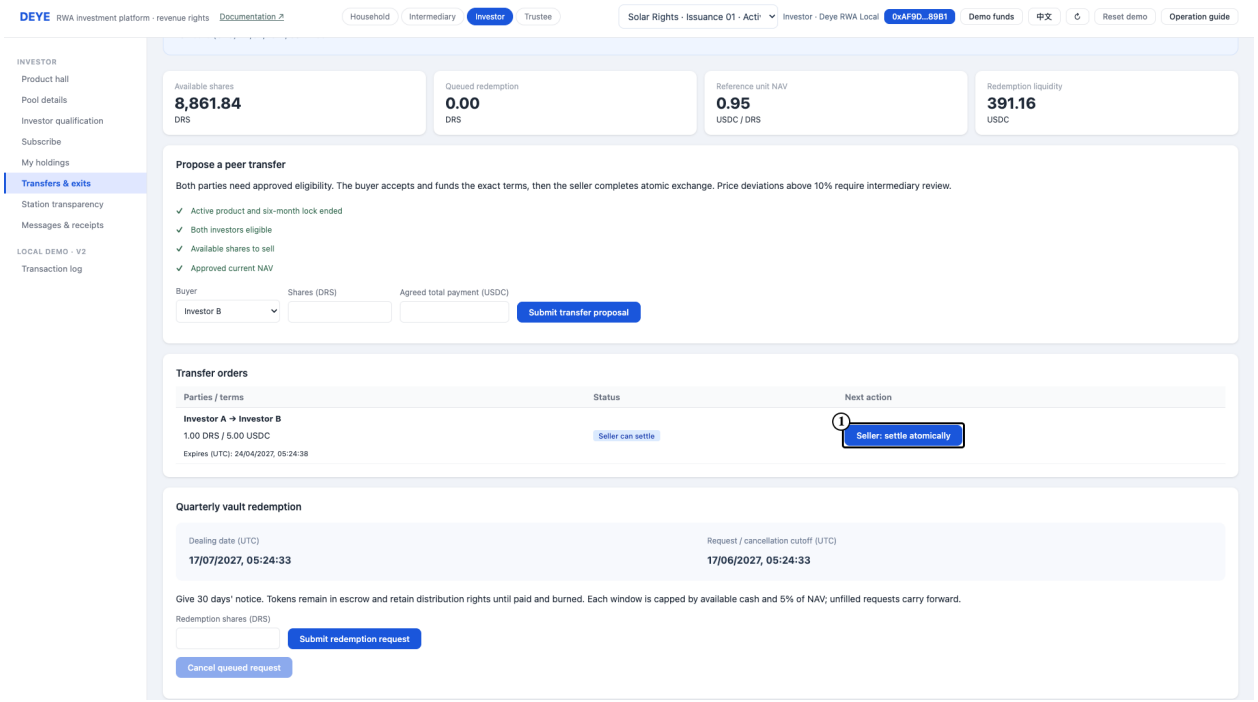
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



17.4 F4 Atomic peer settlement

This step is performed by Investor A as seller. Check buyer consent, allowance, eligibility, lock and required review, then confirm settlement. A successful transaction transfers shares to the buyer and USDC to the seller together. Verify both token transfers in Blockscout.

After this step, both investors review their updated positions.



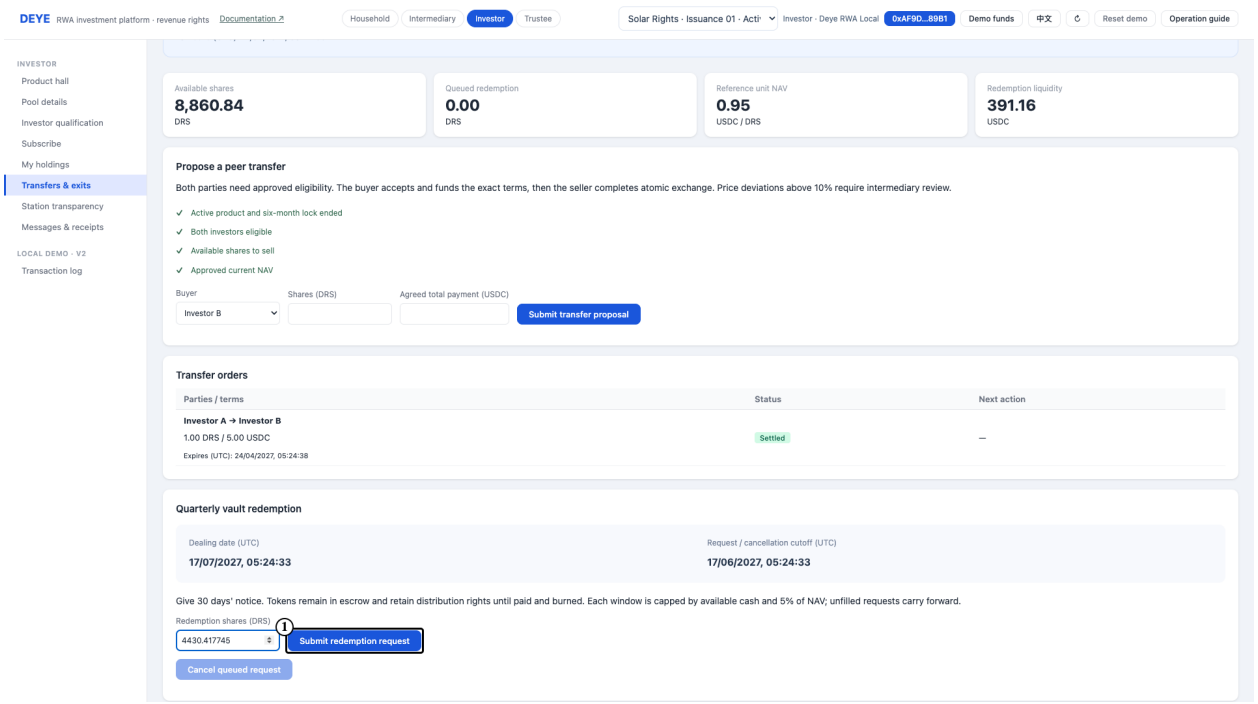
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



17.5 F5 Quarterly redemption request

This step is performed by Investor. Before the notice cutoff, enter the shares to redeem and confirm the request. The shares enter escrow and remain entitled to distributions until paid. The request is not an immediate cash redemption at today's reference NAV.

After this step, the intermediary processes an eligible quarterly dealing window after the lock.



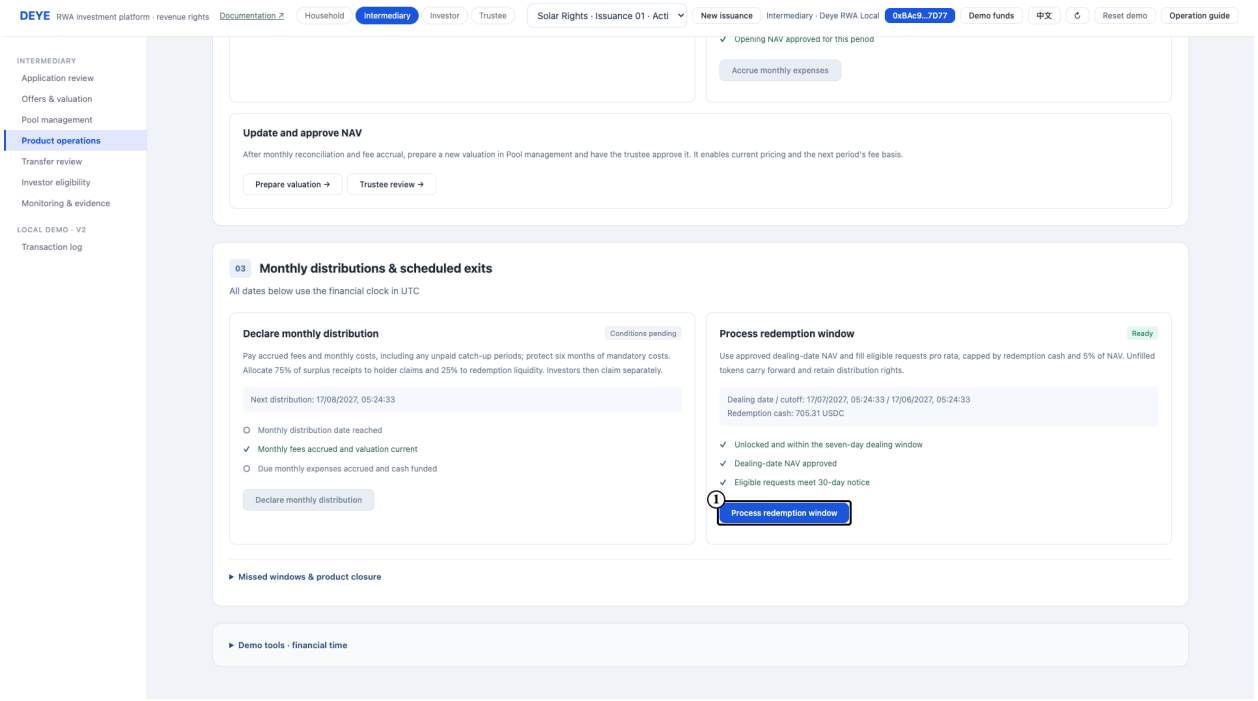
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



17.6 F6 Proportional redemption processing

This step is performed by Intermediary. At the eligible dealing date, reconcile and approve the dealing NAV before processing requests. Payment is capped by the lower of redemption liquidity and 5% of pool NAV. The contract pays and burns only filled shares and carries the unfilled balance forward.

After this step, the investor checks payment, burned shares and the remaining escrow request.



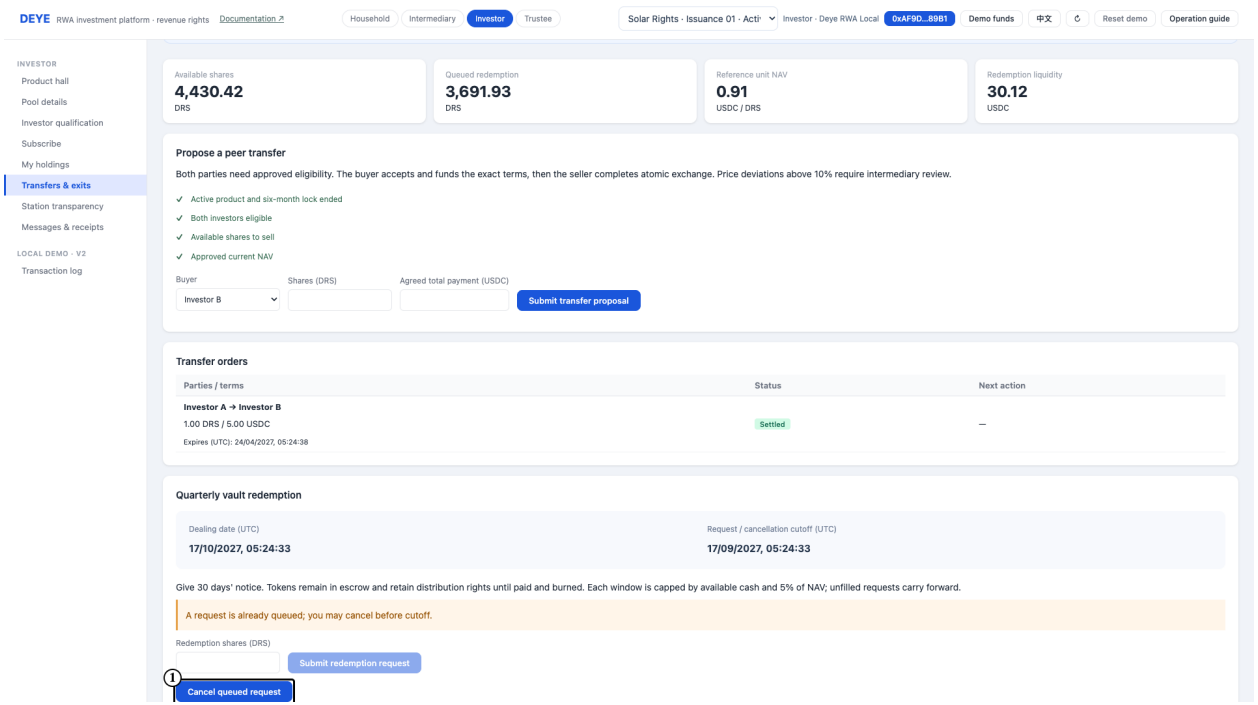
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



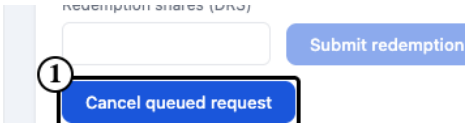
17.7 F7 Cancellation of the unfilled request

This step is performed by Investor. Before the next cutoff, cancel the outstanding unpaid portion of the redemption request. Confirm that these escrowed shares return to the wallet. Already paid and burned shares cannot be cancelled.

After this step, continue holding or submit a later request if appropriate.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

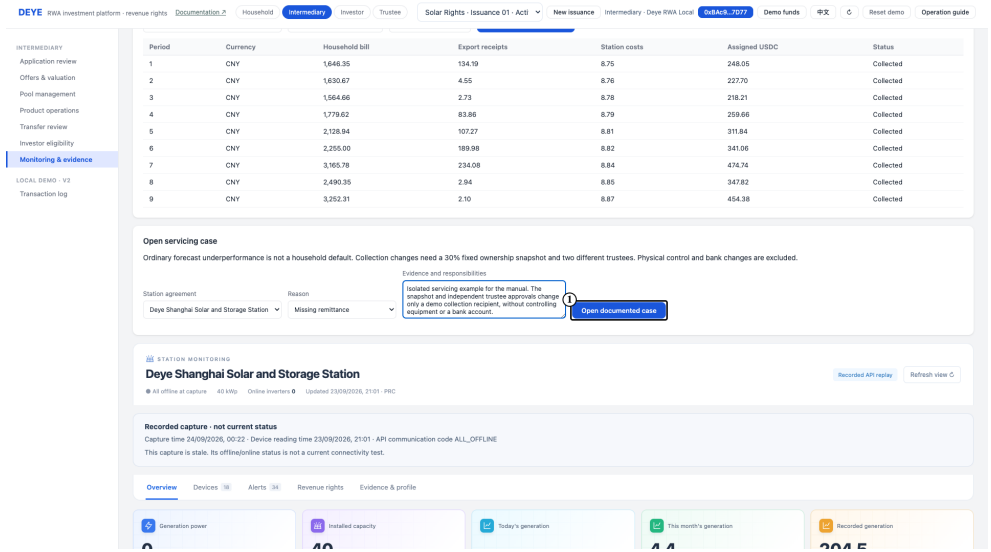


18 Servicing cases and trustee intervention

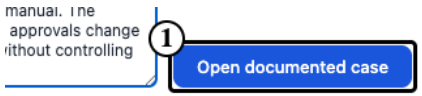
18.1 G1 Servicing case creation

This step is performed by Intermediary. Open Monitoring and evidence and document the reason, supporting evidence and proposed collection-recipient change. Create the case and inspect its fixed ownership snapshot. Ordinary forecast underperformance is not automatically a household default.

After this step, investors review the case and may support it using snapshot weights.



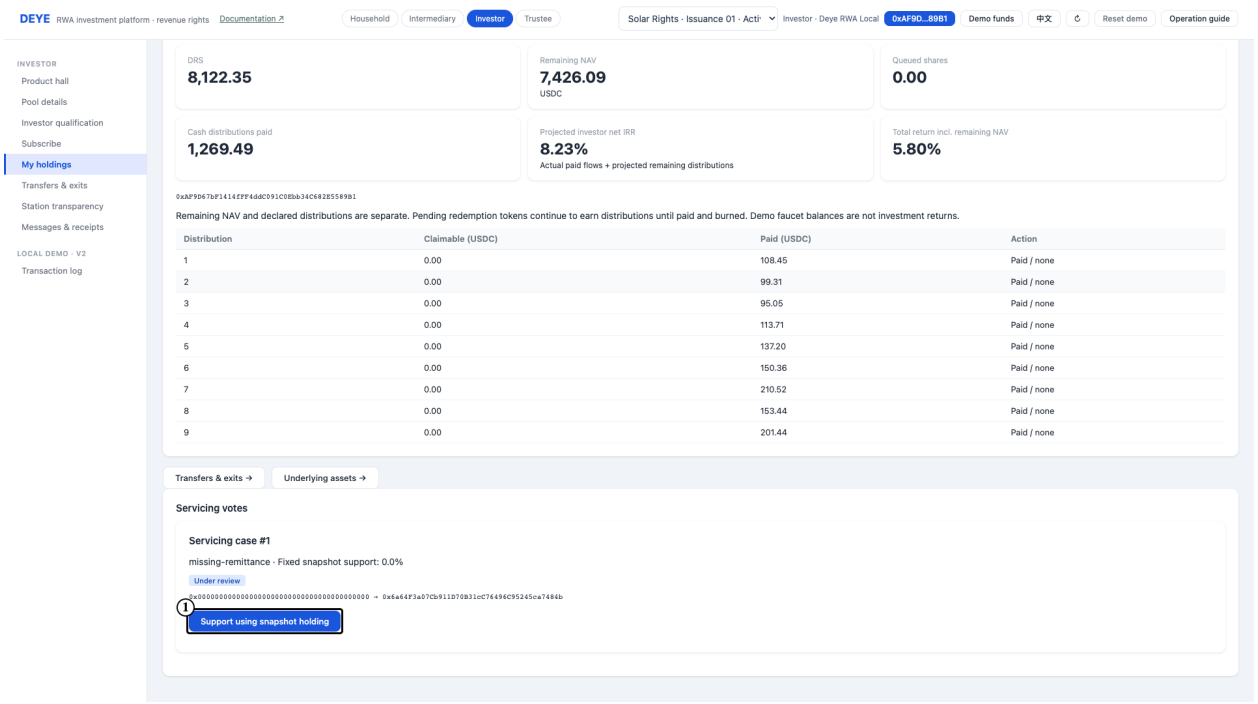
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



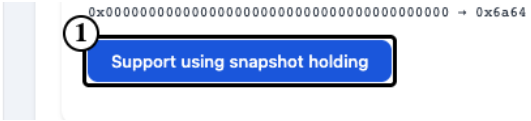
18.2 G2 Snapshot-weighted support

This step is performed by Investor. Read the case and the proposed recipient, then confirm support with the investor wallet. The contract uses the fixed snapshot and requires 30% support before trustee action. Later transfers cannot create duplicate voting weight.

After this step, trustee A independently reviews the eligible case.



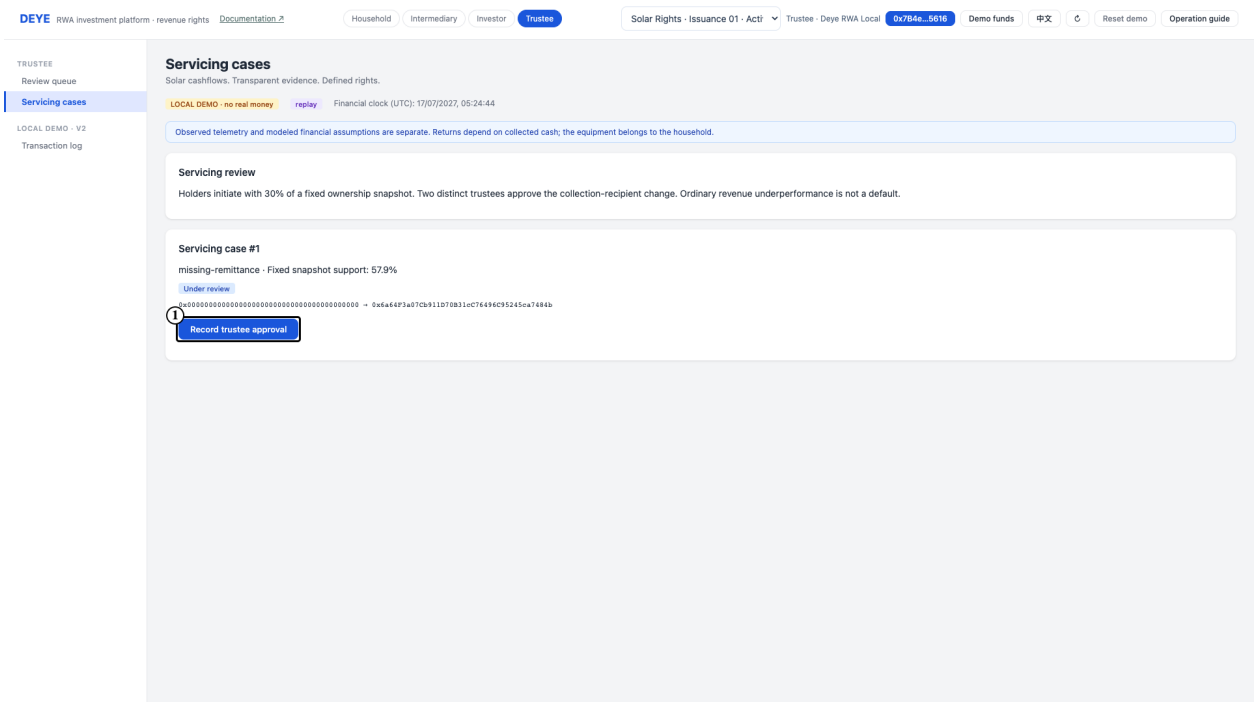
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



18.3 G3 First trustee approval

This step is performed by Trustee A. Open the case, check evidence, snapshot support and recipient, then approve with trustee A's wallet. The contract records this distinct address but does not yet execute the recipient change.

After this step, trustee B performs a separate review of the same case.



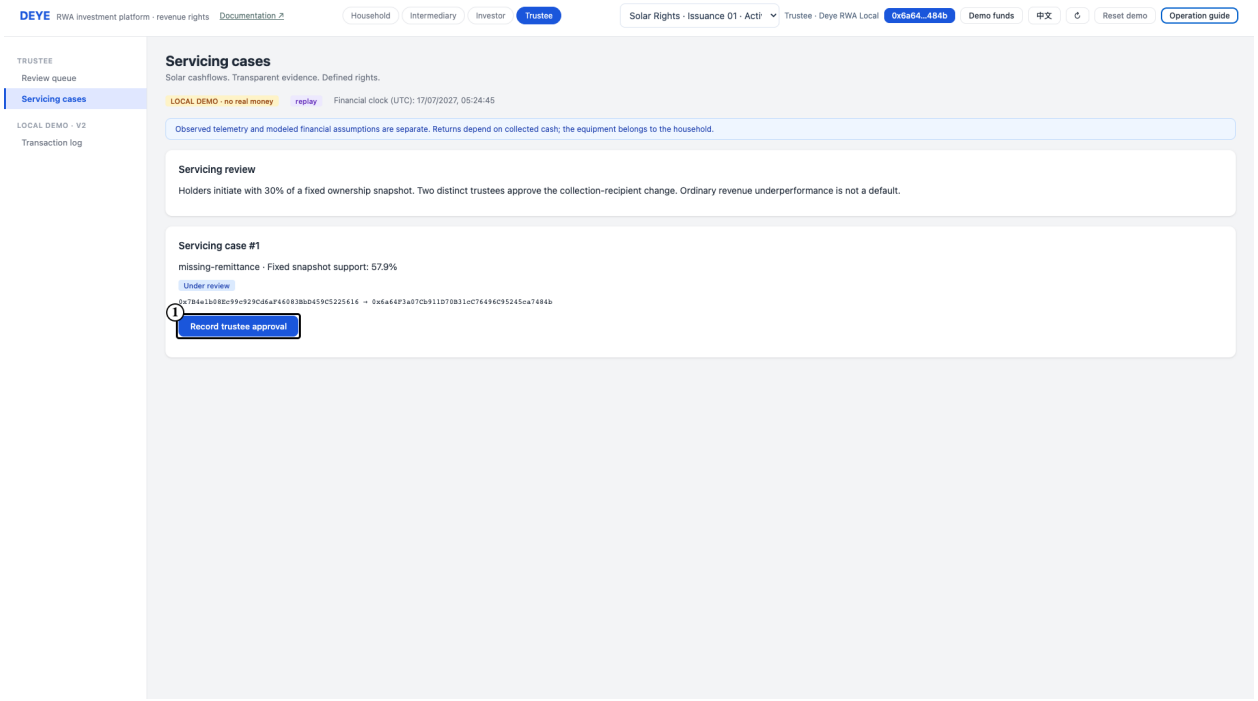
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



18.4 G4 Second trustee approval and execution

This step is performed by Trustee B. Switch to trustee B's configured wallet and sign in again. Review the same case independently and approve it. Two different trustee addresses are required to authorize the demo collection-recipient change, which does not control equipment or a real bank account.

After this step, the intermediary checks execution and continues servicing.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.

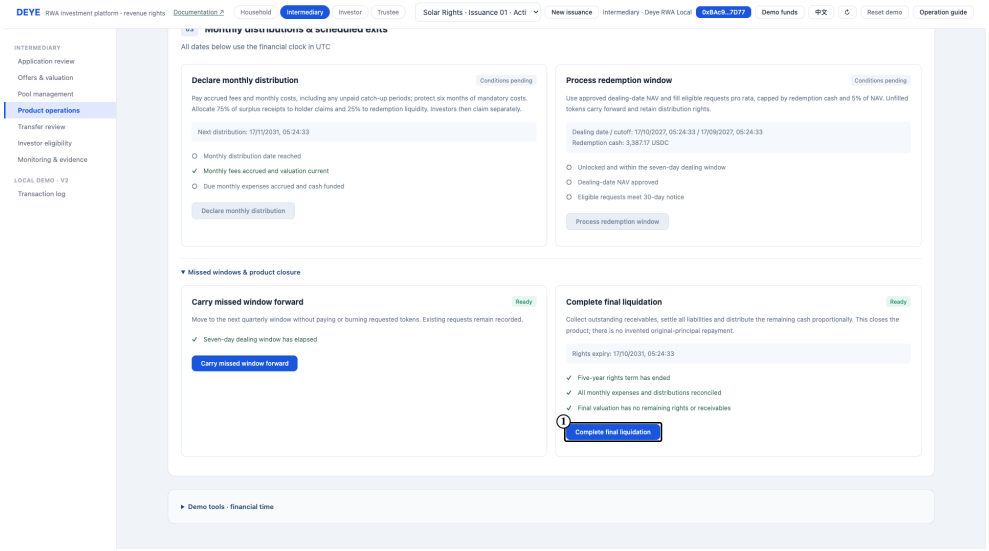


19 Expiry, closure and credential release

19.1 H1 Product closure at rights expiry

This step is performed by Intermediary. After all 60 service months, reconcile receivables, accrued liabilities and distributions, then obtain the final required valuation approval. Confirm Close product when every condition is satisfied. Remaining cash becomes a final distribution rather than an invented repayment of original principal.

After this step, investors claim remaining amounts and the owner retrieves the station credential.



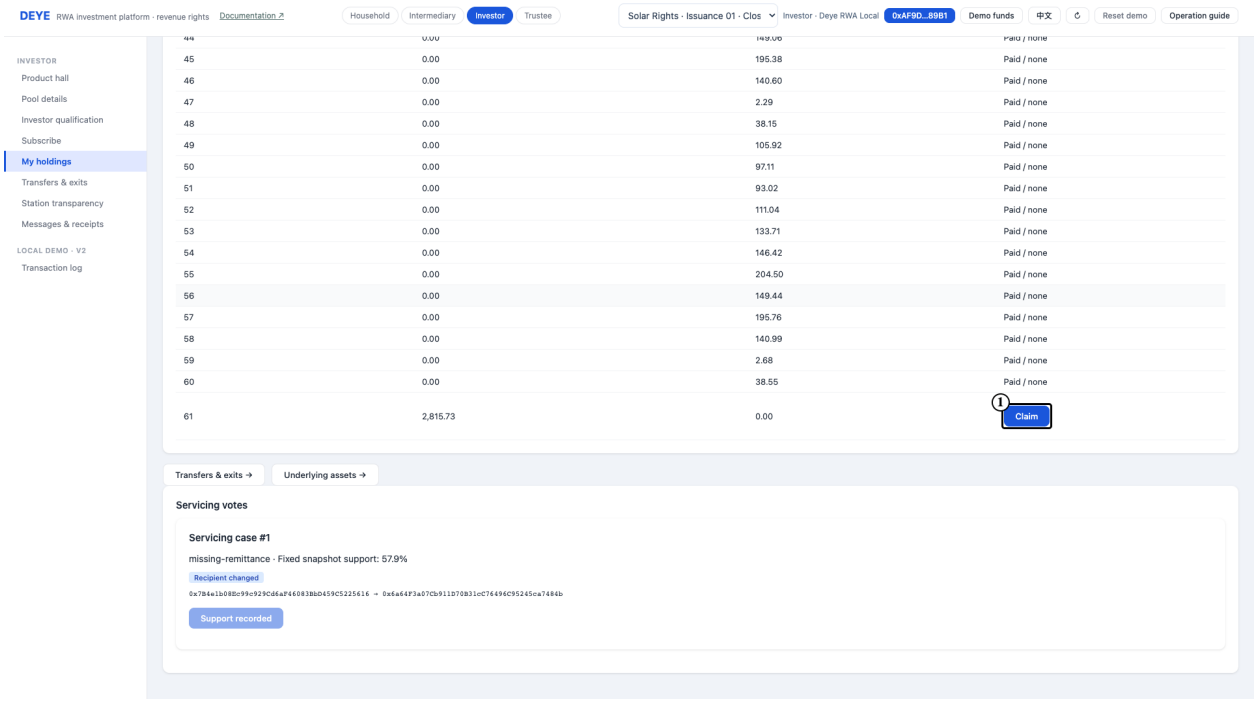
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



19.2 H2 Final and outstanding distribution claims

This step is performed by Each investor. Review every unpaid distribution in My holdings and claim the amounts belonging to this wallet. Check the confirmed USDC transfers. Previously paid distributions are not paid again, and a remaining historical DRS balance does not extend expired revenue rights.

After this step, the owner completes station credential release.



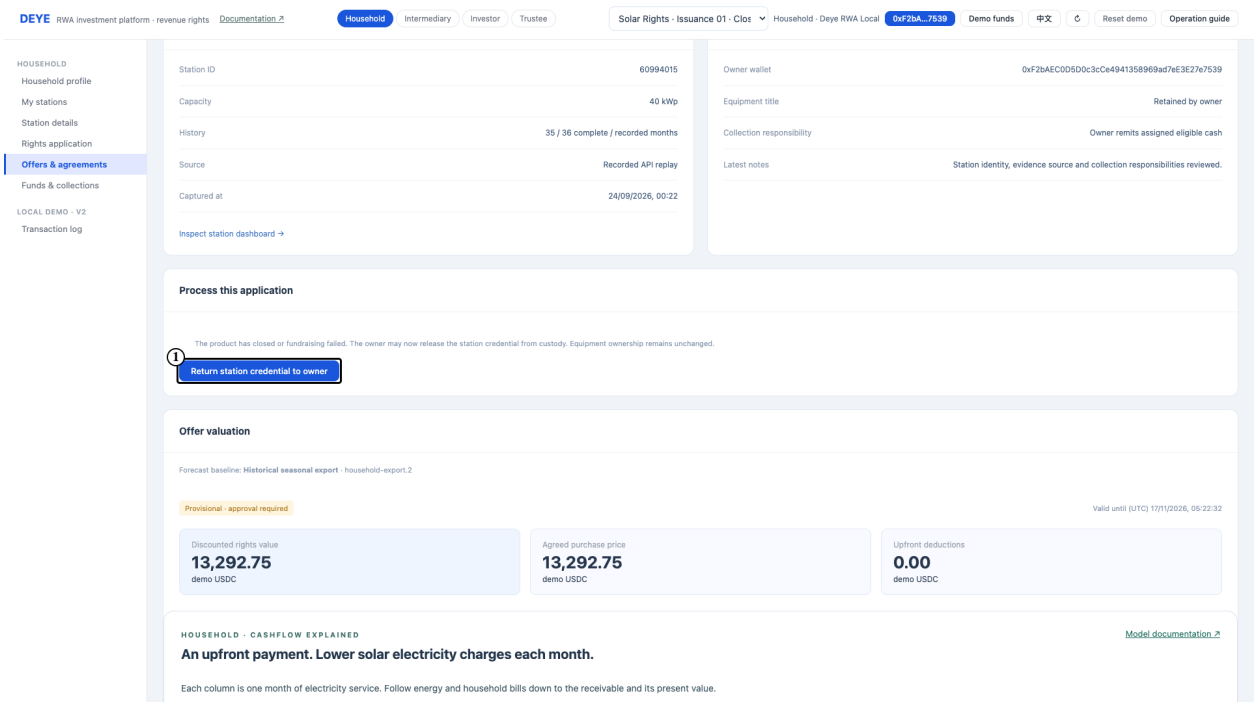
Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



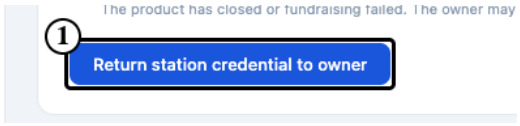
19.3 H3 Station credential release

This step is performed by Owner. Open the expired agreement's release task after product closure. Confirm the release transaction and check that the station NFT returns from the vault to the owner. Equipment ownership remained with the owner throughout the agreement.

After this step, the product lifecycle is complete and its transaction history remains queryable.



Marker 1 identifies the control or information to inspect for this step. Amounts and dates refer to the isolated demonstration.



20 Cashflow calculations and valuation

The forecast uses up to 36 available historical months to form calendar-month seasonal baselines, then applies degradation, availability, export behavior and tariff schedules. Fewer than 12 complete months requires a labeled provisional estimate and review. Imbalanced generation and export samples require investigation rather than silently capping export at 100% and treating the result as valid self-use.

$$S_t = \max(G_t - E_t, 0)\, q_t, \qquad B_t = S_t\, p_t^{grid}\, d_t.$$

G denotes forecast generation, E is validated eligible export and q is the self-use delivery factor. S is delivered self-use and B is the owner's electricity payment at discount factor d. The grid-price comparison is S times the grid tariff. The difference is the saving on this delivered solar energy, not a claim that the owner's entire grid bill falls by the same percentage.

$$C_t = (B_t + E_t p_t^{export} - K_t)\, f_t\, a_t, \qquad V_0 = \sum_{t=1}^{60} \frac{\widehat{C}_t}{(1+r)^{\tau_t/12}}.$$

K represents agreed station costs borne by the rights, f is the demo FX conversion into USDC and a is the signed assignment proportion. New agreements default to 100% assignment, with the owner retaining a benefit through the discounted tariff. Discounting uses expected cash-collection dates, so tau includes receipt delays. The default 10% nominal annual discount rate is a pricing assumption, not a promised return, and 6% and 14% sensitivities are also shown.

20.1 Discount-rate meaning and pricing basis

The discount rate converts each expected future net receipt into a value on the valuation date. A higher rate reduces present value for unchanged cashflows. It differs from the owner paying 80% of the grid tariff, the pool retaining 10% of initial cash at settlement, and the investor's eventual net IRR. The current 10% is a demonstration pricing assumption, not yet fully calibrated against this product's realized collections and comparable transactions. The 6% and 14% cases are sensitivity scenarios, not a probability range.

$$r_{\text{rights}} \approx r_{\text{benchmark}} + \pi_{\text{payment}} + \pi_{\text{operations}} + \pi_{\text{legal/liquidity}}.$$

Start with a low-risk benchmark in the valuation currency with a duration comparable to the receipts. Assess household payments and grid export collections, generation and operations, data integrity, contract enforceability and illiquidity. The equation is a calibration framework requiring evidence, not a way to generate one objectively correct rate. The intermediary should document each premium using agreement terms, payment history, grid settlement, station performance, exit limits and comparable transactions; the trustee reviews the rationale. Correlated risks cannot be priced by arbitrarily adding percentages.

When forecast cashflows already deduct expected generation loss, expenses or nonpayment, the same expected loss must not be charged again in full through the risk premium. For example, if 100 USDC is billed but only 90 USDC is expected to be collected, the report must state whether it discounts 100 or 90 and what residual risks the rate covers. Here, nominal means cashflows and rates are not adjusted for inflation; the system compounds the annual rate over actual days, rather than multiplying a simple monthly rate by 12. Tax treatment, cashflow currency, price growth and expected receipt dates must also be consistent.

20.2 Currency convention and current system limitation

The current system converts CNY household bills, export proceeds and station costs into USDC using a fixed demonstration FX assumption, then discounts them at one 10% annual rate by expected collection date. The rate should therefore be read as the required return on USDC-denominated rights cashflows. US Treasury yields can be a low-risk reference, but USDC, cross-currency settlement and enforceability risks still require assessment. Fixed FX does not imply a constant future exchange rate. A CNY valuation should instead discount CNY cashflows with a CNY rate and convert the present value on the valuation date; applying a CNY bond yield directly to USDC cashflows is inconsistent.

For 41.888 USDC expected exactly one year later, present values at 6%, 10% and 14% are approximately 39.52, 38.08 and 36.74 USDC. Each monthly receipt is discounted using the actual days from valuation to expected collection; five years of receipts are not divided by 1.10 just once. A higher discount rate changes the quote, not the owner tariff or the contractual investor cashflows automatically.

20.3 Comparables and approval record

As of 12 November 2026, the US Treasury five-year yield was 4.81% on 10 November and the ChinaBond five-year government yield was 1.329% on 12 November. Subtraction from 10% gives residuals of 5.19 and 8.67 percentage points respectively, but the currencies cannot be mixed and neither result validates a premium. The IEA reported nominal after-tax WACC of 4.4% to 5.4% for revenue-supported Chinese utility-scale solar projects in 2019 and 6.4% to 6.9% with merchant risk. These historical financing costs refer to a different scale and rights structure. Centrifuge now follows product-specific NAV policies; the 3% rate in its legacy Tinlake explanation is only an illustrative calculation, not a protocol-wide rate.

Public sources IEA (2021); Centrifuge NAV; Tinlake; ChinaBond; US Treasury.

Each formal quote should record its valuation currency, benchmark curve and observation date, evidence for the premiums, expected losses already deducted from forecast cashflows, compounding convention, present values at 6%, 10% and 14%, reviewer and validity period. Product-specific calibration requires sufficient realized household payments, grid proceeds and comparable trans-

action prices. A revised rate takes effect through a new quote or revaluation, never by silently rewriting a signed agreement.

20.4 Comparison of the three valuation baselines

The three methods change the forecast baseline for generation or export allocation in each future service period. The formulas for discounted household payments, export receipts, agreed costs, assignment and cashflow timing remain common. These baselines are not three different rights agreements, and the valuation discount rate is not a guaranteed investor return.

Method	Generation baseline	Export allocation and selection conditions
Long-run historical seasonal	Up to 36 complete metered months, averaged by matching calendar month and adjusted for availability, degradation and remaining time.	Uses the export-to-generation ratio from eligible history. Short, missing or inconsistent evidence requires a provisional label and review.
Recent export ratio	Generation still uses the long-run historical monthly mean; one recent day is never projected over the whole term.	Uses at least 7 and at most 30 eligible complete days before capture. An inconsistent recent energy balance disables this option; the choice requires review.
PVGIS irradiation climate baseline	Converts 2019–2023 hourly modeled power to calendar-month specific yield using recorded or explicitly provisional coordinates, DC capacity and array design.	Export allocation still uses eligible Deye history. The report discloses parameter status, complete hourly records and source digest; this is a historical-weather scenario, not a future-weather forecast.

As of 12 November 2026, the private-chain, original Base Sepolia and self-hosted Oracle applications all include the PVGIS baseline. The Shanghai station can load the provisional demonstration inputs below to preview a new quote. Their status remains “provisional demo estimate”; they are not an operator-confirmed as-built design, and signed agreements and published products are not automatically repriced.

20.5 PVGIS irradiation-based climate valuation

The third method uses PVGIS 5.3 ERA5 historical weather. The server reads modeled hourly power for 2019–2023, checks the complete series and the returned coordinates, tilt, aspect, crystalline-silicon type and units, then derives monthly mean kWh/kWp. The intermediary records the source of each station input and distinguishes API values from provisional estimates.

20.6 Provisional Shanghai station inputs

Input	Value	Source and limitation
Coordinates	31.173725° N, 121.48858° E	Recorded Deye API locationLat and locationLng; no independent site survey.
DC capacity proxy	40 kWp	API installedCapacity normalizes to 40 kW but does not distinguish DC from AC; the same numerical value is used provisionally.
Tilt and aspect	30°, −2°	PVGIS model-optimal fixed angles at the recorded location, slightly east of south; not measured as-built angles.
Additional first-year loss	14%	Demo scenario assumption, not a Deye loss measurement. The hourly PVGIS request uses zero loss, followed by one deduction.

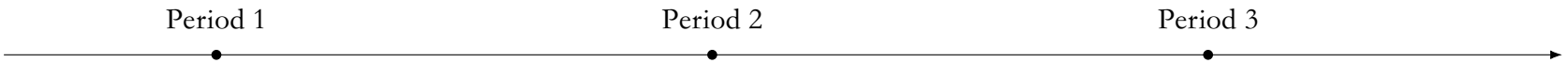
The replayed Deye observation was captured on 24 September 2026 China Standard Time. The complete 43,824-hour PVGIS series for 2019 – 2023 yields a modeled mean of 1,465.70 kWh/kWp annually. At 40 kWp, 14% additional loss and the existing 98% availability assumption, illustrative first-year generation is 49,411.78 kWh. This is not a Deye meter reading or proof of the DC nameplate. The raw PVGIS response SHA-256 is c2779962f6bc066525f9556b05ade82bcba9e8495c6eeb30c1cf31f20eef067c. Intermediary and trustee review remains necessary; tariffs, expenses and export allocation are separate inputs.

$$G_t^{\mathrm{PVGIS}} = \Big(\sum_{m\in t} \bar{Y}_m \frac{\Delta t_m}{D_m}\Big) C_{\mathrm{DC}}(1-L)A(1-d)^{(t-1)/12}, \qquad E_t = G_t^{\mathrm{PVGIS}} r_t^{\mathrm{Deye}}.$$

Y is modeled PVGIS monthly specific yield for 1 kWp with API loss set to zero. Delta t over D is the fraction of each calendar month actually covered by the service period. C is confirmed DC nameplate capacity or an explicitly provisional numerical proxy. L is separately reviewed first-year additional system loss applied once, and A and d are availability and degradation. Export ratio r comes from eligible Deye history. The common formulas then produce household payments,

export receipts, costs and discounted value. The server archives the raw PVGIS response with SHA-256 and separately binds its raw and complete-parameter digests. This scenario does not forecast future weather or prove Deye-metered generation or grid payment; intermediary and trustee review remain required.

20.7 Owner cashflow timeline for the first three periods



Calculation	Period 1	Period 2	Period 3
Generation kWh	3,861.20	4,307.49	3,732.04
Export kWh	279.01	308.98	67.61
Delivered self-use kWh	3,223.97	3,598.66	3,297.99
Grid-price comparison CNY	1,934.38	2,159.19	1,978.80
Owner electricity payment CNY	1,547.51	1,727.36	1,583.04
Owner saving CNY	386.88	431.84	395.76
Export income CNY	97.65	108.14	23.66
Station costs CNY	8.75	8.76	8.78
Assigned net receivable USDC	229.10	255.74	223.71
Discounted present value USDC	225.48	249.68	216.64

These values come from one isolated product report and are modeled examples. The full report covers 60 periods. Owner bills, export income and costs are calculated separately and discounted at their expected collection times.

20.8 Investor cashflow timeline for the first three months



Calculation	Month 1	Month 2	Month 3
Forecast cash collected USDC	0.00	229.10	255.74
Accrued annual-fee portion USDC	6.04	6.09	6.06
Mandatory monthly pool costs USDC	1.00	1.00	1.00
Pool distribution USDC	0.00	166.51	186.51
Pool cash after distribution USDC	1,469.93	1,525.43	1,587.60
Forecast payout per 1,000 USDC	0.00	11.27	12.63

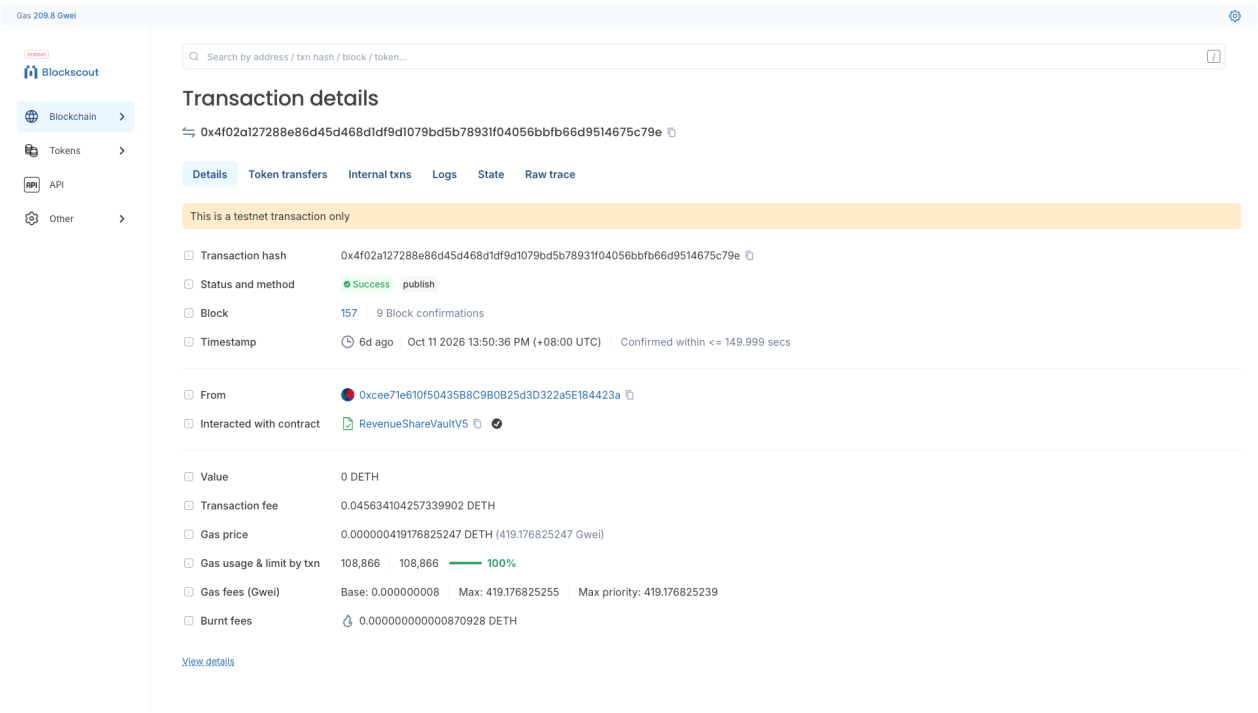
The vault automatically retains 10% of initial subscription cash and uses the balance to purchase rights. Monthly declarations require the relevant collections, expenses and valuation conditions and do not guarantee a positive payout each month. Investors must still claim. Monthly distribution does not remove the six-month transfer lock or change quarterly redemption windows.

Investor NAV consists of cash, collectible receivables and remaining rights, less liabilities and applicable future costs. Declared distributions are separated from dealing NAV to prevent double counting. Projected net IRR uses the initial investment, expected distributions and final cash. Assess paid cash, remaining NAV and total return together.

21 On-chain record reconciliation

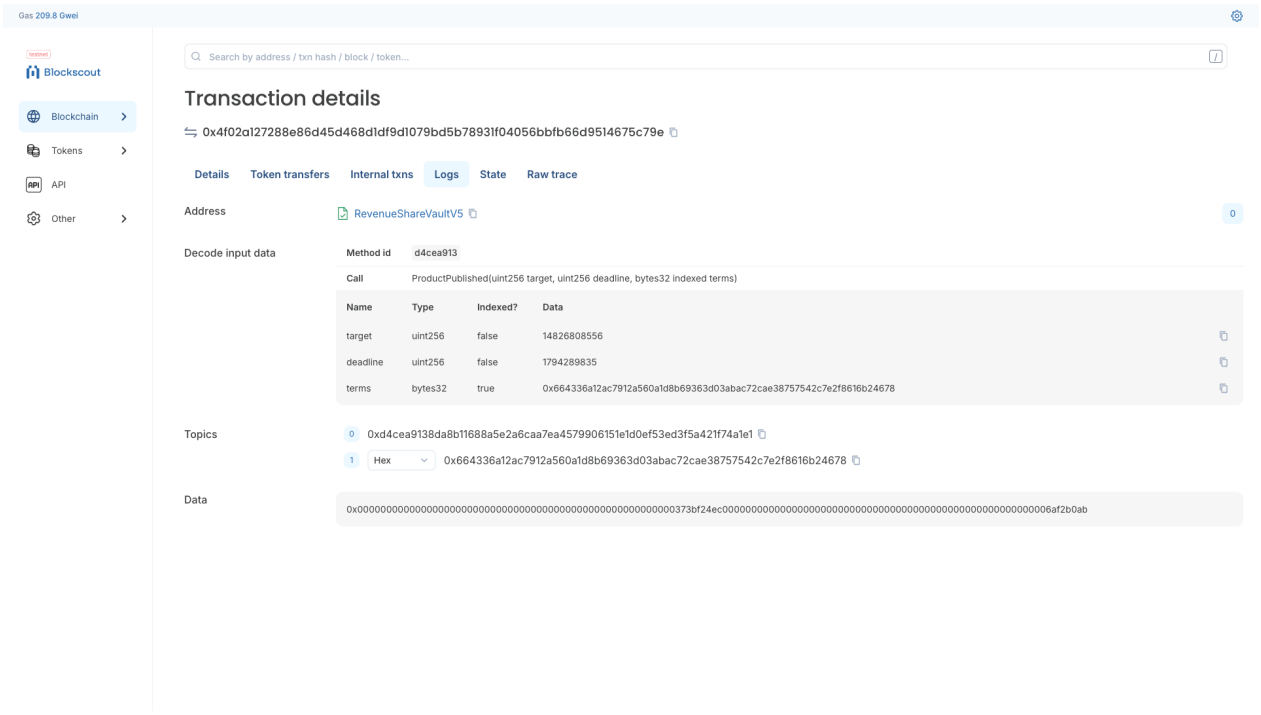
For each monthly invoice, reconcile BillPublished, HouseholdPaid, ExportFunded and ReceiptConfirmed separately. Household payment must come from the named owner to the product billing contract for the invoiced amount. The intermediary signs receipt confirmation. Check net proceeds into the vault and station costs paid once. Paid but unconfirmed amounts remain in the billing contract, separate from vault cash.

For NFT minting, check the station token ID and owner. For approval, check the authorized vault. For admission, check custody, agreement digest and purchase price. For valuation, check report digest, reviewer and validity. Distinguish subscription transfers from settlement, and reconcile cumulative escrow, issued DRS, owner payment and retained cash.



Explorer transaction example. Check execution status, sender, target contract and block.

Declaration and claim are different transactions. Declaration creates an entitlement, while claiming transfers USDC to the investor. A peer settlement should contain both the DRS and USDC legs. A redemption should reconcile the fill, payment, burn and remaining escrow. A servicing case should show its snapshot, support weight and two distinct approval addresses.



Events and token-record example. A digest links the off-chain record without storing the full underlying material on-chain.

22 Troubleshooting and test boundaries

22.1 Wallet or role mismatch

Check the full address, Chain ID and selected role. Switch the wallet account and sign in again with the intended role. Repeated role-button clicks do not grant permissions.

22.2 Insufficient settlement funding

The contract automatically retains 10% as initial cash. Minimum funding typically covers the purchase price divided by 90%. Add the displayed shortfall and have the intermediary review the new total rather than editing accounting records.

22.3 Missing valuation or fee prerequisites

Use Locate operation to open the specific remedy. The intermediary reprepares an expired or changed report and the trustee approves it. Reaching the cutoff does not itself approve NAV.

22.4 Distribution or balance not updated

Check whether the current period was declared, then distinguish claimable amounts from paid history. Wallet USDC, escrow and investment NAV are separate. If confirmation precedes indexing, retain the hash and refresh later.

22.5 Rejection, pending transactions and recovery

After rejection, review and retry the original action. If a hash exists, inspect its receipt before sending another payment. Refresh and backend restart should recover the original operation. A complete test-chain rebuild requires the operator to notify users about clearing stale test-network transaction caches.

22.6 Remote environment unavailable

The GitHub homepage remains readable, but the app, RPC and Blockscout depend on the Mac, Docker and tunnels. The operator first checks unified:status, then checks the two backend services and tunnels separately. If a tunnel address changes, update the unified entry and GitHub static configuration.